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NUMBER 1

Elementary Gravity And Magnetics For Geologists and Seismologists

By L. L. Nettleton

society of exploration geophysicists

MONOGRAPH SERIES

Paul C. Wuenschel, Editor

NUMBER 1

ELEMENTARY GRAVITY AND MAGNETICS FOR GEOLOGISTS AND SEISMOLOGISTS

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SOCIETY OF EXPLORATION GEOPHYSICISTS

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PREFACE

In 1961 the writer made a distinguished lecture tour for the American Association of Petroleum Geologists. Much of the material for these lectures was published as "Gravity and Magnetics for Geologists and Seismologists" in the Bulletin of the AAPG (Nettleton, 1962).

In 1968 and 1969 the writer participated in the Continuing Education Program of the Society of Exploration Geophysicists and gave a series of lectures over a total period of from 6 to 10 hours in several days at each of six local groups or sections of the Society. The subject matter for these lectures was rather largely expanded from that which had been given previously in the AAPG tour. The present monograph has been prepared to make readily available the material given in the SEG talks.

To the writer an important factor in participation in the SEG program and in preparing this monograph has been that it may lead to a better understanding by the nonspecialists in potential methods of the place which these methods should have in overall petroleum exploration planning and practice. Therefore, the material included which was selected from the extensive body of published material and from the writer's experience of over forty years in this field, has been chosen to give the basic principles of operations of the potential methods and to illustrate their effectiveness. Technical details and mathematical derivations are reduced to those considered desirable to clarify the applications.

If this material contributes to a better appreciation of the role of the potential methods and brings about their greater application in petroleum exploration which this writer firmly believes is well deserved, the purpose of this effort will have been achieved.

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INTRODUCTION

The purpose of this work is a general review of the gravity and magnetic methods of geophysical exploration as applied in the search for petroleum. This material is not designed for the gravity and magnetic specialist but rather for the geologists and seismologists who may not have a thorough appreciation of the applications of these methods in the overall exploration picture.

A subtitle for this monograph might well be "The Other Five Percent." This is because the seismic method and its associated data processing account for some 95 percent of the total expenditures for petroleum exploration geophysics so that whatever application is made of the gravity and magnetic methods comes out of the other 5 percent. This does not mean that these methods make a proportionately small contribution to the overall exploration effort. Because of the relatively rapid rate of progress in the field, particularly by airborne magnetics, the total area covered by gravity and magnetic surveys may be greater than that covered by the much greater seismic expenditures. As a very rough rule-of-thumb, the relative cost per unit area of magnetic, gravity and seismic field work with data processing stand in the ratio of 1 to 10 to 100. It is the hope and purpose of this monograph that a better appreciation of the value of the potential methods and understanding of their applications may be brought about so that they can be applied with proper perspective in the overall exploration picture.

From the beginning of geophysical exploration in the petroleum industry in the 1920's, three basic physical principles were used: i.e., the measurement of small variations in the magnetic field, the measurement of small variations in the gravitational field, and the propagation of elastic waves through the earth. These three and only these three physical principles are the basis for practically all of the geophysical work up to the present time. Many other methods have been conceived and tried in the field in a limited way, but none has persisted to the extent that field operations are carried out on a scale at all comparable with that of the three primary methods listed above.

The seismic method, of course, usually is much more direct in its relation to geology than the potential methods. Reflection zones or horizons frequently are directly correlative with geologic strata and give relatively accurate measures of their depth and form. In many cases, however, correlations with geology may be uncertain or misleading. In such cases, gravity and magnetic data may contribute by establishing bounds on possible correlations and provide lithologic information.

To some degree, the dominance, and occasional over-application of the reflection seismic method may be attributable to the relative simplicity of the basic principles applied. The concept of initiating a disturbance at the surface, detecting a return signal from a reflecting layer and determining the depth of that layer by knowing the speed of the wave propagation is very simple. On the other hand, the magnetic and gravity methods are both "potential" methods and depend on action at a distance. This means that as measurements are made farther from the disturbing anomalous magnetic or density contrast, the effect becomes more and more attenuated and smoothed out so that details are less evident. The effect is a little like that of the covering of the roughness of objects or other irregularities of the ground surface by a heavy snow fall. Features on the ground which are immediately recognizable under a thin snow cover become more and more rounded out as the snow gets deeper until their effect becomes only rounded bumps on the surface. Even a body as large as an automobile could be covered to such an extent that it is not immediately recognizable as such without additional information. For instance, a series of such rounded bumps in a line along the side of a street, would be recognized as cars parked along the curb. Discrimination among them would be difficult; a Volkswagen could be distinguished from a Cadillac but not a Ford from a Chevrolet. In a somewhat similar way the interpretation of magnetic and gravity surveys is inherently ambiguous and any quantitative control on the nature of the body causing the disturbance must come from additional information. Such information can come from drilling contacts, seismic results or reasonable geologic limitations. In spite of such ambiguities and lack of precision, magnetic and gravity surveys can impose very definite limits on geological interpretations and can thereby make specific and useful contributions to the overall exploration picture.

PART I
THE GRAVITY METHOD

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CHAPTER I

SOURCES OF GRAVITY ANOMALIES

All gravity anomalies come from horizontal variations in density. If the earth materials were in layers of horizontally uniform density, there would be no gravity anomalies no matter what the vertical variation in density might be. If layers with different densities are disturbed, anomalies in mass concentration are produced. In Figure 1, let us assume that the layers, 1, 2, 3, and 4 have successively higher density values, d_1, d_2, d_3, and d_4. The flat-lying strata at the margins of the figure will produce no gravity anomaly there because there is no horizontal variation. Over the central part of the figure, these layers are disturbed by a structural uplift which produces the density contrasts indicated by the various hachured areas. In the upper part of the figure, layer 2 is uplifted into the normal area for layer 1 and produces a density contrast $d_2 - d_1$ as indicated. Other density contrasts would be produced as indicated on the diagram. In this instance, where each layer is of successively greater density, all the density contrasts are positive and the sum of all of their effects will produce the positive anomaly indicated by the gravity profile.

From this example, it is evident that any disturbance of normally flat-lying layers of different densities will produce some sort of density contrast which will cause a gravity anomaly. The magnitude and form of the gravity effect depend

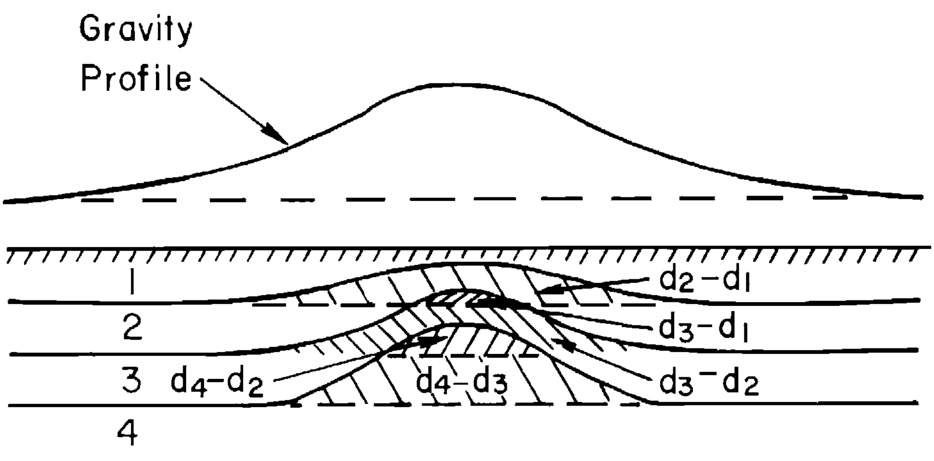


Fig. 1. Density layers, density contrasts, and gravity anomaly.

on the details of the densities involved, their magnitudes, vertical relief, depth, and horizontal extent. On the other hand, if the layers are of uniform density, no gravity anomaly would be produced no matter how strongly the layers may be disturbed or deformed. Thus, the primary property of the rocks which relates gravity anomalies to geology is the density of the various components of the geologic column and the resulting density differences among the rocks produced by structures developed in these rocks. It is the structure of these rocks that is the objective of the gravity survey.

ROCK DENSITIES

The densities of rocks are controlled by three factors; the grain density of the particles forming the rock mass, the porosity, and the fluid in the pore-space.

For the common rock-forming minerals, the grain density is not widely variable. For pure quartz (SiO_2) the density is 2.65, for calcite ($CaCO_3$) the density is also about 2.65. The clay minerals are more variable but are generally in the range 2.5 to 2.8. Thus for the more common rocks the densities of the basic materials of which they are formed do not vary widely and their density is controlled, to a very large extent, by their porosity. For most rocks below the surface water table we can assume that the pore-space is filled with water. The density of the water probably is somewhat greater than unity because of salt and other minerals in solution. Also lighter fluids, oil, and gas may be present. However, these factors affect only the pore-space portion of the rock volume and usually there are so many other variables that, when densities are estimated from porosity, the pore-space is considered as filled with water with density 1.0.

Figure 2 is a simple chart which gives the rock density as a function of porosity. Each line represents the density of the rock with the grain density indicated and with the pore-space dry (lower part) and saturated (upper part). The numerical values of Figure 2 are derived very simply from the relations:

$$d_r = d_g(1-P) \text{ for dry rocks}$$

and

$$d_r = d_g(1-P) + P \quad \text{for rocks saturated with water of unit density}$$

where:

d_r = bulk density of the rock
d_g = grain density of the rock-forming minerals
P = fractional porosity

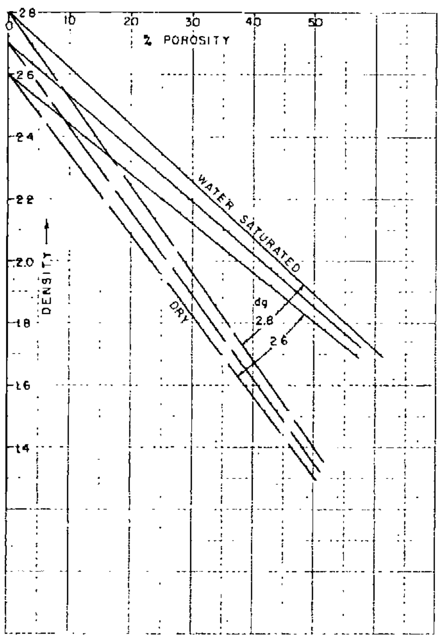


Fig. 2. Relation of density to porosity for dry and for water-saturated rocks. Curves are shown for grain densities of 2.6, 2.7, and 2.8.

The above discussion and the chart are valid for the common sedimentary rocks. There are some important exceptions which may be encountered in gravity surveys and which are due to the chemical nature of the rock material. The most common and important are those encountered in salt domes. Rock salt (halite) has a very low grain density, about 2.165 for pure sodium chloride, and practically no porosity; it forms the general mass of the dome. Quite commonly the salt is overlain by a cap rock, usually (in the Gulf Coast) composed of a lower layer of anhydrite ($CaSO_4$) with a density of about 2.9, and an upper layer of limestone. Anhydrite and dolomite, with grain densities of 2.8 to 3.0, and often with very low porosity, are common components of chemical (evaporite) sedimentary deposits. Some minerals (e.g., magnetite) have densities up to 5.0 but are not a factor in petroleum gravity surveys. The lowest density material affecting gravity surveys, of which this writer is aware, is diatomite (in the southwestern part of the San Joaquin Valley, California) which can have a density as low as 1.0 or slightly less.

SOURCES OF DENSITY INFORMATION

Since densities and density contrasts are the fundamental factors controlling gravity anomalies, it is important to have as much factual information on densities in the geologic section involved as is feasible. Quite often, especially in newly explored areas where the lithology of the section is quite unknown, it is impossible to obtain factual density data, and possible variations in density must be inferred from such information as is available or from the nature of the gravity anomalies themselves.

Several sources of factual density data may be available in different areas and under different circumstances and with widely differing reliability. Any possibility of their use should be pursued in connection with the geologic interpretation of gravity surveys.

Cores

Density measurements on core samples can give good measurements especially in consolidated sediments but are rarely available for any but very limited segments of the total geologic column.

Cuttings from drilling

Density measurements can be made on cuttings but the results are not very satisfactory. There is a tendency for the sampling to be selective in that the harder and more resistive parts of the sections cut by the drill are those which tend to be preserved in chips large enough for measurement. Also, some pore-space is lost. Both of these effects tend to make the values from cutting measurements too high.

Gamma-gamma density logs

This well-logging tool measures gamma rays generated within the formation by Compton scattering of a primary gamma-ray beam from a high-intensity, artificially radioactive source such as irradiated cobalt. The intensity of the secondary radiation is proportional to the density of electrons in the rock material. This, in turn, is a direct measure of the density of the material except for certain relatively small secondary corrections due to the chemical nature of the rocks. For use in connection with gravity surveys, the great detail of these logs is not needed. A convenient way of using the logs is to take an average value for each one hundred feet of depth and plot these on a relatively small vertical scale such as 1000 ft to the inch. A smooth curve through these points, with discontinuities where any abrupt changes in lithology occur, will give the general pattern of density changes with depth. An example from such a plot is given by

Figure 3. These density logs may give quite satisfactory, regular, and reliable values in consolidated sediments. They are often very ragged and unreliable in shallow, unconsolidated sediments as in the upper several thousand feet of section in the Gulf Coast. Part of the irregularity is due to the sensitivity of the measurement to variations in the size of the hole and such variations are much greater in unconsolidated sediments.

Seismic velocity logs

There is an approximate relation between seismic wave speed and density and, in general, rocks with high wave speeds are of high density. Continuous velocity logs (CVL), usually expressed in the reciprocal of wave speed or as microseconds per foot, are considerably more common than radioactive density logs.

Published curves for velocity-density relations have been given by Woollard (1959, p. 1530) and by Drake (see Grant and West, 1965, p. 200). Individual values are widely variable and also these two publications differ materially in general trend in intermediate ranges. For these reasons velocities can be depended on to give only rather general information on variation of density with depth.

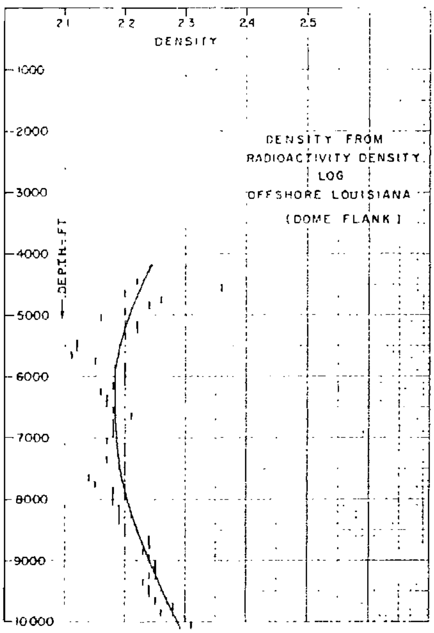


Fig. 3. Density from radioactivity (gamma-gamma) density log, offshore Louisiana.

Borehole gravity meter

Measurement of gravity differences within a borehole gives a direct measure of the average density of the section between the points of measurement. A sample of density derived from such a measurement is given by Figure 4 (Howell et al, 1966). This gives the most direct

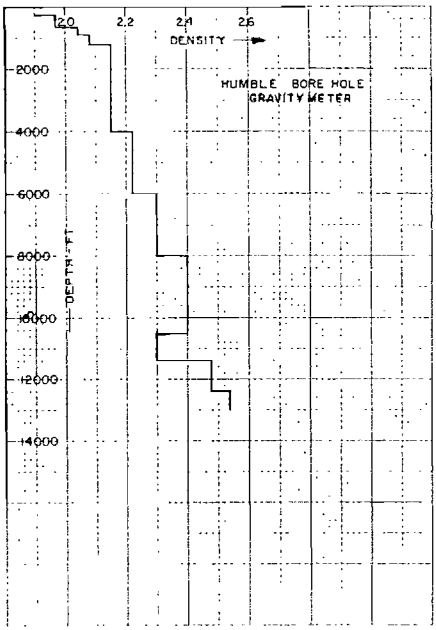


Fig. 4. Density versus depth from a downhole gravity meter. A gravity reading is made at the top and bottom of each vertical line; average density is calculated from the relation:

$$d = \frac{0.09406\,\Delta h - \Delta g}{0.0254\,\Delta h}$$

where Δh is the interval, in feet, and Δg is the corresponding gravity difference, in milligals (replotted from Howell et al, 1966).

measurement of the quantity desired for gravity interpretation which is an average density over the interval between depths of measurement. Instruments for measurements in boreholes are difficult to make and to operate, and such density measurements are relatively rare. This situation is improving as at least one geophysical company is providing borehole gravity measurements as a routine logging service.

Surface sampling by gravity meter

An "elevation factor profile" for determining average density of surface material is often observed in connection with field gravity operations. A line of stations at close spacing is observed over a topographic feature. The measurements are reduced with different elevation factors to find the one which minimizes the correlation of gravity with topography. The corresponding density is an average value for the topographic feature sampled. An example of such a profile is given by Figure 5.

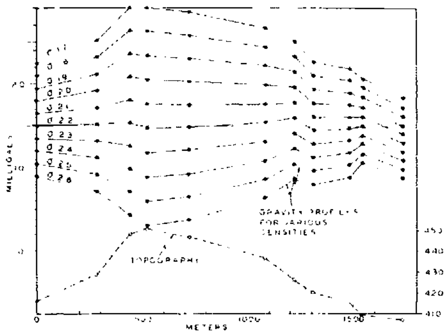


Fig. 5. "Density Profile" for determination of density value for reduction of gravity observations. Each profile is calculated for the density shown. The indicated density equals 2.20.

RANGES OF ROCK DENSITIES

In spite of all the possible ways of determining density contrasts which may be responsible for gravity anomalies, completely satisfactory data on densities is rarely, if ever, available and certain general values and ranges may be all that can be used to serve as a basis for estimating structural relief corresponding to observed gravity anomalies. While the extreme range of possible densities is from about 1.0 (diatomite) to 2.9 or 3.0 (anhydrite, dolomite, or basic igneous rocks), the actual range likely to be encountered is very much less. Unconsolidated sands and shales are likely to have densities in the range of about 1.7 to 2.2 increasing rapidly with depth (Figures 3 and 4). Compact sands and shales which are older (Paleozoic for example) are likely to have densities of around 2.5 to 2.6. Heavy massive limestones, e.g., the Ellenburger, may have densities of around 2.7. While the range of these values is fairly large, a local contrast usually will be only a fraction of this range. If no definite density information is available, a common order of magnitude figure for density contrast is about 0.25 and often such a general figure can be used as a basis of analysis when no definite information is available. However, any really quantitative interpretation of gravity anomalies in terms of geologic structure involves a reasonably reliable determination of the magnitude of the density contrasts involved.

CHAPTER II

GRAVITY INSTRUMENTS

It is not intended to describe the physical details or operation of the considerable number of gravity field instruments which have been developed over the years. Some twenty or more different land gravity meters have had very extensive use in the field.

UNIT OF GRAVITY MEASUREMENT

A body on the surface of the earth has a "weight" which results from the gravitational attraction of the entire earth. If the body is allowed to fall, it is accelerated by this weight. The unit of acceleration is the gal, named after Galileo, and one gal is simply one cm/sec^2. The average acceleration on the earth's surface is about 980 cm/sec^2 = 980 gals (but increases by about ½ percent from equator to pole).

In geophysical prospecting we are interested in "anomalies" in the gravitational field produced by variation in density as defined in the previous chapter. These are only very small fractions of the earth's total field, and a smaller unit is needed. The commonly used unit is the milligal or mgal which is .001 gal. Anomalies from local geologic structures are commonly of the order of one to ten mgals.

The "gravity unit" which is 0.1 mgal has been used quite commonly in gravity field operations and mapping. It leads to some confusion to have two units which differ by only one order of magnitude.

Since anomalies from geologic sources may be as small as 1 mgal and must be measured in a total field of the order of 980 gals = 980,000 mgal, it is evident that a gravity meter must measure variations in the total field to better than one part in a million to be useful for geophysical prospecting. Actually, instruments now in routine use measure to an accuracy of 0.05 mgal or, with very careful use to 0.01 mgal, which is a variation of only one part in one hundred million of the total gravitational field in which the measurements are made.

CLASSIFICATION OF GRAVITY METERS

Most gravity meters can be classified, according to their basic physical principles, as: (1) weight on a spring, (2) torsion fiber, or (3) "zero-length" spring.

Weight on a spring

For a simple weight on a spring the fractional change of length of the spring is proportional to the fractional change of total force which is the same as the fractional change of total gravity. Such instruments require a high optical or mechanical magnification to make changes of the order of one part in a million or less apparent and measurable. In the Gulf (Hoyt) gravimeter the spring is wound from a thin ribbon which converts the change of length to a rotation. The small changes of angle resulting from gravity changes are measured by multiple reflections in the optical system.

In the Hartley instrument the change in length was measured by an optical lever. For the most part these and other direct deflection instruments have had limited field use except the Gulf gravimeter.

Torsion fiber

The usual T-shaped moving system is supported horizontally by delicate torsion fibers attached to the cross bar with the active mass at the end of the cross arm. The torque due to gravity acting on the weight is balanced by the torsion of the elastic supports. The instrument is adjusted to a very critical condition where the torque of the nearly horizontal beam carrying the active mass is equal to the torque of the supporting fibers and the period becomes relatively long, commonly about 7 sec. The position of the sensitive mass is read with a microscope. A balancing or null screw, operating on the moving system through a weak spring and a mechanical linkage, is adjusted at each reading to balance the meter by bringing an index fiber of the moving beam to a fixed reference mark on a scale in the field of the microscope. Differences in gravity are proportional to differences in the position of the reading screw when the system is balanced. A calibration factor or curve is determined by the maker to translate the dial readings of the balacing screw into milligals. The Atlas, Worden, World-Wide, Sharpe, and some other meters are of this type.

Zero length spring

A "zero length" spring is made so that an initial stress is required before the coils begin to separate (like a common screen door spring). By a proper relation of the distance between the points of attachment (i.e., the "length" or total strain) the stress-strain curve, projected, passes through zero length for zero strain. It can be shown that with such a spring a gravity meter can be designed with theoretically infinite period and linear deflection. The La Coste and Romberg instrument is of this style; the null adjustment to bring the deflection back to zero to measure the gravity change is made by a small vertical adjustment of the support of the main spring.

RELATION OF SENSITIVITY TO PERIOD

Instruments of types (2) and (3) gain their sensitivity by being adjusted to a relatively long period. The relation between sensitivity and period can be shown by considering a simple weight of mass M on a spring (Figure 6). The total elongation of the spring is:

$$d = Mg/c,$$

where c is the spring constant, i.e., the ratio of force to stretch. As an oscillator, the period of the system is

$$T = 2\pi\sqrt{M/c},$$

and

$$M/c = T^2/4\pi^2.$$

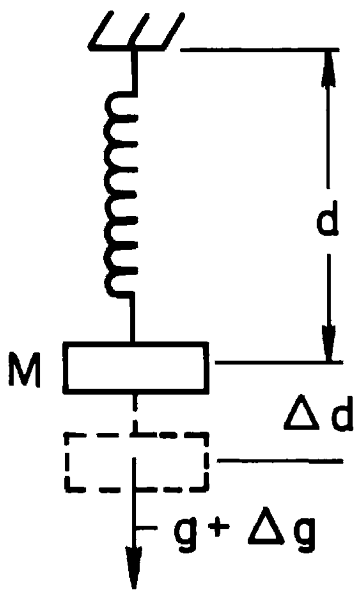


Fig. 6. Diagram for a simple "weight on a spring" gravity meter.

Substituting in the first equation gives

$$d = T^2/4\pi^2 \cdot g.$$

For small changes in gravity Δg, the resulting change in elongation of the spring is

$$\Delta d = T^2/4\pi^2 \cdot \Delta g,$$

and the sensitivity of the system is

$$\Delta d/\Delta g = T^2/4\pi^2,$$

so that the sensitivity is proportional to the square of the period. These considerations give us some idea of what is required to make a sensitive gravity meter.

If it were decided that Δd could be measured to .001 mm and that we would need to measure Δg to 0.1 mg, the ratio of $\Delta g/g = 0.1/10^6 = 10^{-7}$ from which $\Delta d/d$ is also 10^{-7} and the required total elongation d is .001 mm x $10^7 = 10^4$ mm (because normal gravity at the surface of the earth = 980 gals $\cong 10^6$ mgal) or 40 ft. It was from this consideration that an early consultant on the gravity meter problem said that it would be possible to build a meter but it would have to be over 40 ft long. The period of this same system with elongation of 10 m or 1000 cm can be calculated from

$$T^2 = 4\pi^2 d/g = 40 \times 1000 \text{ cm}/980$$
$$\cong 40 \text{ and } T = \sqrt{40} = 6.3 \text{ sec}.$$

If ways other than making the spring very long can be devised to attain the long period, the sensitivity of displacement to gravity change will be the same as for the long spring. The meters of type (2) and (3) have an unstable mechanical system and the meter is built and adjusted to operate very near this unstable position. As that position is approached, the period becomes longer because the force of the principal spring is nearly balanced by a contrary force which comes about from the geometry of the system as the point of instability is approached. If the system can be adjusted to have a period of, say, 7 sec we can still measure 0.1 mgal by measuring a deflection of .001 mm. This long period and resulting sensitivity are attained in moving systems with dimensions of only one or two inches.

GENERAL REQUIREMENTS OF GRAVITY METERS

The problem of building gravity meters is that of devising a system which can be adjusted to be sufficiently close to the point of instability to get the long period required and still be stable and also be reasonably rugged and reliable in routine field use. All instruments have some drift because the spring constant is not completely without change. Also they are sensitive to temperature as the stability of the adjustment depends on the maintenance of the exact dimensions of the moving system and its supports. Therefore all instruments are either within a chamber kept at constant temperature (within .01° F or better) by a thermostat and electrical power source (storage battery) or have a temperature compensating element. For such compensated instruments the entire moving system must be very accurately at the same temperature and must be well insulated, usually in a vacuum flask.

Buoyancy effects from changes in air pressure must be compensated by a suitable sealed cell on the moving system or else the entire moving system must be within a sealed, constant pressure chamber.

In spite of all these difficulties rugged, field-worthy gravity meters have been built with which gravity measurements are routinely made to a precision of less than 0.05 mgal. Many hundreds of such instruments have made millions of gravity observations all over the land areas of the world. With special housing and electronic accessories measurements are extended to the sea-bottom and to operation within moving ships and even airplanes (see following section).

CHAPTER III

SPECIAL GRAVITY INSTRUMENTS

The gravity instruments previously described and referred to in the discussions of measuring and reducing gravity observations are designed for operation on land and on a firm base. Gravity measurements have been extended to the water bottom and to moving ships.

UNDERWATER (BOTTOM) GRAVITY METERS

The underwater gravity meter commonly used is basically a land gravity meter in a water-proof shell, and with mechanical and electronic accessories arranged so that the same operations are done remotely through an electric cable to a control box on the ship as would be done if the instrument were manipulated by hand. The instrument is mounted on gimbals within the waterproof case and small motors move the instrument on these gimbals until remote indicators at the control box show that it is level. In some instruments this function is automatic and the leveling motors are operated through a servo-device to keep the instrument level. The position of the beam of the moving system is indicated at the control box and the instrument can be balanced and read in much the same way as it would be manually. All of the operations of leveling, releasing the clamps to free the moving system, balancing the meter, obtaining the gravity value, and clamping the instrument at the end of the observation are carried out from the control box on the surface in a few minutes. The ship is not anchored but is held within a short horizontal distance from the meter by manipulation of engines and rudders.

In early experiments with underwater meters it was found that, especially if the meter is resting on very soft bottom material as in much of the Gulf Coast offshore, it is subjected to small vertical motions which seriously interfere with making accurate readings. One approach to this problem is that of using the instrument itself as a seismograph and containing the sensitive part within a vertically moving "elevator" within the outer case (Figure 7, LaCoste, 1967). A servo-motor is arranged to move the elevator vertically, and this motor is controlled through a servomechanism which is responsive to the position of the beam between its stops. When the outer case

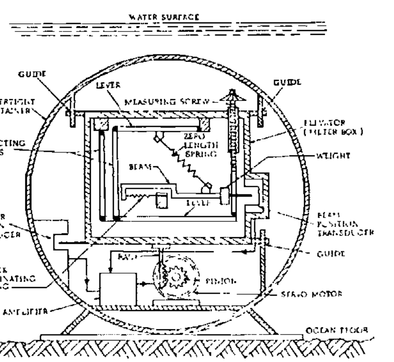


Fig. 7. Schematic diagram of the LaCoste and Romberg underwater gravity meter with "Elevator" to compensate for vertical motion (LaCoste, 1967). *Courtesy AGU.*

moves downward the beam, acting as a stationary weight in a seismograph, tends to move upward; this motion decreases the light on one photocell and increases it on another. These changes act through the beam position transducer to control the elevator motor, so that when the outer case moves downward in response to an external disturbance the elevator moves the gravity meter upward; and if the outer case moves upward the sensitive element is moved downard. In this way the meter itself stays relatively stable in space while the outer case moves up or down, and thus the external accelerations are largely eliminated from affecting the gravity sensitive system. In another system, very heavy damping is used and the rate of change of displacement of the beam rather than the displacement itself is used as a measure of the gravity change (this is an adaptation from the shipborne gravity meter mentioned below).

Underwater gravity meters have now been developed until they are very reliable and can make observations under quite adverse conditions of sea surface and bottom motion. These motional problems decrease in deep water and are most severe in very shallow water (less than about 25 ft) and with soft bottom. The operations of lowering the meter over the side of the ship, making the gravity reading and returning the meter to the surface are now quite routine. With good meter handling equipment and good cooperation of the boat crew and instrument crew, underwater observations can be made in a few minutes so that the rate of station production and the precision of the results are comparable to those obtained with land surveys.

SHIPBORNE GRAVITY METERS

The problem of making gravity observations on a moving ship is primarily one of observing a very small quantity (the desired gravity variation), within an extremely "noisy" environment (the motional accelerations due to movement of the ship). There is no physical principle by which motional accelerations can be distinguished from gravitational accelerations. The horizontal and vertical accelerations due to the movement of the ship, even in rather moderate sea conditions, can be a tenth or more of total gravity. This means that, to make a measurement comparable with that made with land gravity meters, we must detect changes of the order of one milligal or less within a background noise of the order of one hundred thousand milligals (1/10 of normal gravity). The problem is basically one of filtering a low-frequency signal from high-frequency noise. The "frequency" (or, conversely, the "wavelength") of the desired gravity effect is a function of the horizontal extent of the gravity disturbance and the speed of the boat which control the time required to pass over the disturbance. The period of the noise is primarily that of the motion of the ship and is usually in the range of about one to ten seconds depending on the size of the ship and the sea conditions in which it is operating. Also there may be relatively large long period accelerations due to "fishtailing" of the ship when the steering (either automatic or manual) tends to "hunt" about an average course.

The motional accelerations can be divided into their vertical and horizontal components. The vertical components affect the gravity meter directly in the same way as gravity changes would. Horizontal acceleration effects can be almost completely eliminated by mounting the meter on a gyroscopically stabilized platform to maintain the instrument level (i.e., perpendicular to the average vertical). This still leaves horizontal forces due to lateral motions of the level platform. There may be "cross-coupling" effects between these and the vertically sensitive axis of the instrument.

Brief descriptions of different shipborne gravity instruments follow.

The LaCoste and Romberg shipborne gravity meter

The action of the moving system of the instrument can be represented by the fundamental differential equation

$$\ddot{x} + k\dot{x} + cx = g + a,$$

where x, $\dot{x}$ and $\ddot{x}$ are the vertical displacement and its time derivatives, k is the damping, c is the coefficient of the net restoring force on the beam, g is the gravity, and a is motional acceleration. The instrument is built with very heavy damping (see "damper," Figure 8) and is adjusted to very high sensitivity. This makes the damping coefficient k large and the displacement coefficient c small. The left side of the equation is controlled very largely by the damping term $k\dot{x}$.

The action is analogous to that of a heavy ball in a dish of very viscous material (e.g., oil). If the dish is level (left side, Figure 8) the ball does not move. If it is tilted, the ball will move downhill. This movement begins immediately in spite of the high viscosity (heavy damping) of the fluid. In a similar manner, the heavily damped gravity meter system (LaCoste, 1967) responds immediately to changes of gravity by a change in its rate of deflection. By measuring the rate of deflection rather than the deflection itself an instantaneous response is obtained in spite of the heavy damping which suppresses effects of high frequency motions.

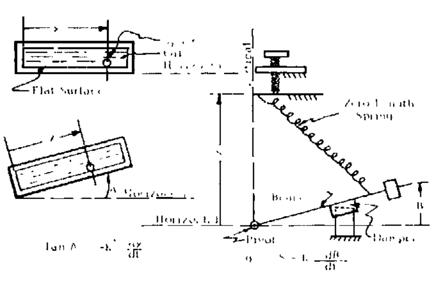


Fig. 8. Schematic diagram of the LaCoste and Romberg shipborne gravity meter and a mechanical analog (LaCoste, 1967). *Courtesy AGU.*

The instrument is kept approximately balanced by a servo-mechanism which adjusts continually the spring tension control (Figure 9, from LaCoste, 1967), much as if an operator of a land meter would turn the reading screw if the field were changing. Then the action of the instrument can be represented by the symbolic relation

$$g + a = st + ks$$

where g = gravity, a = motional acceleration, st = spring tension, s = slope or time rate of change of beam position and k is the slope coefficient.

Horizontal accelerations result from tilting (roll or pitch) of the ship and from its horizontal motions. The tilting effect is very effectively eliminated by mounting the gravity meter itself on a stable table which is held in a horizontal plane (perpendicular to the average vertical) by a pair of gyroscopes (3, Figure 9) controlling the tilt in the transverse and longitudinal directions of the ship. The gyroscopes are referenced to horizontal accelerometers (4, Figure 9) which act as long-period levels so that the vertical axis of the gravity meter is maintained in the direction of the average vertical.

Even though maintained level, the gravity meter is still subject to horizontal accelerations resulting from the horizontal components of the motions of the ship. The mechanical parts of the gravity meter are not perfectly rigid. There can be small distortions of the ligaments spring, or beam of such a nature that horizontal forces can result in small vertical effects. Also, even for an ideal instrument with a horizontal beam,

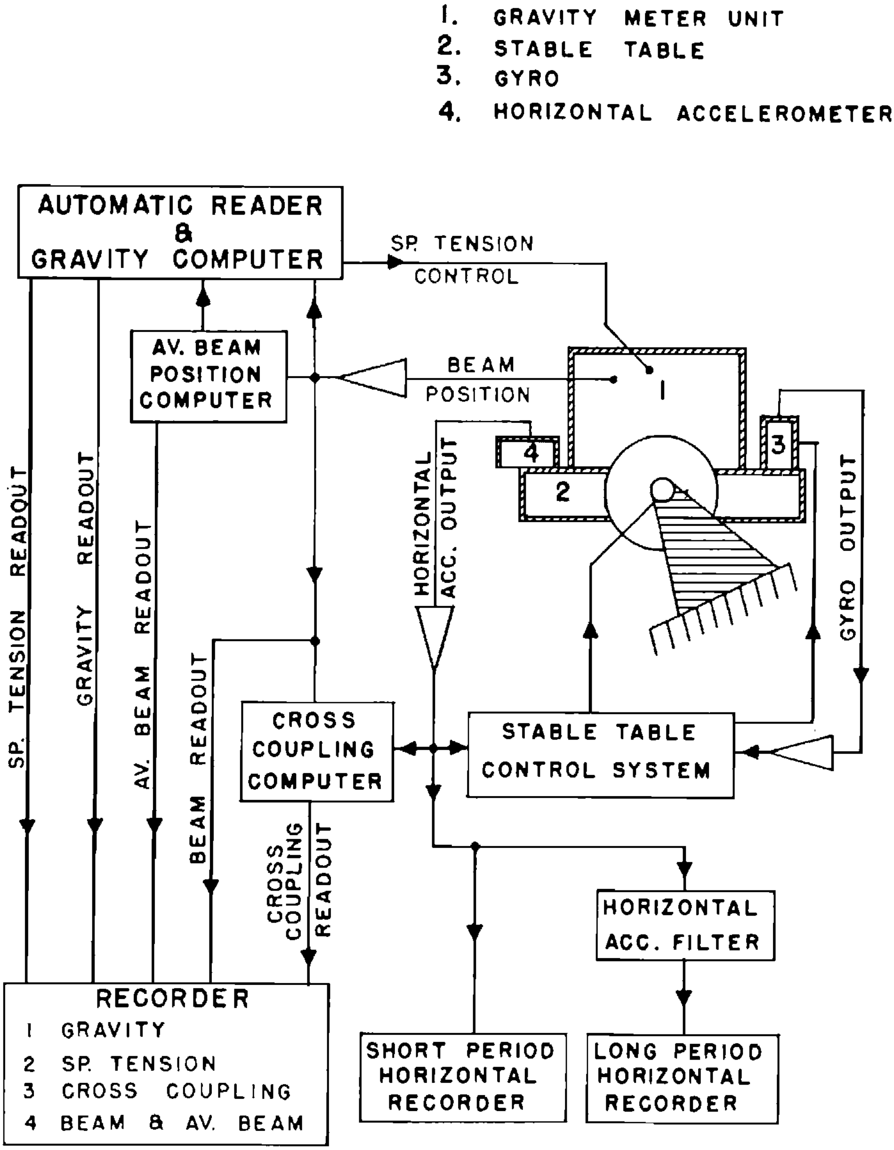


Fig. 9. Block diagram of stabilized platform gravity meter (LaCoste, 1967). *Courtesy SEG.*

there can be coupling of horizontal and vertical components which results from the phase relations between the horizontal and vertical components of motion. These effects are indicated by Figure 10 (LaCoste et al, 1967). The motion of meter is assumed to be circular, corresponding to circular motion of the water in the waves. At time 1 on the diagram, the ship is moving downward, the meter is deflected upward and the horizontal motion is to the left, making

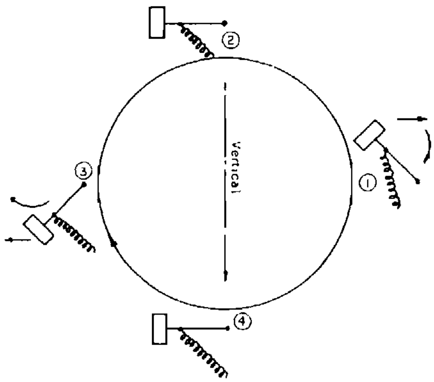


Fig. 10. Principle of the inherent type of cross coupling (LaCoste et al, 1967).

a force to the right and the two combined make a clockwise torque on the moving system. If, at time 3, while the meter is moving upward, the horizontal motion is to the right, the force is to the left and again the torque is clockwise. Thus there are "cross-coupling" effects which depend on the nature of the individual instrument and the magnitude and phase relations of the motions to which it is subjected. These cross-coupling effects can, to a large extent, be determined and measured and proper allowance made for them in determining final gravity values, either by subtracting them from the observations or by incorporating a cross-coupling correction in the instrument itself.

The above description is generally applicable to the LaCoste and Romberg shipborne gravity meter which, so far, has been the most extensively used. There are other instruments which have had some application to measurement of gravity at sea. All are mounted on gyroscopically stabilized platforms.

The Graf-Askania seagravimeter

This is basically a torsion spring type instrument with very heavy damping. (Graf, 1958) The basic moving system of the instrument is a horizontal bar (Figure 11a) supported by horizontal springs. The bar moves in a strong field of a permanent magnet to produce eddy-current damping. A set of 8 ligaments (Figure 11b) stabilizes the system so that its motion is confined to that about the horizontal axis of the springs. This instrument has been used rather extensively by oceanographic institutions in long cruises far out to sea but relatively little, as yet, in petroleum exploration.

The Bell ship gravity meter

The basic sensor used in this instrument was originally developed for inertial navigation systems and was later

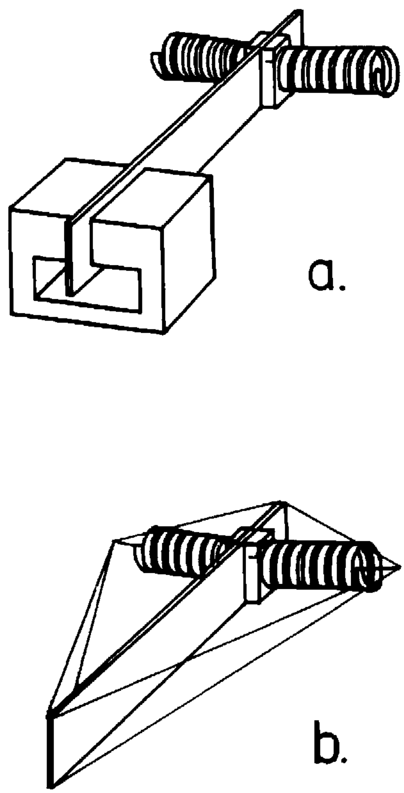


Fig. 11. Principle of moving system of Graf "sea-gravimeter": Upper — Magnetically damped moving system. Lower — Moving system with 8 stabilizing fibers to restrict motion to vertical plane only (Graf, 1958).

adapted for measurement of gravity at sea. The sensitive element is a coil in the radial field of a pair of conical permanent magnets (Figure 12). The variation in weight of the coil and accompanying "proof mass" due to changes in gravity and to motional accelerations are balanced by the force between the varying current in the coil and the constant field from the magnets. The dimensions are very small, since the entire sensor is mounted in a flat, cylindrical case about 1.8 inches in diameter and ¾ inch thick.

The two conical magnets are oppositely polarized (N ends toward each other) to produce the radial field across the gap between them and the surrounding pole pieces (one only, marked S, is shown in the figure) which are part of the return magnetic circuit. The proof mass serves as a form for the coil and is aligned within the gap between the magnet poles by three guide springs, only one of which is shown.

The position of the coil within the gap is sensed by capacitive pick-off rings which are in a capacity bridge. This bridge is balanced at the reference position and any unbalance of the bridge produces a variation in the current in the coil in the direction to restore the balance.

The sensor is contained within a very carefully controlled constant temperature oven, as the field of the permanent

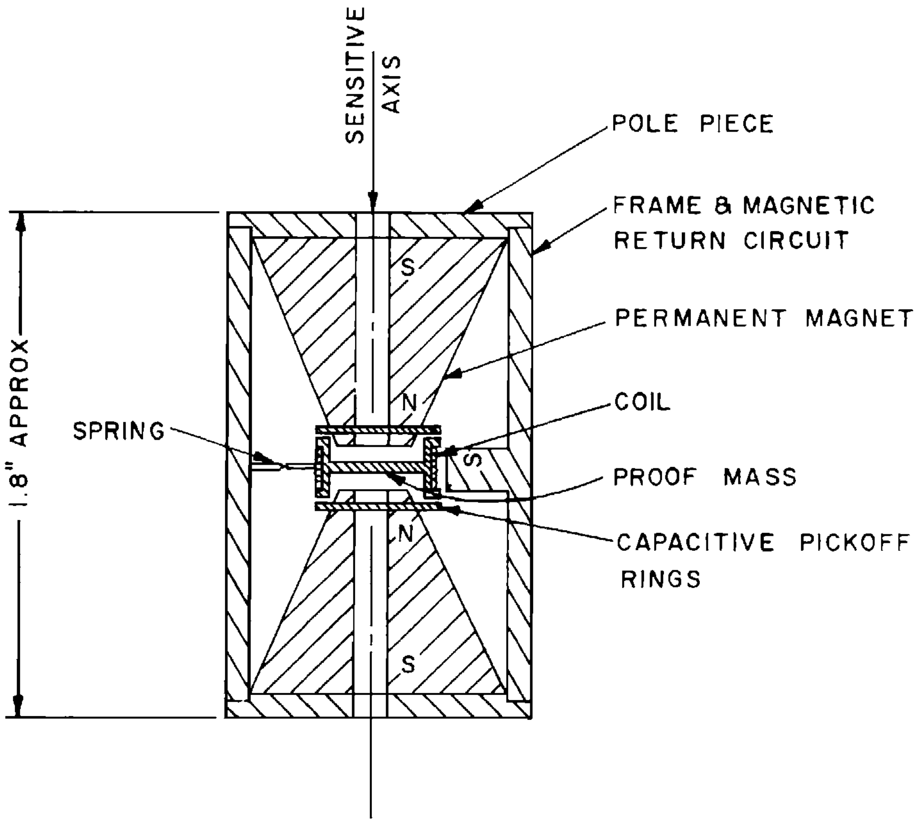


Fig. 12. Schematic diagram of acceleration sensor of Bell ship gravity meter.

magnets is strongly temperature sensitive. To make the assembly sensitive to only vertical accelerations it is mounted on a stabilized platform. Cross-coupling effects from horizontal accelerations are negligible.

The coil carries a biasing current which is controlled very accurately by a unit of the external electronics system. This current, in effect, takes the place of the main spring in most gravity meters.

The variations in the balancing current, which measure the vertical component of all motional accelerations as well as changes in gravity, are converted to digital form. A digital filter separates the desired changes of gravity from the motional accelerations.

The vibrating string acceleration sensor

Originally developed as an accelerometer for inertial guidance by Aerospace Systems of AMBAC Industries, this instrument has been adapted to shipborne gravity measurement (Wing, 1967; AMBAC, 1967).

The acceleration sensor (Figure 13) includes two vibrating strings (actually thin ribbons), and two weights, one above the other. A relatively large, constant external tension is applied to the over-all linear system. The strings are in a magnetic field and are driven at their resonant frequency. A change of acceleration (including any change in gravity) changes the tensions in the two strings, but in opposite directions. Thus an increase in gravity increases the weight of Mass 1 which increases the tension, and thereby increases the natural frequency of string 1. The effects are reversed for Mass 2 and string 2. Thus a change in gravity (or total vertical acceleration) is indicated by a change in the difference between the natural frequencies of the two strings. The electronic system measures this difference frequency which includes, of course, any short period effects due to the vertical component of the motion of the ship. The sensor is carried on a gyrostabilized platform to keep the sensitive axis of the instrument in the average vertical. Effects of horizontal accelerations (cross-coupling) are eliminated quite completely by careful design of the cross-support ligaments which keep the center of mass of the two masses accurately aligned in the sensitive axis.

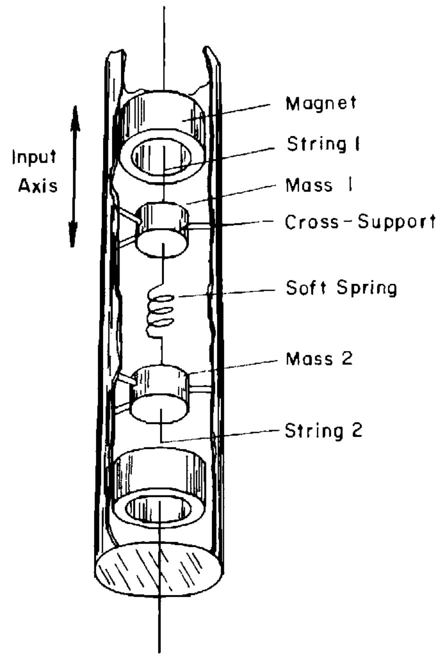


Fig. 13. Schematic diagram of acceleration sensor, vibrating string shipborne gravity meter (AMBAC, 1967).

Test data indicate that this may make a successful instrument for measurement of gravity at sea, but extensive tests from actual operations on ships are not yet available.

THE EÖTVÖS EFFECT

When gravity is measured on a moving vehicle it is subject to an acceleration due to the rotation of the earth. This is the vertical component of the Coriolis acceleration. The effect can be visualized by considering the outward acceleration from the centrifugal force due to the earth's rotation, which is a maximum at the equator and decreases to zero at the poles (see Figure 14). At the equator this outward force tends to decrease gravity amounts by about 3000 mgal. If one were in an airplane flying

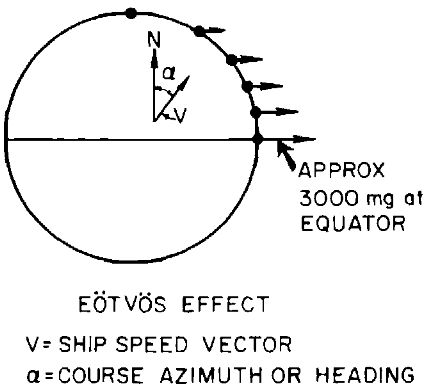


Fig. 14. Schematic diagram of Eötvös effect showing a variation of centrifugal acceleration as a function of latitude and nomenclature for calculating Eötvös correction.
V = ship speed vector
α = course azimuth or heading.

to the west at about 1000 miles per hour, or as fast as the earth turns to the east, this effect would decrease to zero and apparent gravity would increase by 3000 mgal. If the plane were going east at the same speed, the outward (negative) effect would be double. At lower speeds the same effect is present to smaller degrees and its magnitude depends on the east-west component of speed of the ship and on the latitude. At the equator the effect is 7.49 mgal/knot for an east (or west) direction. For a ship at another latitude and moving on any other course the effect is

$E = 7.49\, V \cos\varphi \sin\alpha$,[1]

where V is the ship's speed in knots, φ is the latitude and α is the ship's heading with respect to north (Figure 14). The effect is negative when the ship is headed easterly and positive when headed westerly, and therefore the correction for this effect, which is of the opposite sign, is positive for easterly courses and negative for westerly courses.

The Eötvös effect must be taken into account very carefully. For example, at a common geophysical boat speed of ten knots and at latitude 30° the Eötvös correction is about —60 mgal for an east course and +60 mgal for a west course. On an exact north or south course the correction would be zero, but, at a speed of ten knots, if the course were one degree off true north, the effect would be one milligal. Thus on courses near east or west, the Eötvös effect is very sensitive to ship speed and on courses near north or south it is very sensitive to ship direction.

The magnitude of the Eötvös effect and its sensitivity to both course and speed of the ship mean that its proper determination depends very much on the precision of the location of the ship. For really accurate determination of the Eötvös effect (within less than 1 mgal) it is necessary to know the speed within about 0.1 knot and the course within about 1 degree. Data necessary to determine these factors with this precision can be obtained with accurate radio positioning systems such as Shoran,

[1]The more exact expression is

$$E = 7.487\, V \cos\varphi \sin\alpha + 0.00415\, V^2$$

The V^2 term represents the centrifugal acceleration from moving over the curved earth's surface, irrespective of direction. It is negligible at usual ship speeds (0.4 mgal at 10 knots) but becomes important at airplane speeds, e.g., 166 mgal at 200 knots.

Lorac, Raydist, Toran, etc., when well operated and free of lane jumps. The sensitivity to the Eötvös effect means that a ship should be operated at as nearly constant speed and course as possible to avoid sudden changes in the gravity recording due to changes in Eötvös effects. Extensive experience with measuring gravity at sea has shown that much greater inaccuracies in the final results are due to inaccurate knowledge of the correct positioning and resulting uncertain Eötvös effects than to any other component of the operation.

CHAPTER IV

FIELD MEASUREMENTS AND DATA REDUCTION

Only the fundamentals of field operations are included here. The basic gravity measurement is similar to that of measuring elevations in that gravity differences are determined from one or more starting points or controls where gravity values were previously established or are assumed, just as levels are determined from previously established bench marks. Closure errors of the network are adjusted in much the same way that adjustments of closure errors are made for elevation surveys. The network of gravity differences is adjusted for meter drift by observations at control stations several times a day. The result of the field operation is a network of points at which "station gravity" has been determined. Before being compiled into a map, these values must be corrected for certain well-known effects.

THE LATITUDE CORRECTION

This correction is made to remove the effect of the increase of gravity from the equator to the poles which is due primarily to the rotation of the earth and the fact that the equatorial radius is thirteen miles greater than the polar radius. The total gravity increase is 5172 mgal. The gradient of this effect is 1.307 sin 2φ, mgal/mile where φ is the latitude. The maximum gradient, at 45° latitude, is 1.307 mgal/mile and is more than one mgal/mile for all latitudes between 25° and 65°. This latitude effect is calculated for each station, usually from the geodetic formula for gravity on the International Ellipsoid for which the value of gravity at sea level, γ_0, as a function of the latitude, φ, is given by the equation:

$$\gamma_0 = 978.0490\,(1 + 0.0052884 \sin^2 \varphi - 0.0000059 \sin^2 2\varphi).$$

Published tables of the latitude effect are available from which the corrections are readily determined from the latitudes of the stations as read from the station location map.

THE ELEVATION CORRECTION

An elevation correction must be made to take account of the fact that gravity changes with elevation, because a higher station is farther away from the center of the earth than a lower one, and that gravity is affected by the attraction of material under the station (see Figure 15). Usually the elevation correction reduces the gravity to what it would be if the measurements were made on a sea level surface (or a level surface at another reference elevation). The correction has two components:

The "free-air" effect is that due to the vertical gradient of gravity, i.e., the decrease with elevation as the point of measurement is farther away from the center of the earth. The magnitude of this effect is 0.09406 mgal/ft.

The "Bouguer" effect is a measure of

the attraction of the material between the station and sea level or some other reference elevation. In Figure 15, station no. 1 is subject to the attraction of the sheet of material with thickness h_1 between the surface and sea level which would not be present if the station were made at sea level. Similarly no. 2, at the greater elevation h_2 is subject to the attraction of a thicker sheet of material which would not be present if the measurement were made at sea level. The magnitude of the Bouguer effect is

$$B = 0.01276\,dh \text{ (mgal/ft)},$$

where d is the density of the material.

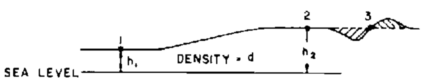


Fig. 15. Topographic section illustrating Bouguer and terrain corrections.

The total elevation correction combines the free-air and Bouguer effects (since both are multiplied by the elevation) and is

$$E = (0.09406 - 0.01276d)\,h,$$
in mgal/ft

and thus depends on the density d of the rock material within the range of topography.

Figure 16 gives elevation corrections as a function of density.

TERRAIN CORRECTION

Terrain corrections are made when there is substantial local topography around the station. The Bouguer correction, as outlined above, is calculated on the assumption that the sheet of material between the station elevation and the reference level (usually sea level) is flat, as the magnitude of the effect is calculated on the basis of an infinite horizontal slab. If there is substantial topographic relief in the vicinity of the station, as at station no. 3 in Figure 15, the simple Bouguer effect is incomplete. If there are areas below the station elevation, as on the left side of station 3, they are taken into account when they should not be, the calculated Bouguer effect is too large and the reduced value is too small. If there are areas with higher elevations, as on the right side of station 3, they have an upward attraction which is not taken into account and which also decreases the measured value. Thus, for both high and low topography, gravity values with simple Bouguer corrections are too small and the terrain correction is always positive. The effect varies with the distance from the station and the density of the material and is approximately proportional to the square of the average elevation difference of the area considered with respect to the station. To make the correction, a topographic map is divided by zones and radial sectors into compartments and the effect from each compartment is calculated by a suitable table. This is a rather tedious process. For a large volume of such calculations an electronic computer can be very effective. The topography is digitized (divided into regular units by a suitable grid with average elevation for each grid square estimated) and the effects are calculated from differences of the coordinates and elevations of the station from those of the digitized grid.

TERRAIN CORRECTION FOR GRAVITY AT SEA

When gravity is measured on the sea surface above the level of the sea bottom, the terrain effect is quite different. The Bouguer and terrain effects are calculated as if the water were replaced by rock.

The condition for a simple vertical cliff is shown by Figure 17. For a mea-

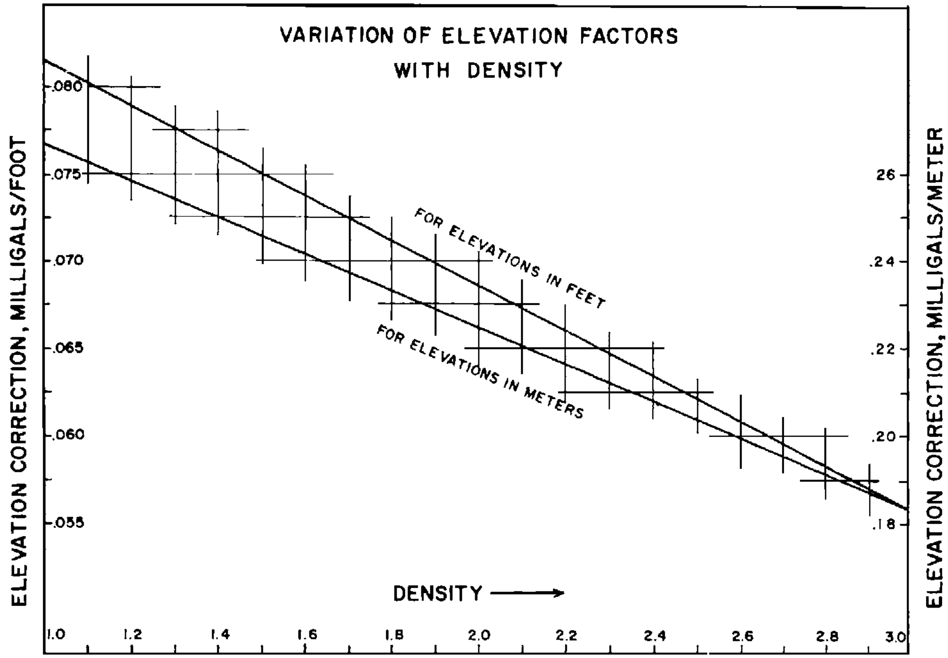


Fig. 16. Variation of elevation correction with density.
$f = (0.09406 - 0.01276\sigma)h$ (mgal for h in ft)
$= (0.3086 - 0.04185\sigma)h$ (mgal for h in meters)

surement at position 1, the Bouguer effect is calculated as if the depth h_1 extended continuously and does not take into account the deeper water to the right of the cliff. Therefore the calculated Bouguer effect is too small and the terrain correction is positive. At point 2, the Bouguer calculation does not see the higher topography to the left of P, the calculated effect is too large and the correction is negative. In the ideal case shown, the change increases to the point P and then suddenly reverses as shown by the terrain correction curve in the upper part of the diagram. In actual topography the cliff would not be vertical and the two sharp peaks are smoothed out, but the curve is otherwise similar.

Because the instrument is above the topography rather than on the surface, as in land measurements, the effects are much larger. For a survey off the southern California coast, with bottom depth relief of several thousand feet, terrain corrections of the order of ±25 mgal were calculated.

The same considerations would apply to gravity measurements in the air, with even larger effects because the density contrast is that of rock relative to air, rather than with rock relative to water as in marine measurements.

THE BOUGUER GRAVITY MAP

The gravity map after the application of latitude, free-air, Bouguer, and terrain corrections, if used, is the "Bouguer" map. This is the ordinary "observed" gravity map on which the primary results of a gravity survey are presented.

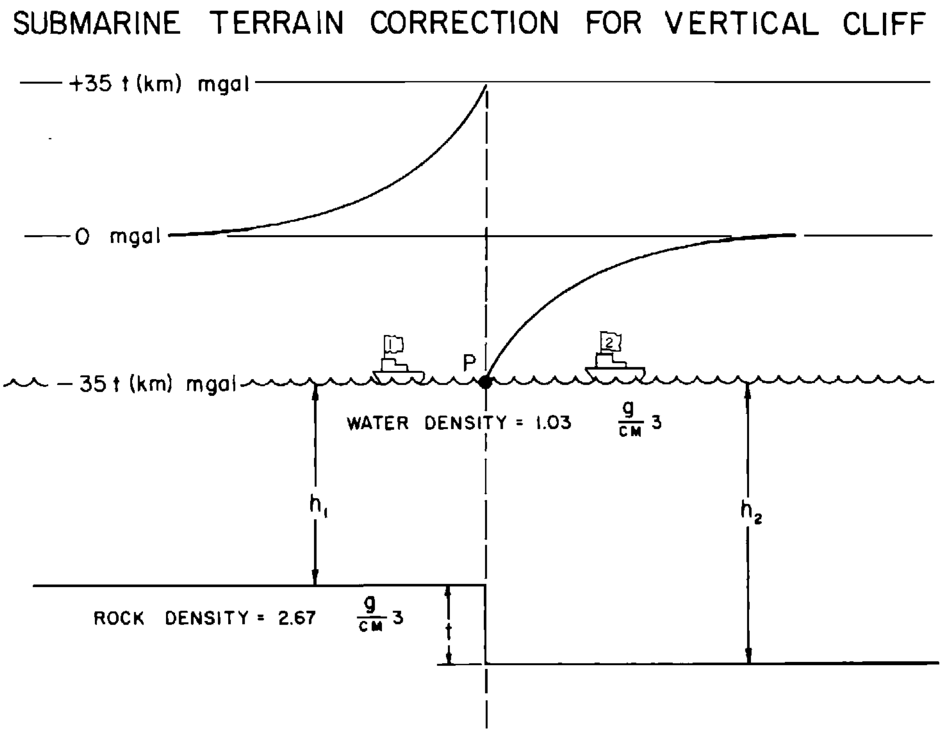


Fig. 17. Terrain correction for gravity measurement at sea surface.

Occasionally, for special purposes, the gravity data are presented as a "free-air" map in which the Bouguer correction has not been made. This is usually for maps made on the ocean; any "free-air" map from land observations will show a strong correlation with local topography.

The nature and appearance of a gravity map depend very greatly on the nature of the station pattern, the quality of the field work, and the underlying geology. The largest single source of error in gravity maps comes from elevation errors. Maps from high quality field operations in areas of low relief usually are quite smooth and the station-to-station differences are regular. Occasionally there are near-surface geological disturbances, solution sinks for instance, which form shallow geological anomalies and may make a rough map even with the best quality of field work. Gravity maps in areas with large and relatively sharp topographic relief are rarely smooth, even if the elevations are accurate, because of imperfect terrain corrections and local inhomogeneities not taken into account in the density used for the Bouguer correction. A few gravity surveys in connection with mineral exploration have been made in quite rugged areas and with great detail which have produced regular and reliable results. These have been done with very careful attention to the elevation control and with detailed terrain corrections and are of a much higher order of difficulty and cost than the ordinary gravity maps used for petroleum exploration.

CHAPTER V
ANOMALY SEPARATION

Each gravity measurement determines, at the station location, the sum of all effects from the grass roots down. A gravity map is almost never a simple picture of a single isolated disturbance but practically always is a combination of relatively sharp anomalies which must be of shallow origin, of anomalies with intermediate dimensions which may be those most probably indicative of geologically interesting sources and from very broad anomalies of a "regional" nature which have their origin far below that of the section within which the geological interest lies. Therefore, gravity interpretation frequently begins with some procedure which separates the anomalies of interest from superficial disturbances on one hand and the smooth, presumably deep "regional" effects on the other. The anomaly separation procedure may consist of the removal of a smooth regional by either of two methods; by an intuitive graphical method or by an analytical method involving a numerical procedure applied to an array of values usually on a regular grid. The analytical process is basically a filtering process intended to emphasize or enhance certain components of the gravity field and suppress others.

The proponents of these two systems have been termed "smoothers" and "gridders." The "smoother" draws smooth curves on profiles or as contours on a map. These curves represent the component of the gravitational field which is to be removed. This "regional" is subtracted from the observed gravity map and the resulting "residual" contains the components of the field which, presumably, are caused by mass irregularities representing geological disturbances of interest. This "regional" is a gravitational field which is determined entirely empirically and is based on the judgment of the operator as to the nature and sharpness of those components of the field which he wishes to discard. While the process is entirely empirical, it can, if skillfully applied, lead to a very effective isolation of anomalies and gives results which are simply and directly amenable to quantitative analysis.

It was once said, facetiously, by a geologist, that "the regional is what you take out to make what is left look like the structure." While not intended as such, this remark points a way toward useful control on the selection of a regional. Simple calculations can be made from features with the depth, form, and density contrast of geologically interesting objectives and the resulting width and gravity amplitude can help determine the nature, particularly the rate of curvature, of the profiles or contours of the regional map which would leave such features as residuals.

The "gridders" have developed a large variety of processes, all of which are readily handled by digital computers and are therefore much more rapid than the smoothing processes. Also, they can be readily contoured mechanically so that the complete process, from the input of gravity data by location coordinates and gravity values on punched cards, to final contoured maps can be done automatically including, if desired, computations with a number of different parameters.

Once set up the various grid systems (including some for which the input can be gravity stations at irregular or random locations) are fixed and automatic and therefore are not empirical and subject to human judgment in the same way that the "smoother" systems are. However, it must be kept in mind that all of these analytical systems contain elements of choice in determining the parameters of the system and in choosing an elemental grid spacing on which the calculations depend. A common feature of all second derivative and grid residual systems is that they give a zero value for a region that is plane (i.e., has no curvature) within the area considered and therefore are not handicapped by very steep regional gradients. Such steep gradients may make serious problems for a "smoother" as it is difficult to recognize anomalies which are expressed as small departures from a uniform background with very close contours.

A number of somewhat different grid systems have been developed and used in producing various interpretation maps. Some of those more extensively used are described briefly in the following paragraphs.

THE CENTER-POINT-AND-ONE-RING SYSTEM

This was one of the earliest developments toward mechanical anomaly separation. In this system, the average value from a number of points (commonly 8 for a rectangular array or 6 for a hexagonal array) on a circle about the point of calculation is taken as the regional and the difference between this regional and the value at the center of the circle is the residual. In spite of its ambiguity and simplicity, this system can be very effective in outlining the position of a local anomaly. It is very sensitive to the choice of the radius of the circle. The optimum enhancement of an anomaly will be given when the diameter of the circle is approximately the same dimension as the width of the principal part of the anomaly. A convenient and commonly applicable definition of this dimension is the "half-width" which is the minimum diameter of the contour at half the maximum amplitude of the local feature. Since a given map may contain anomalies of interest with a considerable range of horizontal dimensions, it is common practice to carry out the operation with two or more radii for the outer circle.

ANALYTICAL CALCULATIONS OF DERIVATIVES

These are mathematical operations which have been developed as a refinement over the simple empirical grid system. There are a number of systems which calculate the second vertical derivative by the application of potential theory to the gravity field around the point of calculation. The rather involved mathematical formulations have been reduced to practical schemes of calculation using values from a regular grid of points to determine averages around circles of different radii from the central point as indicated by Figure 18 (Nettleton, 1954). A number of analytical systems of this nature have been published which have led to different parameters and coefficients by which the averages at different distances are multiplied to give the second derivative values. All these mathematical systems involve assumptions or empirical choices at certain stages in reducing them to practical numerical use. These are, in effect, variations in the weighting factors by which the gravity differences between the value at the center point and the average around a particular ring are multiplied and added to derive a second derivative value.

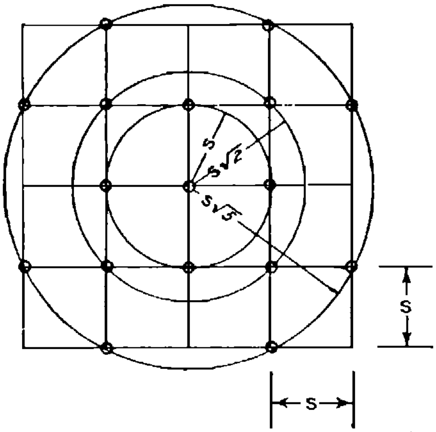


Fig. 18. Array of grid points used for calculating 3-ring second derivative (Nettleton, 1954). *Courtesy SEG.*

All of the published second derivative systems are of the form

$$g_{zz} = C/S^2\,[W_0 g_0 + W_1 g_1 + W_2 g_2 + \ldots + W_n g_n]$$

where g_{zz} is the derived (i.e., second derivative or "residual") value at the point 0, C is a constant for the particular system, S is the grid spacing, W_0, W_1, . . . are weighting factors, g_0 is gravity at the point of calculation and g_1, g_2, . . . are average values around the successive rings. The weighting factors are of positive and negative sign and the sum of the factors, $W_0 + W_1 + \ldots + W_n = 0$.

A summary of several of the published formulas, modified to be readily compared, is given in the table, Figure 19

TABLE 1. Equivalent formulas with unit weight for center point

Formula no.	db	Factor K	W_0 $r=0$	W_1 $r=s$	W_2 $r=\sqrt{2}$	W_3 $r=2s$	W_4 $r=s\sqrt{5}$	W_5 $r=s\sqrt{9.23}$
1	0	1	1	−1				
1-a	12.0	4	1	−1				
2	15.8	6.185	1	−1.354	+0.354			
3	15.6	6.000	1	−1.333	+0.333			
4	16.9	7.000	1	−1.523	+0.571	−0.48		
5	0.56	1.067	1	−0.125	−0.250		−0.625	
6	−4.8	0.571	1	+0.500	0		−1.500	
7	−2.8	0.710	1	+0.364	−0.273		−1.091	
8	1.2	1.156	1	+0.221	−0.385		−1.175	0.339
9	9.5	3.000	1	−0.750	−0.333		+0.083	

The factor $K = CW_0$; in order to "normalize" the formulas to make the weighting of the first term unity, all the weights are divided by W_0.

To calculate derivative values, we must include a factor $1/S^2$ where S is the grid spacing. For instance, if S is the grid spacing on a map of scale $1{:}k$, then $S = kr$. A common unit for gravity is gals/cm^2; with this unit, second derivative values are usually of the order 10^{-15}. Other units are mgal/mile^2, gamma/mile^2 (for magnetics).

Fig. 19. Table of normalized equivalents of second derivative calculation formula (modified from Nettleton, 1954, p. 7). *Courtesy SEG.*

(Nettleton, 1954). The original equations have been "normalized" by dividing through by the first weighting coefficient (W_0) so that the relative weight of the g_0 term is unity. The decibel (db) value is the decibel equivalent of the multiplying factor K and will be used in the discussion of frequency analysis.

It might be expected that the different formulas would give similar second derivative values. Actually, the calculated values vary widely. Some of the formulas which give relatively more accurate values for ideal, theoretical fields tend to be unsatisfactory for actual field data because they depend on small gravity differences and emphasize irregularities. Others, which give better looking maps do not reproduce the calculated values for ideal fields very well. In general, these differences come about primarily because a theoretically continuous gravity field is represented by a series of discrete points at the grid corners and the different formulas vary in the manner in which the field is represented by the limited number of points. This leads to different choices for the values of the weighting factors. For instance, if the value for W_1 is relatively large and negative, it will give strong emphasis to differences between the center point and the first ring and will be very sensitive to small errors or irregularities of the map. On the other hand, if W_1 is relatively small and positive and the outer rings have negative coefficients, the derivative value depends on the difference between the average over the central part of the gravity field and that of the outer circles. The resulting map will be much smoother and will tend to give a filtering effect to suppress the local variations within the area taken into account in each calculation.

The dimension of the grid spacing has a large effect on the nature of the resulting map; calculations made with a smaller grid spacing emphasize smaller details and those with a larger grid spacing tend to suppress smaller features and emphasize broader ones. All of these grid calculations act as filters and the degree of emphasis with which different features of the map may be brought out is a function of the relation of the horizontal dimensions of the map feature to the diameter of the more heavily weighted rings of the calculation array.

Since the second-derivative-type calculations have weighting coefficients which add up to zero the resulting map is balanced between positive and negative areas. A simple positive anomaly on an observed map will, on the derivative map, have a central positive anomaly centered at the same place but will be accompanied by a flanking negative anomaly not present on the original map.

In the general mathematical expression for a second derivative (equation p. 26) the term C/S^2 converts the weighted gravity differences to second-derivative units with the dimensions of gravity divided by the square of distance. If the C/S^2 term is omitted, the weighted gravity differences add up to a quantity which is a modified residual in the dimensions of gravity (i.e., milligals).

The choice of map units, either second-derivative values or gravity differences, is dictated by the attitude and preference of the user. Making maps in second-derivative units (g/z^2) is dimensionally correct in terms of the numbers used and to the degree (which may not be close) that the formula used determines true values. For a user interested primarily in anomaly separation the gravity difference units may be preferable, even though they have no firm mathematical foundation, because they are in dimensions more readily understood (by geologists) than the more esoteric second-vertical-derivative quantity. Since the second-derivative values are derived from the weighted differences between the center point and averages around successive rings multiplied by C/S^2, the zero contour and the maximal and minimal closures will be in the same places on second derivative or on simple difference maps. By appropriate choice of contour interval the two maps will be identical.

CONTINUATION SYSTEMS

For certain purposes, it is of interest to consider the form which a potential field would have if it were measured at a higher or a lower level. In principle the field can be computed at a different level if there is no disturbing body (mass or magnetized material) within the range of the continuation.

Upward continuation gives a smoother field, downward continuation a sharper field than that at the surface of measurement. Either process is similar to the second derivative process in that it requires a surface integration over the measured field, carried out on a regular array of values. The continuation formulas usually involve values over a considerably wider area than second derivative schemes; some use as many as ten rings of values about the center point.

Downward continuation is applied for anomaly separation, in particular the resolution of overlapping effects of sources close together. In gravity or magnetics, if two separate but nearby sources are under homogenous overlying material, the downward continuation could be made to a level closely above the sources where the effects would be clearly separated whereas they would be confused by overlap at the level of measurement. If continuation is carried to depths greater than the source the continued field will begin to oscillate; this oscillation could be a criterion of depth. These results are not often obtained clearly from actual fields because the continuation process depends on emphasizing small effects at the measured surface. If such effects are present from shallow sources and are continued below their depth of origin, they contribute large false components to the continued field. For this reason, downward continuation has more application in magnetic than in gravity interpretation because airborne measurements made at substantial elevations above the ground and therefore remote from possible sources are relatively smooth and can be operated upon by a system of sharp coefficients without false emphasis of shallow sources.

Upward continuation always gives a smoother map than the original. It has been used as a form of smoothing as a basis for anomaly separation. In airborne magnetics, upward continuation has been used to calculate the field at a higher level and the result compared with measurements actually made at that level. When control is adequate, the upward continued and measured values are almost identical.

COMPARISON OF GRAPHICAL AND GRID SYSTEMS

A qualitative comparison of the grid residual and graphical systems of anomaly separation is indicated by Figure 20 (Nettleton, 1954). The line of close dots in the upper part of the figure is a hypothetical gravity profile that includes two obvious local anomalies super-imposed on a much larger feature.

The nature of the "grid residual" calculation is indicated by the short straight lines which "bridge" segments of the gravity profile. The ends of these lines correspond to the diameter or outer circle of a grid residual calculation array such as that of the center-point-and-one-ring system. The arrows at the center

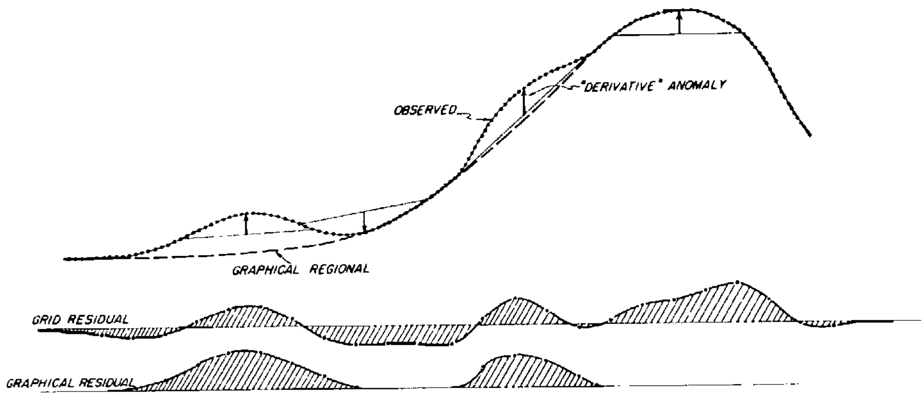


Fig. 20. Hypothetical gravity profile with graphical regional; "Grid residual" and graphical residual below (Nettleton, 1954). *Courtesy SEG.*

indicate the magnitude of the difference of the center point from the average around the circle or at the ends of the "bridge." This arrow is upward if the center point is higher than the average around it and downward if its value is lower. If we take such bridges at regular intervals along the anomaly curve and plot the length of the arrows, we get the result shown by the "grid residual" profile with the shaded areas marking the departures or local differences from the observed curve. It will be noted that these differences are positive over the central parts of a local anomaly and negative on its flanks.

The graphical residual process is illustrated by the "graphical regional" shown along the profile. For a relatively simple situation such as is illustrated here, the local departures are quite obvious and the construction of a graphical residual is simple and nearly any experienced interpreter would draw this in about the same way. The differences between this "regional" and the observed curve are plotted as the graphical residual at the bottom of the diagram. Note that for the two local anomalies, the positions of the peaks of the anomalies are very nearly the same as for the grid residual curve.

The principal likenesses and differences between the anomalies as determined by the two systems are: 1) Both systems show the two local anomalies and their centers at the same locations; 2) the two local anomalies are both shown as simple maxima by the grapical residual and as narrower maxima with flanking minima by the grid residual; 3) the positive amplitude is considerably greater for the graphical residual; for the grid residual, the total amplitude from the bottom of the flanking minima to the top of the maximum is approximately the same as the positive amplitude of the graphical residual. This relation is very much dependent on the choice of a radius appropriate to the size of the anomalies. 4) The grid residual system shows a comparatively large positive anomaly over the apex of the regional maximum whereas, with the graphical smoothing, this feature is entirely included in the regional and no residual anomaly is shown. This illustrates the manner in which a choice can be made in constructing a graphical regional which will include or exclude a feature more or less completely. Such a choice can never be made automatically by any grid type calculation unless the difference in the width (or "wave length") of the anomalies is quite large.

An example of the application of graphical and grid residual techniques to the same actual gravity map is shown by Figures 21 through 24.

The gravity map, Figure 21 (Nettleton, 1954), is for an area immediately south of Houston, Texas which contains a small salt dome with oil production on its westerly flank. The gravity field is dominated by a relatively strong northerly gravity increase of about 3.5 mgal within the limits of the map.

Figure 22 shows this regional change expressed as smooth, east-west trending contours at 1.0 mgal interval. The residual remaining after subtraction of this smooth regional is indicated by the residual contours at 0.2 mgal interval. These show the fairly circular residual minimum with a relief of a little over 0.6 mgal, which is fairly well, but not exactly centered over the dome as indicated by the lines of dots corresponding to the 4500 and 6000 ft contours on the salt. This residual gives a relatively simple anomaly which would be amenable to comparison with effects which might be calculated from an assumed form of the dome and

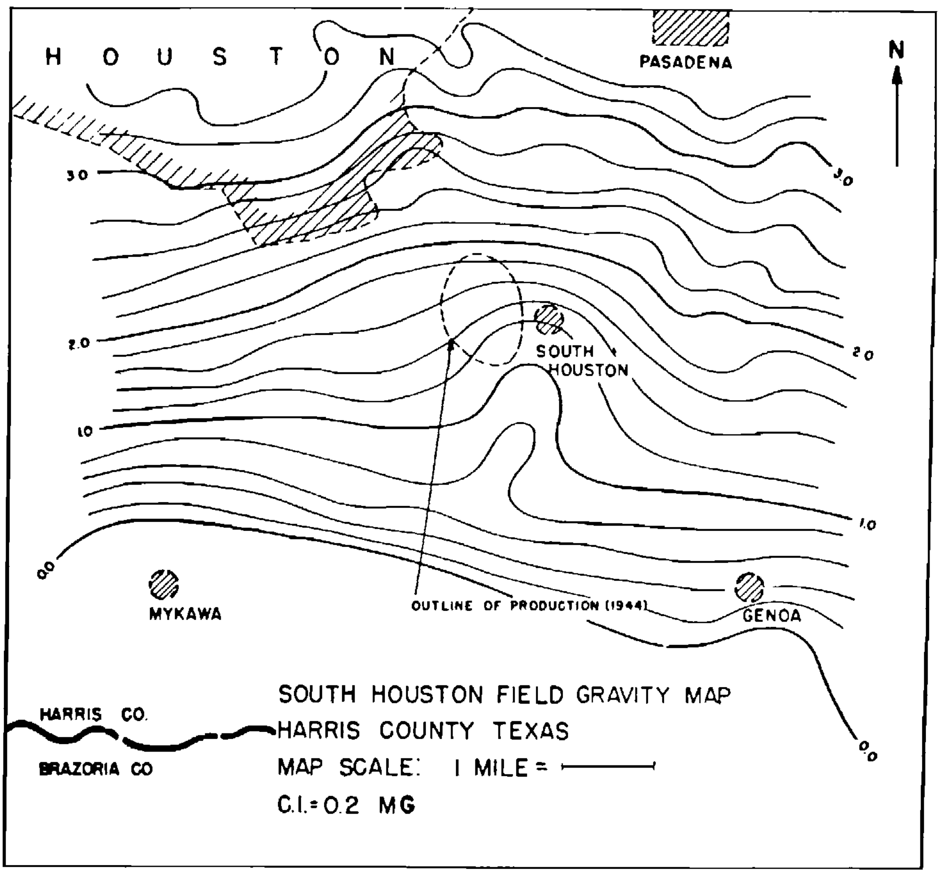


Fig. 21. Observed gravity map, South Houston area, contour interval 0.2 mgal (Nettleton, 1954). *Courtesy SEG.*

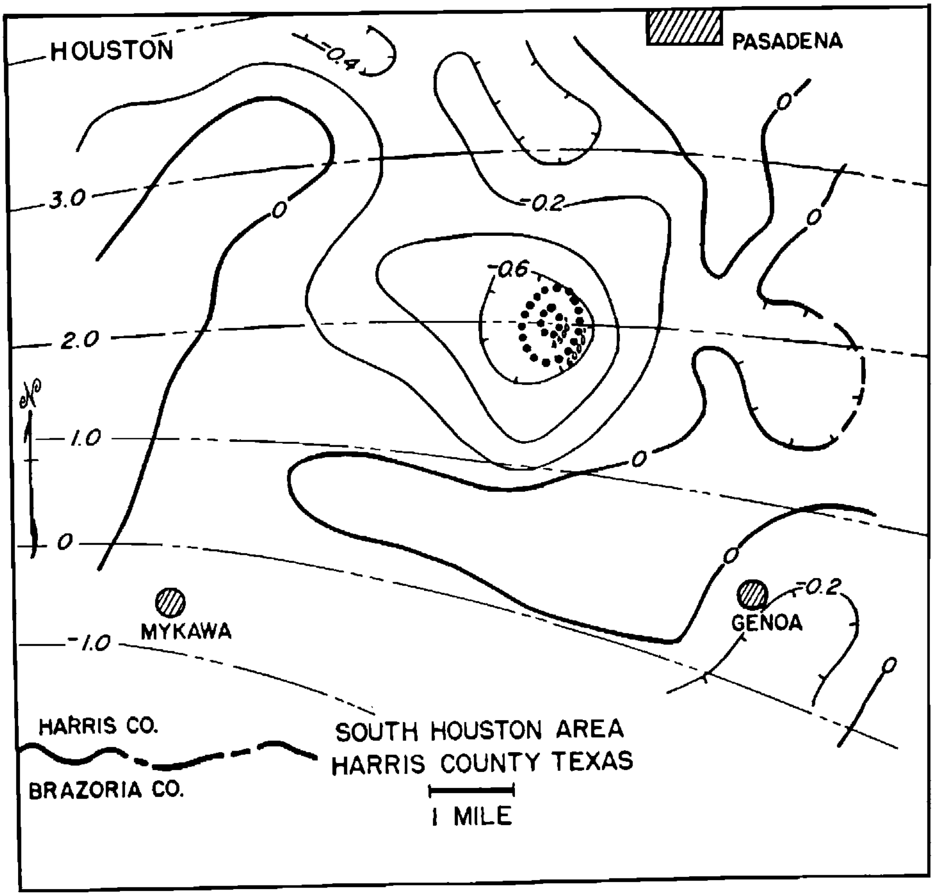


Fig. 22. Regional gravity contours, at 1.0 mgal interval, and graphical residual contours, at 0.2 mgal interval; residual contours obtained by graphical subtraction of the regional from the contours of Figure 21 (Nettleton, 1954). *Courtesy SEG.*

which could be used to determine the approximate size of the salt column.

The contours of Figure 23 give values from a second derivative calculation carried out on gravity values interpolated on a one-half mile grid from the contours of Figure 21. This map shows a quite sharp negative anomaly over the dome and the contour pattern is considerably more complicated than in Figure 22. Note that the negative anomaly is quite well centered over the dome. Note also that this negative anomaly is surrounded by a peripheral zone of positive values indicated by the dashed line. This is the flanking reversal which is introduced into the result by the fact that the positive and negative values add up to zero as is indicated by the profile of Figure 20.

Figure 24 is calculated with a simple center-point-and-one-ring system but with the radius expanded by the factor $\sqrt{5}$ (see Figure 18), so that its outer ring is at the same distance as the outer ring of the formula used for Figure 23. This makes the effective outer limit of the area of calculation comparable with that of the heavily weighted outer ring of for-

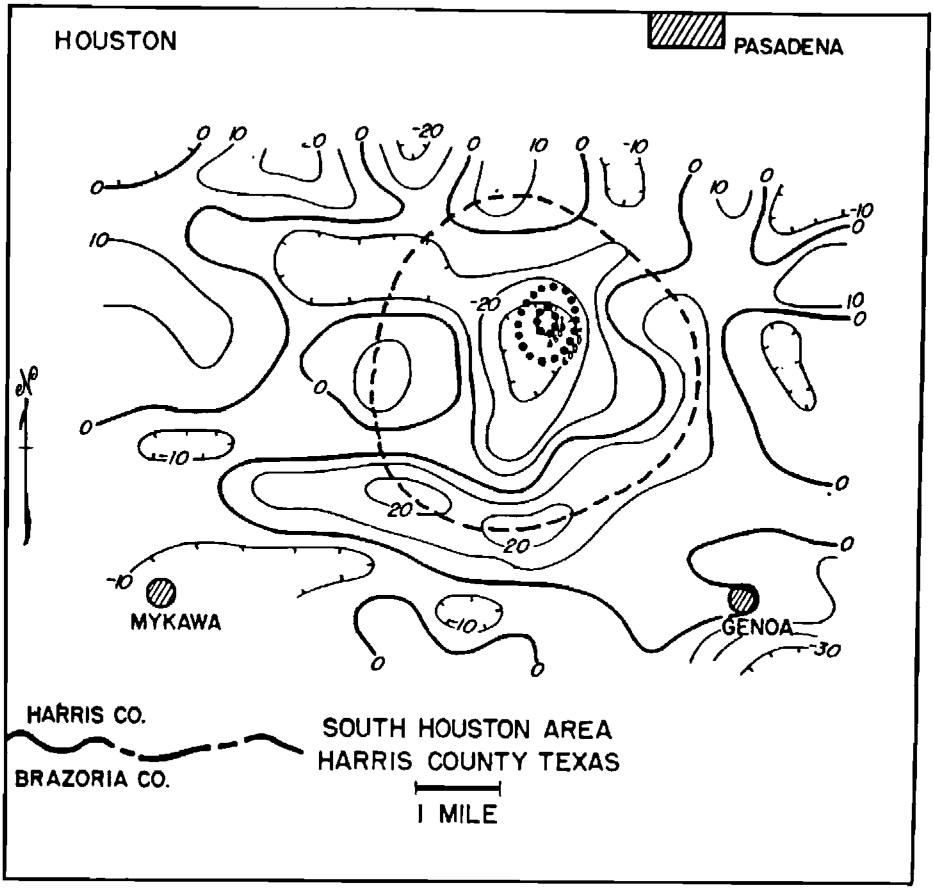


Fig. 23. Second vertical derivative contours, calculated with ½ mile grid spacing from map of Figure 20, with weighting coefficients of formula 7, Figure 19. Contour interval 10×10^{-15} cgs units (Nettleton, 1954). *Courtesy SEG.*

mula 7 and therefore the resulting contour pattern is very similar.

Calculations of gravity from mass anomalies could be made readily to fit the graphical residual but would not be directly applicable to the grid residual because of its positive and negative parts. As mentioned above, the effectiveness of the grid residual system is quite independent of the general slope of the field and grid residuals can be calculated readily where the background is very steep and where choice of a regional and subtraction to form a graphical residual might be very difficult (see example from Los Angeles Basin, Figure 33).

SURFACE FITTING METHODS

A quite different approach to the anomaly separation problem is the application of two-dimensional surface fitting calculations. In this method, a least squares operation is carried out which determines a potential "surface" which is fitted to the observed gravity map. The closeness of fit depends on the "degree" or "order" of the calculation.

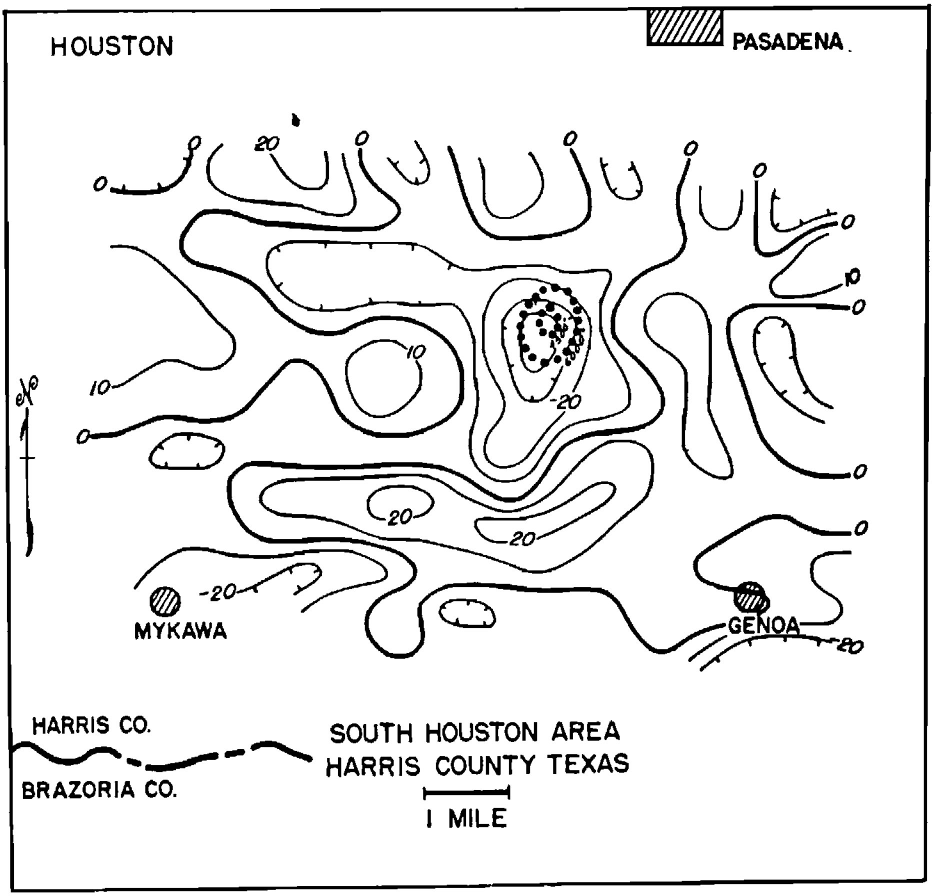


Fig. 24. Second derivative map, calculated with grid spacing ½ .$\sqrt{5}$ mile, from map of Figure 20 with weighting coefficient from formula 1 a. Table 1; contour interval 10 x 10^{-15} cgs units (Nettleton, 1954). *Courtesy SEG.*

The general idea of the surface fitting technique is illustrated schematically by Figure 25. The illustration is for a single profile line only, whereas the operation is carried out in two dimensions to fit a surface rather than the single line. Observed gravity is represented by curve G. A first degree fit would be a straight line, no. 1. A second degree fit could have one reversal and two line crossings as indicated by line no. 2. A third degree curve could have two reversals and three line crossings as shown by line no. 3 and a fourth degree curve could have three reversals and four line crossings as shown by line no. 4. In each case, the calculated curve is fitted so that the sum of the squares of the differences between it and the observed curve is a minimum. The differences between the calculated and observed values are the residuals. As higher orders are calculated the fit between the calculated and observed curves becomes closer and the residuals become smaller. A series of maps can be prepared for different degrees of fit. At higher degrees, the smaller and sharper residuals emphasize smaller details of the

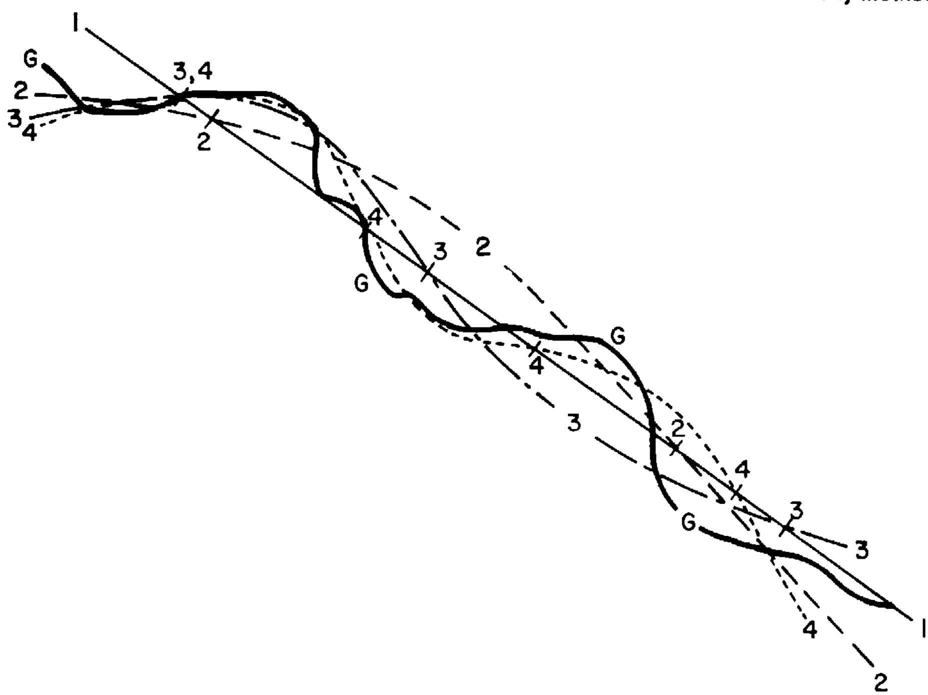


Fig. 25. Schematic curves illustrating least squares surface fit technique. Curve G represents an observed gravity profile. Curves 1, 2, 3, and 4 represent fits of successively higher degrees. The "residual" for a given order is the difference of the observed from the corresponding surface fit.

original map. It is possible also, if desirable, to map the difference between a higher and a lower order fit. By using different combinations, a great variety of maps can be made depending on the use desired and the nature of the features chosen for emphasis.

The rather involved mathematical procedure of surface fitting has become feasible only with the application of high-speed digital computers. With the computer it is possible to carry out the entire operation, including the contouring of the maps, quite rapidly. The initial input can be the horizontal coordinates and gravity values at the individual stations on a set of punched IBM cards.

An example is available from an application of the procedure to the Mid-Continent gravity high (Coons et al, 1967). This is one of the strongest gravity anomalies in the United States, with very sharp relief and with amplitudes of well over 100 mgal. It is not a particularly good example from the standpoint of petroleum exploration but serves quite well to illustrate the application of surface fitting.

Figures 26 and 27 illustrate the process as carried out over a portion of the survey in north-central Iowa. In Figure 26 the set of four maps shows the original Bouguer anomaly map and the surface fits calculated for the 7th, 10th, and 13th orders. It will be noted that, as the order becomes higher, the complexity of the pattern increases. The axis of the central maximum, as drawn on the Bouguer anomaly map, is repeated on each of the surface fit maps. With the higher orders, the details of the maps

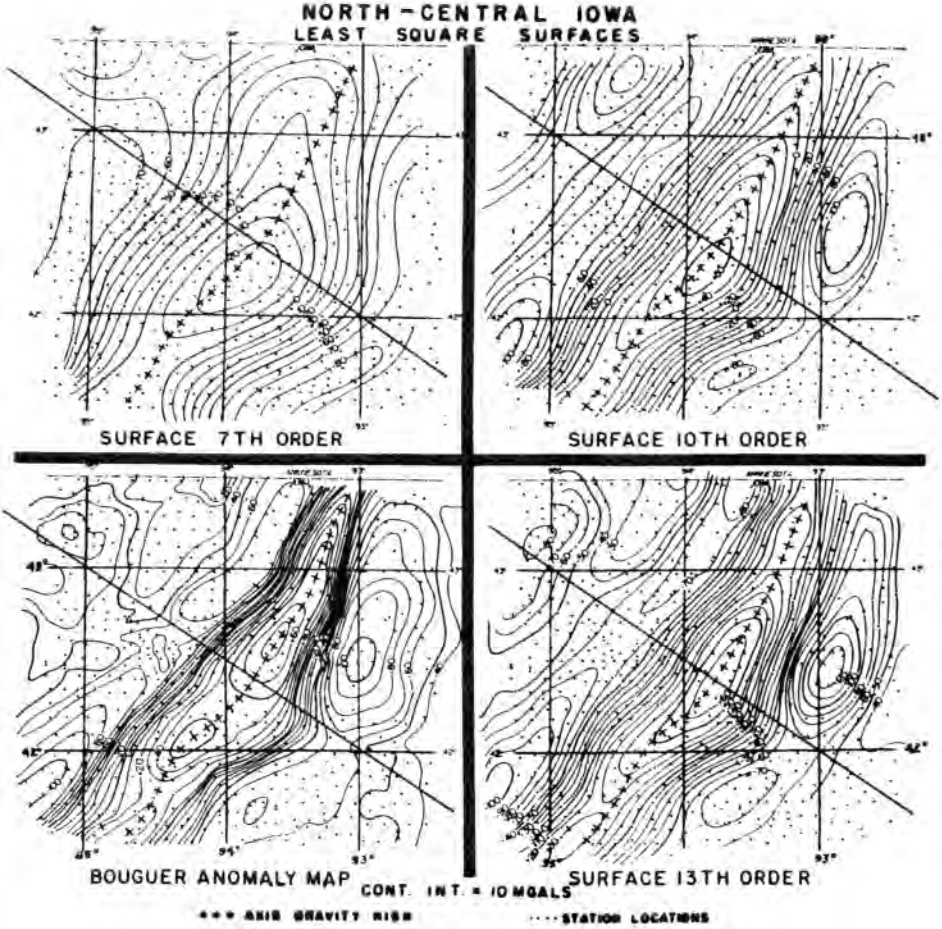


Fig. 26. Example of least square surface fit; lower left map is Bouguer gravity. Other maps show gravity calculated with successively higher orders of fit (Coons et al, 1967). *Courtesy AAPG.*

conform more closely with this axis.

The four maps of Figure 27 shows the Bouguer anomaly map repeated and the residuals for the three different orders. These residuals are the differences between the surfaces shown in Figure 26 and the Bouguer anomaly map. It will be noted that, as the order of fit increases, the complexity of the residual increases but the total relief of anomalies decreases. The 7th order residual is composed largely of the central maximum and its flanking minima with relatively little detail within these features. In the 13th order residual, there is a great deal of internal detail and many local features are brought out which are not apparent in the 7th order map. The anomalies marked A and B on the 13th order map are local minima which develop from quite inconspicuous saddles on the Bouguer map.

Profiles from diagonal lines across the maps of Figures 26 and 27 are shown by

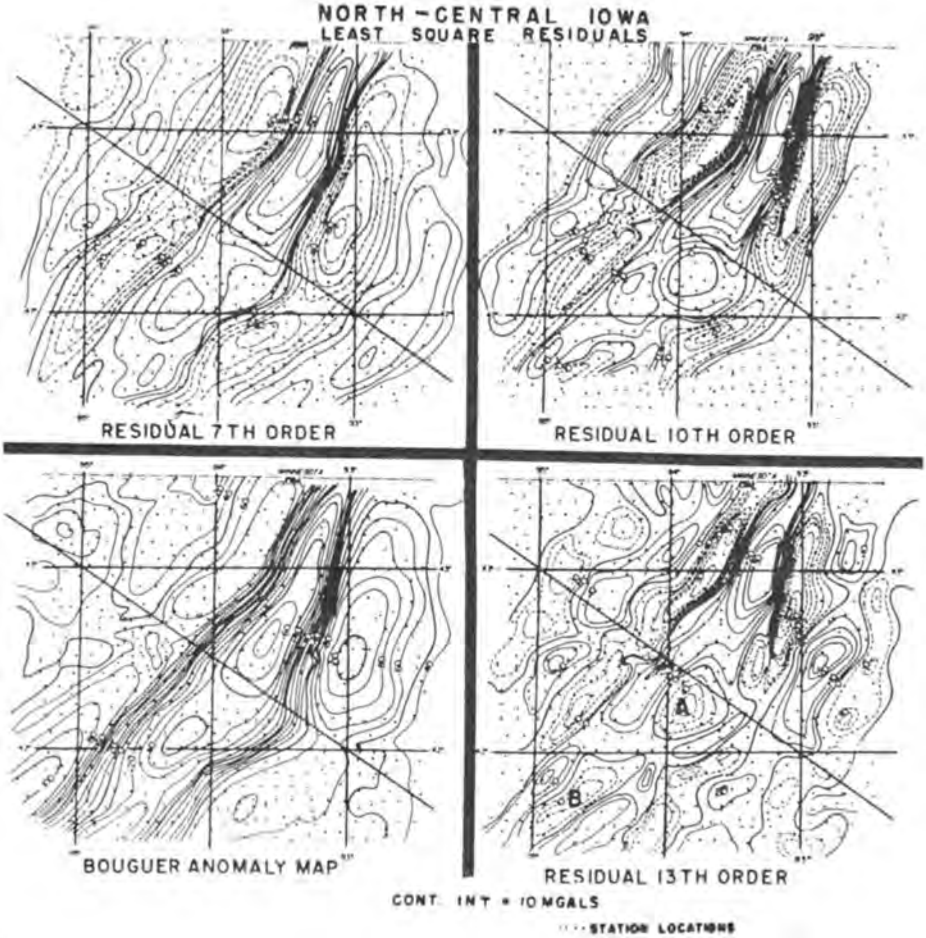


Fig. 27. Example of least squares residual; lower left is observed gravity. Others are residuals calculated by subtracting successively higher orders of fit from observed gravity (Coons et al, 1967). *Courtesy AAPG.*

Figure 28. Each set shows the observed gravity (the same for all three sets) the fit profile (from Figure 28) and the residual profile (from Figure 27). There is a tendency for the system to introduce residual periodic features as the "wavelength" of the order attempts, but does not completely succeed, in fitting the observed gravity. On the 13th order residual, the local features 1 and 2 are of this nature, where the fit does not completely match the flanks of the large central anomaly.

Since the surface fit technique is a least squares process, the fitting surface, which is the "regional," is a surface which will have both positive and negative differences from the observed; the residual maps are therefore balanced between positive and negative areas. This is generally similar in its result to the calculation of second derivative or "grid residual" maps with central maxima and flanking minima as shown by Figure 20. For this reason, the residual anomalies from the surface fitting techniques can-

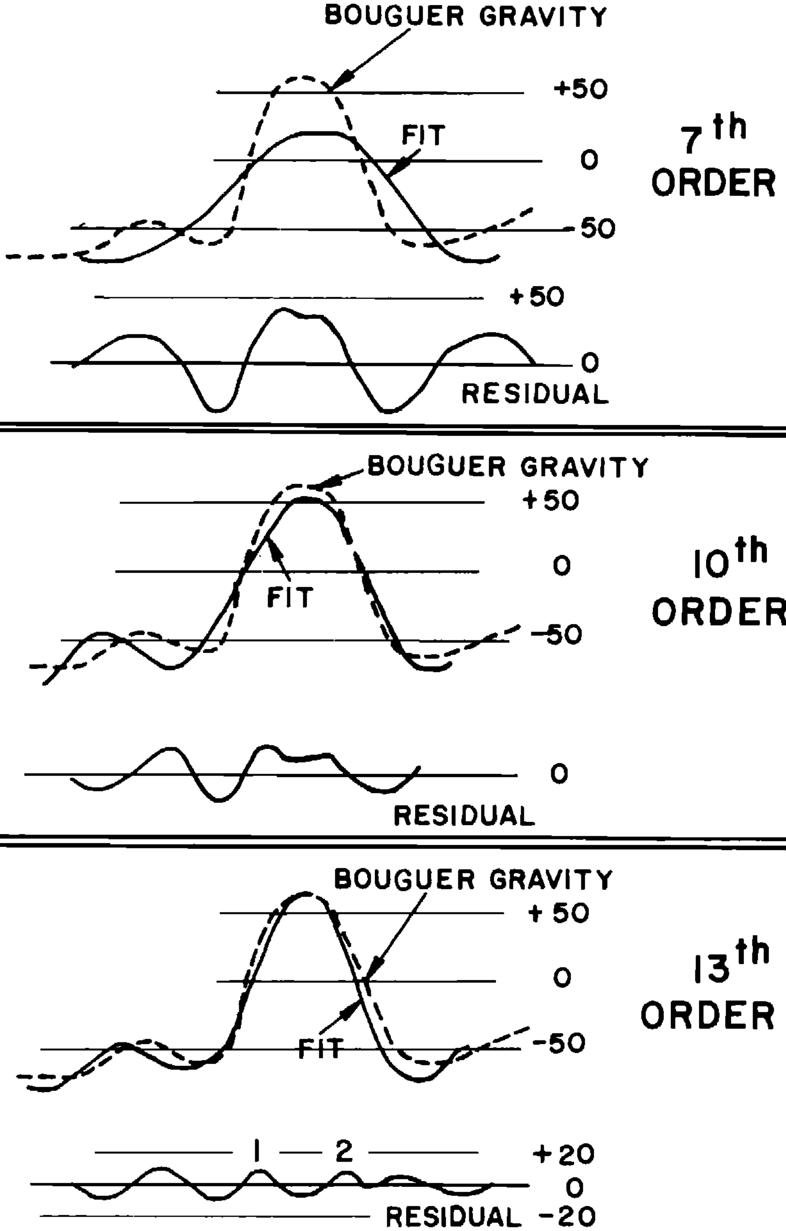


Fig. 28. Profiles from Figures 25 and 26 showing increasing closeness of fit and smaller residuals with higher degrees of least squares surface fitting calculations.

not be used directly for quantitative geological interpretation by calculating gravity effects to compare with the observed anomaly.

As shown by the example, the system is quite effective in picking out local anomalies. It is not completely objective because the results depend to a considerable degree on the operator's judgment and discretion in deciding which size or type of anomalies it is wished to emphasize and the corresponding "order" of the operation. Since the process is so highly automated, it is largely a question of computer time and the efficiency of the program which determine how many and which maps to produce and use. In this way the system is quite comparable with the second derivative or "grid residual" type calculation in that, for those applications also, the general nature of the resulting maps depend very greatly on a human judgment of the spacing and the weighting of coefficients used. Second derivative maps calculated with a suitable choice of grid spacing and coefficients would give results very similar to those shown by the least squares residuals of an order corresponding to a similar size of unit area and would have almost exactly the same centers of maximal and minimal closures.

By an appropriate choice of the order of the residual it is possible, by the least square fit technique, to isolate or emphasize anomalies of almost any magnitude desired within the limits of the definition by the spacing of the original data. If location and approximate amplitude of anomalies is the principal objective of the operation the system can be very effective. If quantitative evaluation of particular anomalies by calculation from geological interpretations or assumptions is to be carried out it is necessary to go back to the original Bouguer gravity map and perform some sort of a graphical regional-residual operation to determine more precisely the magnitude and form of the isolated anomaly which is to be accounted for. The maps from the surface fitting technique may be very helpful in choosing the particular anomalies to analyze and also in providing some control on the choice of a regional as this may be approximately indicated by a low order surface fit.

FREQUENCY ANALYSIS OF POTENTIAL FIELDS

A somewhat different treatment of potential fields is through frequency analysis. This involves transforming the details of a gravity map into the frequency domain through the application of the Fourier transform. In this concept the observed gravity map is approximated by the sum of a series of harmonic waves in the x-y plane that have different spatial frequencies, azimuths, amplitudes and phase relations. This set of waves is called a spectrum and constitutes the frequency domain. A filtering system can then be devised which will pick out certain components and eliminate others in a manner which is quite analogous (in one dimension) to a ordinary electrical filter which will pass certain frequencies and reject others. Thus the filter characteristics can be chosen to pick out desirable components and reject undesirable ones.

The frequency characteristic of a filter (Fuller, 1967) can be expressed in the two-dimensional x-y frequency field as contours of attenuation expressed in decibels. Figure 29 is such a chart and corresponds to the weighing factors and grid distances of formula 7 of Table I (Figure 19). The "map" scale, horizontal and vertical, is in cycles per data interval. The lowest numbers, at the lower left corner of the diagram, correspond to long wavelengths; the highest num-

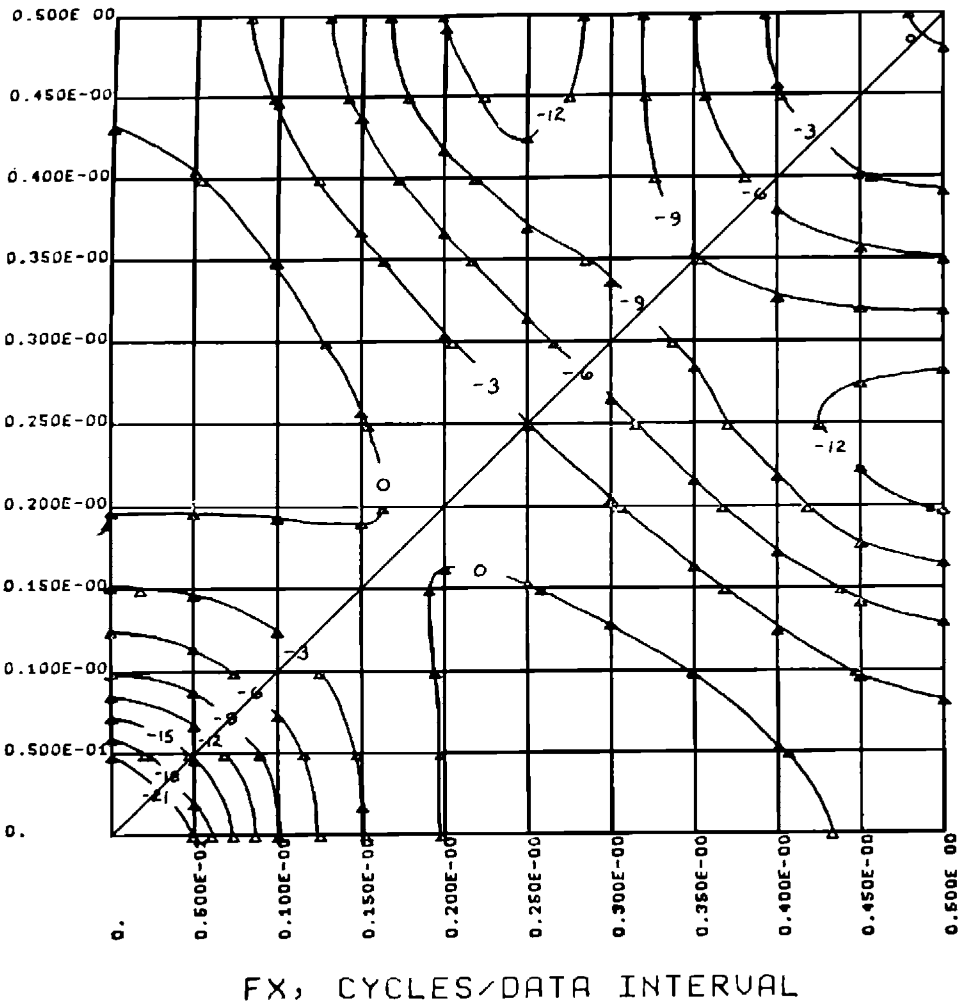


Fig. 29. Sample of Fuller frequency response chart showing contours of equal attenuation. This chart is for equation 7 of the table, Figure 19. Profile 7 of Figure 31 is plotted from the contours on the diagonal line across this chart (Fuller, 1967, p. 667). *Courtesy SEG.*

bers (0.50) at the upper right corner of the diagram correspond to a "wavelength" of only two grid data intervals which is the minimum which can be expressed by the grid.

The general ideas of grid operators as frequency filters can be expressed by plotting the relative weights of the coefficients as a function of their radial distance from the center point of calculation as shown in Figure 30. Here the relative weights are shown as taken from Table 1 (Figure 19). Points are plotted for formulas 1, 3, and 7 of the table. It will be noted that the curve for formula 3, with alternating negative and positive weighting factors, has a relatively sharp form corresponding to a shorter wavelength and this formula, therefore, tends to "fit" the anomalies

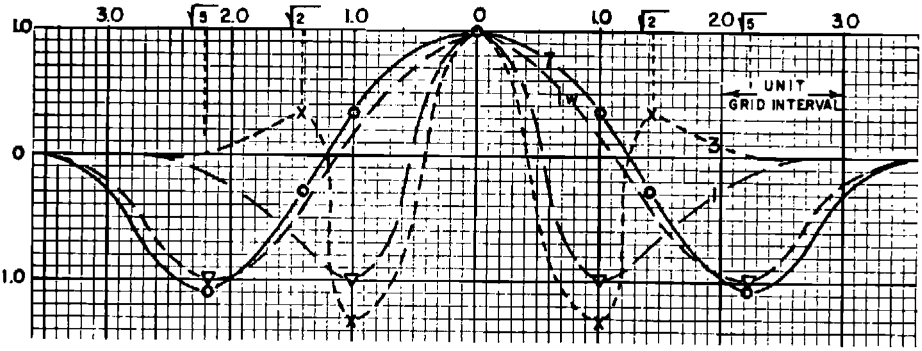


Fig. 30. Profiles of second derivative formula with normalized weights, from the table, Figure 19. The normalized weights are plotted against the radius of the circles to which they apply.

with a diameter of about two grid units and consequently emphasizes relatively small features. On the other hand formula 7, which has positive coefficients in its central part and negative in its outer part gives a much broader waveform and tends to emphasize features with a diameter of about four grid units.

On the Fuller chart (Figure 29), this optimum emphasis corresponds to the maximal zone represented by the zero contour; the axis of this maximum is at coordinates of approximately 0.25 cycles/data interval which is the reciprocal of the four grid units mentioned above.

Curves 1 (for the center-point-and-one-ring system) and 3 (for the Henderson formula with the first coefficient strongly negative) are similar and would produce similar maps. Curve 1W of Figure 30 is the same as curve 1, but with the horizontal scale widened by $\sqrt{5}$. This makes its circle have the same radius as the outer circle of curve 7, to give a very similar curve on Figure 30, and calculations produce similar maps. This similarity of maps is demonstrated by Figure 23, calculated with formula 7 and Figure 24, calculated with the center-point-and-one-ring system with radius expanded by $\sqrt{5}$.

To illustrate other frequency characteristics, Fuller has made diagrams similar to Figure 29, by applying the Fourier transform to a number of the grid operators in Table 1 (Figure 19). By plotting profiles of the frequency characteristics from the Fuller diagrams it is possible to get a feeling for the nature of the various operators. This has been done in Figure 31 for the operators corresponding to formulas 1, 1a, and 7 of the table. In each case the frequency response is plotted from the contours of the Fuller diagrams on a profile diagonally from the lower left to the upper right corner as illustrated by Figure 29 which is the frequency response pattern for formula 7. The profile on this diagonal line is shown by curve 7 of Figure 31. To compare these frequency diagrams in a manner similar to that used for comparing the second derivative formulas of Table 1 they can be "normalized" by making the first co-efficient equal to unity. This means that the numbers from the contours of the Fuller diagrams must be raised or lowered by the decibel equivalent of the factor K of the table which is determined from db $= 20 \log_{10}K$ (Fuller, page 662). Thus profile 7 of Figure 30 can be normalized by increasing it by 2.8 db to give the curve shown as $7n$. The curve for formula 1 (Fuller's Figure 11) which corresponds to a "normalized"

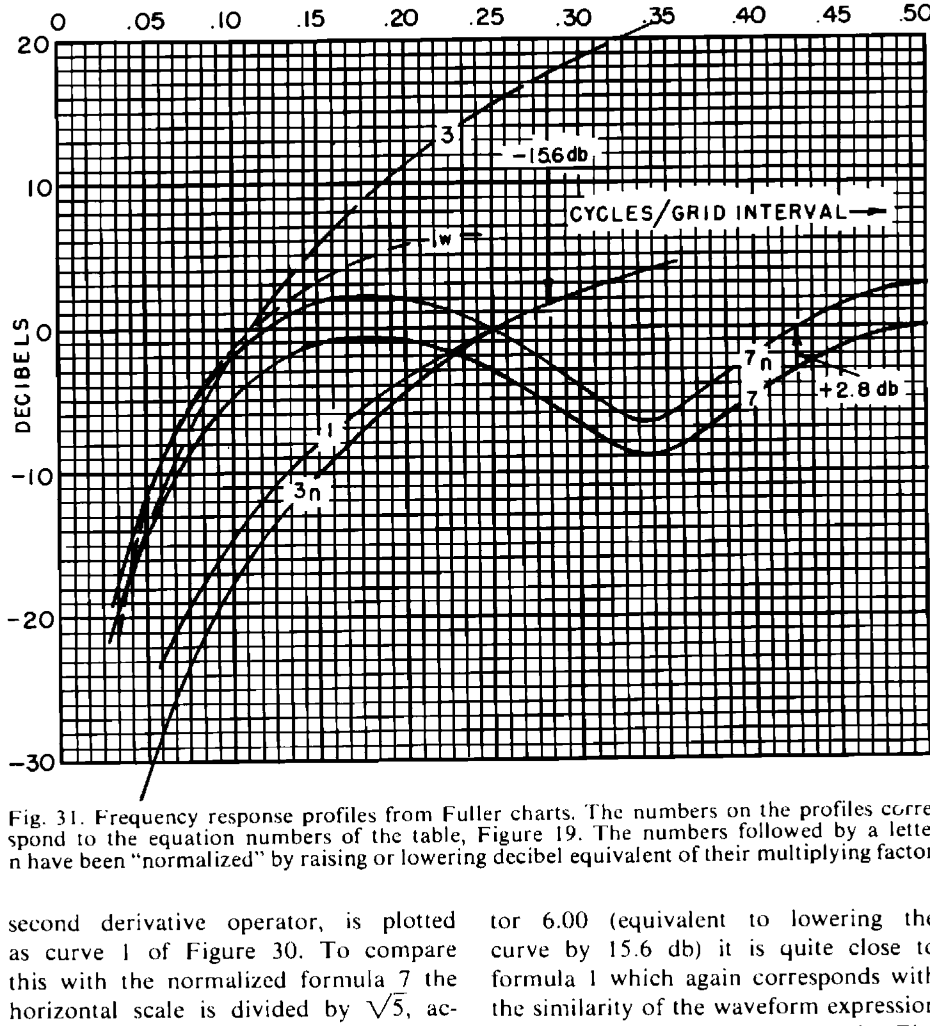


Fig. 31. Frequency response profiles from Fuller charts. The numbers on the profiles correspond to the equation numbers of the table, Figure 19. The numbers followed by a letter n have been "normalized" by raising or lowering decibel equivalent of their multiplying factor.

second derivative operator, is plotted as curve 1 of Figure 30. To compare this with the normalized formula 7 the horizontal scale is divided by $\sqrt{5}$, accounting for the smaller numbers of cycles per data interval, to give the curve 1W shown by the dashed line of Figure 31. Thus these two normalized curves (i.e., $7n$ and $1W$) are very close together, which is another way of expressing the similarity of the profiles for equations 1W and 7, as shown by Figure 30 and by the similarity of the maps, Figures 23 and 24. Similarly the frequency response profile for formula 3 is plotted as curve 3, Figure 31. When it is normalized by dividing by the factor 6.00 (equivalent to lowering the curve by 15.6 db) it is quite close to formula 1 which again corresponds with the similarity of the waveform expression of these two formulas as shown by Figure 30.

The above illustrations indicate in a general way the transformation from the grid domain to a frequency domain. The transforms can be applied to the design of operators which have selected frequency characteristics. These principles can be used not only to separate anomalies but also, in one form of "trend analysis," to favor features having certain directions (see Fuller's Figures 39 to 44).

The frequency response approach can be a useful alternative means of analyzing gravity anomalies. It may have some advantages over the least squares approach in that there may be more flexibility in designing the operators. The frequency analysis may or may not balance positive and negative differences, depending on the design of the operator; this result is inherent in the least squares system. It may give residuals more nearly approaching those which would be obtained by an empirical smoothing operation. This is true if there is a wide difference in frequency between the anomalies "passed" and those "rejected" by frequency analysis, but, as in any filtering operation, the degree of discrimination depends upon the frequency differences involved.

No amount of computer processing will clearly separate anomalies unless their horizontal dimensions are distinctly different. With a ratio of, say 5 to 1 in dimensions, the selection could be fairly complete; with a ratio of 2 to 1 it would be very incomplete.

EXAMPLE OF ANOMALY RESOLUTION

An example of gravity data which illustrates both the geological significance of a regional anomaly and the recognition of subtle but significant local anomalies is available from the Los Angeles Basin.

Figure 32 (Nettleton, 1962) is a geologic map of the basin on which the regional gravity of 5 mgal contour interval has been superimposed. The general configuration of the gravity pattern is in accordance with expectations from the general form of the basin itself. The minimum is over the area of deepest sediments, and the gravity increases very rapidly southwest toward the basement outcrop in the Palos Verdes Hills. The sediments are all Tertiary clastics and of substantially lower density than the Franciscan schist of the basement, and quantitatively, the 60 mgal southwesterly gravity rise from the center of the minimum to the Palos Verdes Hills is consistent with that which would be expected from reasonable estimates of the total thickness of sediments and of the densities of the sedimentary and basement rocks. (See calculation, page 52.)

Figure 33 is a profile across the basin along line AA′, of Figure 32, and shows a geological section together with the gravity profile from Figure 32. This profile shows the bottom of the gravity minimum corresponding almost exactly with the deepest sedimentation. The very strong gravity maximum over the Puente Hills structure almost certainly means that the basement rocks are involved in this structure; although not so shown on the geologic section, which extends downward only into the Tertiary sediments. On the southwest side of the basin, we see that the Dominguez structure does not produce a noticeable gravity anomaly.

Figure 34 shows a detailed gravity survey, with one-half mile station spacing, over a part of the southwest flank of the basin which includes the areas of the Long Beach, Dominguez, parts of the Rosecrans, Wilmington, and Torrance oil fields. On this map, casual inspection would suggest that there is no gravity expression at all of these oil-field structures. However, if one looks carefully, he would notice that, for instance, on the northeast side of the Long Beach field the contours are a little closer together than over the field itself, and, similarly, on the northeast side of the Dominguez field the contours are crowded more closely than they are over the oil-field area. These differences are significant as shown by the second

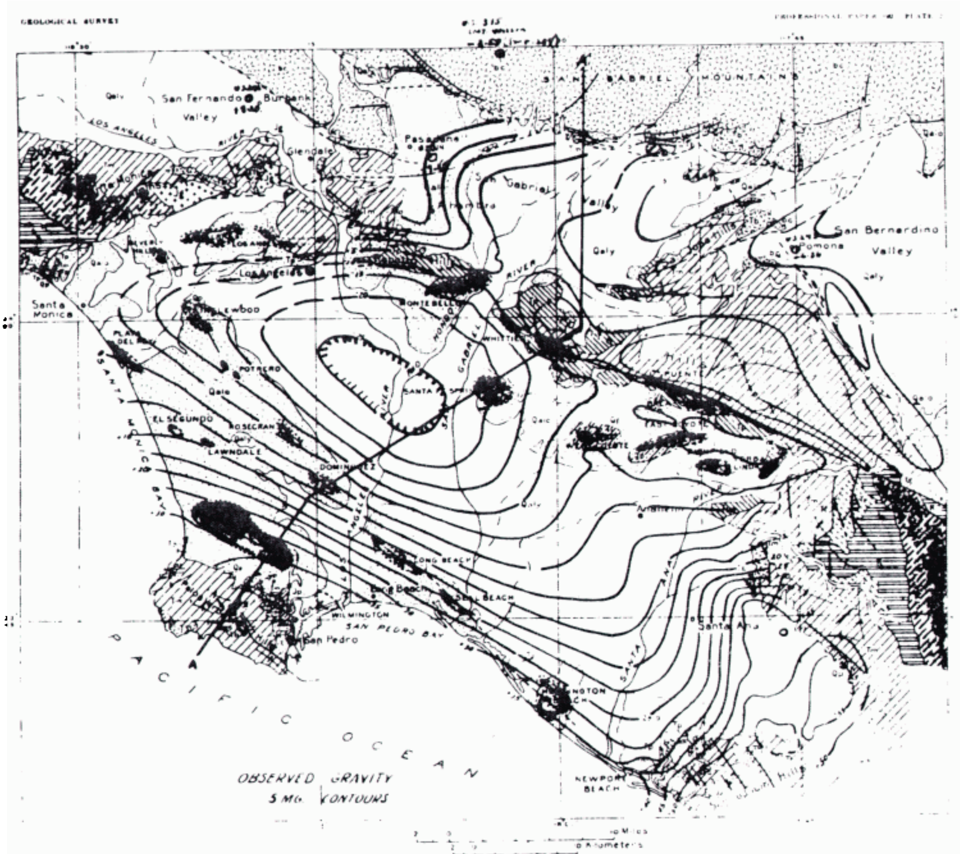


Fig. 32. Surface geologic map, Los Angeles Basin (modified from USGS Professional Paper 190) with gravity contours at 5 mgal intervals added (Nettleton, 1962, p. 1831). *Courtesy AAPG.*

derivative map for this area (Figure 35). This map was calculated by formula 7 of Figure 19 with a one-half mile grid spacing from the observed map of Figure 34. This second derivative map develops a very strong and definite positive trend which extends through the Long Beach, Dominguez, and Rosecrans fields, and also another positive trend which includes the Wilmington and Torrance fields. The second derivative calculation has brought out these features which were very obscure in the observed gravity map because of the very strong "regional" gradient. Of course, they must be present on that map to be developed by the derivative calculation, and their presence is indicated by the previously mentioned variations in contour spacing. It should be pointed out also that the negative flanks of the positive trends are parts of the total anomaly in the same way that previously mentioned negative aspects of second derivative maps have been pointed out.

While a test has not been carried out, it is almost certain that residuals from a least squares fit or from anomalies from a frequency filter, with optimum choice of parameters, would clearly resolve the same features.

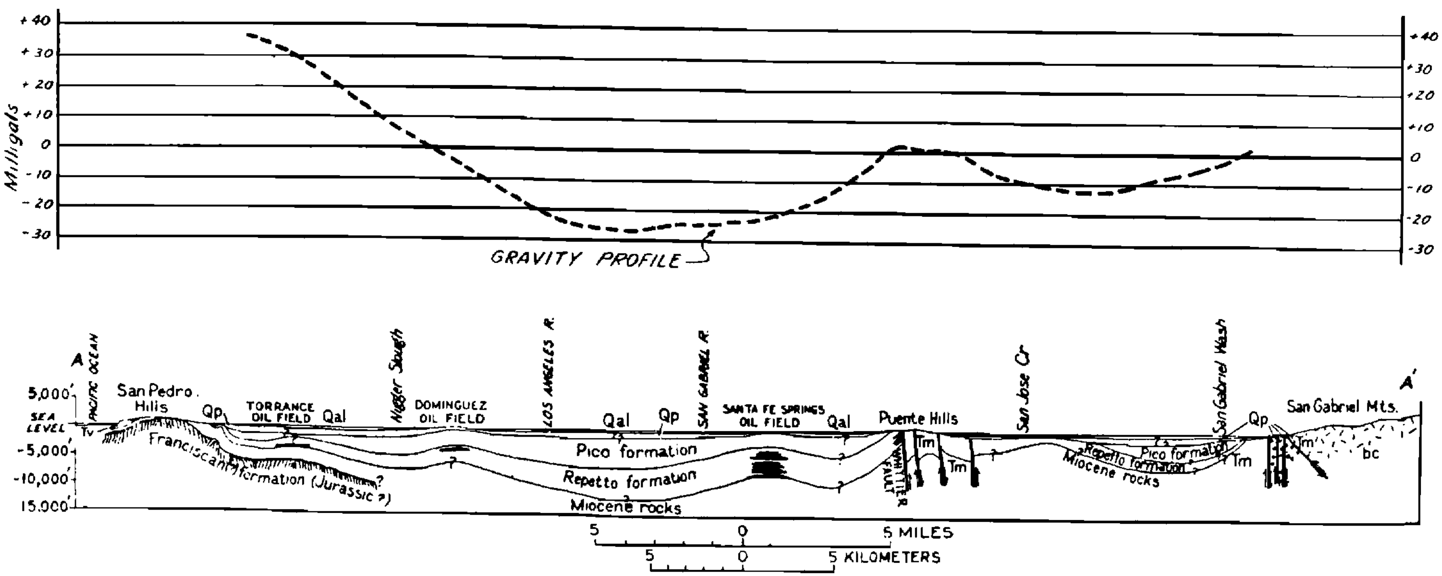
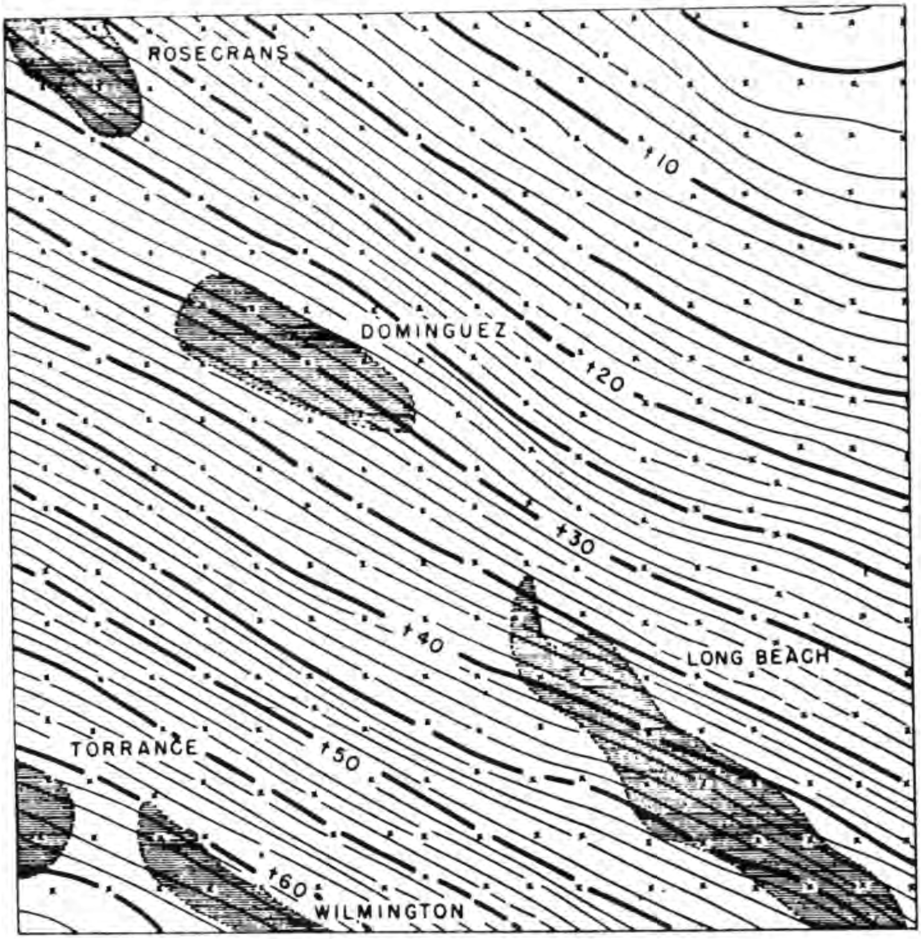


Fig. 33. Cross-section, line A A', Figure 31 (modified from Guidebook, XVI International Geological Congress, 1932) and gravity profiles from contours of Figure 32 (Nettleton, 1962, p. 1832). *Courtesy AAPC.*



I MILE

Fig. 34. Gravity map, part of southwest flank, Los Angeles Basin; contour interval 0.2 mgal, stations on regular grid with ½ mile spacing (Nettleton, 1962, p. 1833). *Courtesy AAPG.*

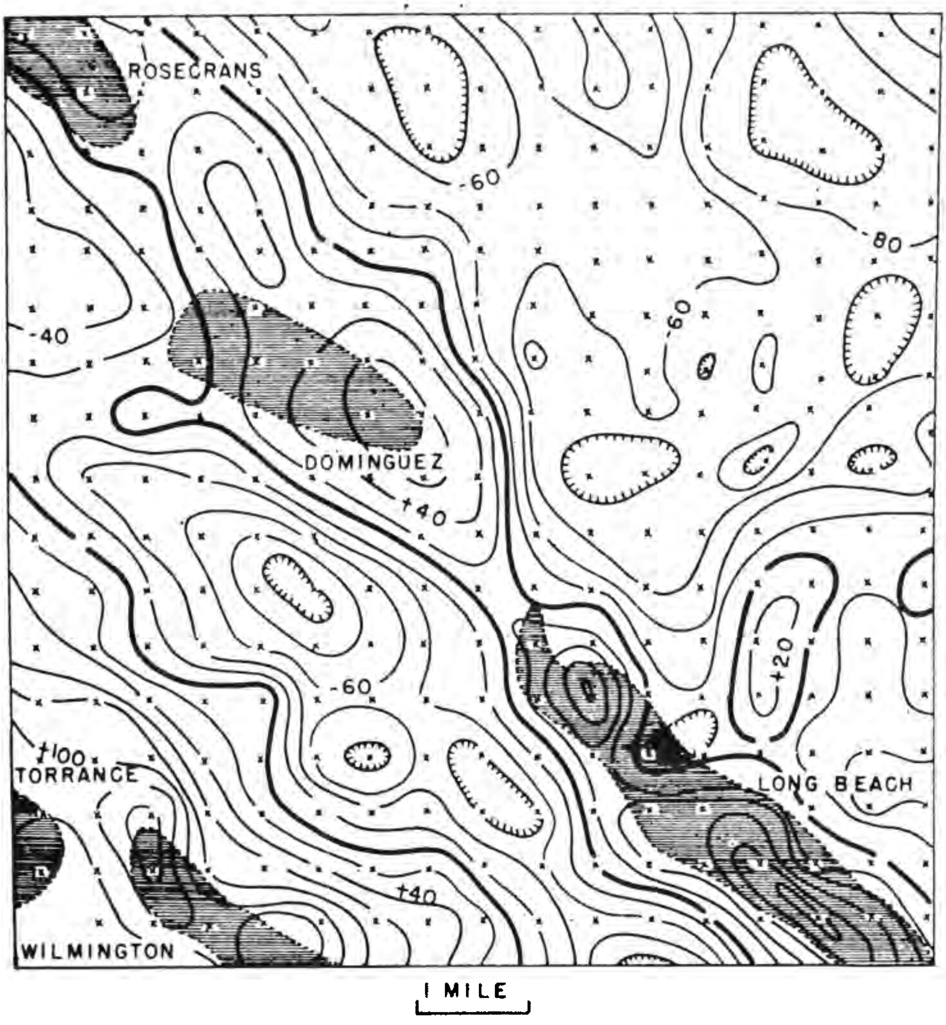


Fig. 35. Second vertical derivative map, Los Angeles Basin; calculated from gravity of Figure 35 by formula 7 of Figure 19; contour interval 20×10^{-15} cgs units (Nettleton, 1962, p. 1834). *Courtesy AAPC.*

CHAPTER VI

THE INTERPRETATION OF GRAVITY RESULTS

The instruments, field operations, data reduction, and mapping described above produce a gravity map which, ideally and theoretically, is free of planetary effects of the earth as a whole and from the visible topography. If the earth's crust were composed of uniform horizontal sheets with no horizontal variations in density, the gravity map would be featureless. All of the features which it shows have their origin in horizontal discontinuities at any depth from the grass roots down. The objective of gravity interpretation is to deduce the geological character of the subsurface from variations in the gravitational field.

The interpretation is inherently ambiguous. In the first place, there is no single mathematical solution to the determination of the source of a gravity field no matter how precisely that field may be mapped. Figure 36 shows a simple positive gravity anomaly which has been calculated as the gravity effect from a sphere (or point mass) shown as body no. 1. The same anomaly could be accounted for completely by a thin lenticular body, no. 3 in the figure, at a very shallow depth or by an intermediate narrower, and thicker body such as no. 2. There is a "cone of solutions" as indicated by the dotted lines. The spherical mass, no. 1, is the deepest possible solution as any deeper mass would produce a broader anomaly but solutions at all shallower depths to the surface are possible. Also combination solutions are possible for which part of the total mass is in different bodies. The one thing that the several possible solutions have in common is that their total mass (i.e., the product of the volume times the density contrast of the anomalous material) must be the same, theoretically, for each solution.

As will be shown later there is a close relation between depth and sharpness or width of gravity (and magnetic) anomalies. For a given breadth of anomaly there is a maximum depth, but, as indicated above, the source with an appropriate modification of shape, can be shallower and broader. Systems of gravity analysis with calculations of varying degree of detail, are sometimes presented as representing disturbances from a limited band or zone of depths. It is width rather than depth which is the basis of discrimination and which

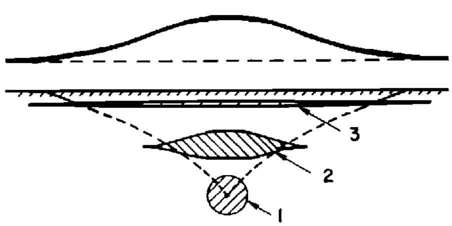


Fig. 36. "Cone of Sources;" the sphere (1) is the deepest body which can approximately account for the gravity anomaly shown. Shallower and broader bodies, as 2 and 3, also could account for the anomaly. All would have the same total mass anomaly.

should be kept in mind in any use of such maps.

It is evident from the above paragraphs that no unique determination of the depth and form of the mass anomaly causing a given gravity anomaly is possible without control other than the gravity field alone. On the other hand, possible sources can be sharply limited if other information is available such as the probable depth to a major density contrast, a drilling contact to give one or more points of control on a principal surface of density contrast, seismograph, or other geophysical data (such as a depth to the basement from a magnetic interpretation) or, in the absence of any factual control, empirical estimates of the probable lithology, the nature and depth to possible density contrasts and general geologic reasonableness. Thus, in spite of the theoretical ambiguity and the lack of mathematical precision, gravity anomalies can be interpreted to give very useful geologic information.

CALCULATIONS IN GRAVITY INTERPRETATION

A quantitative analysis of gravity results and evaluation of detailed relations to geology requires some determination of reasonable gravity effects which would be produced by a proposed interpretation and comparison of these calculated effects with those observed. In most cases such a calculation presupposes some sort of anomaly separation since it is often a particular feature on a map or along a profile which is the subject of the interpretation problem. For such cases the anomaly under investigation must be expressed in gravity, i.e., milligal units rather than as a second derivative because the anomaly resulting from such an analysis is in different numerical units and also has both positive and negative parts. This means that the anomalies subjected to interpretation control by calculation usually are those separated by some sort of smoothing operation. The other types of anomaly separation (grid operator, least square residual or frequency analysis) may serve well to locate the area and, in some cases, give some control on a reasonable regional, but they do not give features for which an interpretation may be readily checked by calculation of gravity effects from simple geometric forms.

Three simple geometric forms, the sphere, the horizontal cylinder, and the semi-infinite horizontal plate, have very simple mathematical expressions for their gravity effects. These effects can be used as approximations to a wide range of geologic bodies so that they are very effective controls on proposed solutions for a geologic body causing the anomaly. Thus, the simple calculations can often give useful control on reasonable values of depth, density contrast, and relief before undertaking a more complete interpretation operation, such as setting up a model and calculating its gravity effects by curves or charts or by a computer procedure.

The sphere

Gravity anomalies which are roughly circular in pattern, i.e., with roughly equal horizontal dimensions in different directions (up to a limit of a ratio of approximately 2 to 1 in their maximal and minimal directions) can be accounted for approximately by the gravity effect of a sphere. Calculations are somewhat better if made on a profile and section across the shorter dimension of the gravity pattern.

The gravity effect of a sphere is the same as that of the same mass concentrated at its center and is $g = \frac{\gamma m}{r^2}$

(Figure 37), directed from the point of calculation, P, toward its center. γ is the universal gravitational constant with value 6.67 x 10^{-8} cgs units. But we are always interested in the vertical component,

$$g_z = g\cos\theta = \frac{gz}{r} = \frac{\gamma mz}{r^3}$$

Putting in the numerical factors, including the gravitational constant and expressing distances in kilofeet (Nettleton, 1940, p. 103), we have

$$g_z = 8.52\sigma\left[\frac{R^3 z}{(x^2 + z^2)^{3/2}}\right].$$

To "normalize" this relation, it is expressed in units of depth by dividing by z so the independent variable is the ratio x/z. Then

$$g_z = 8.52\sigma R^3 \frac{z}{z^3(1 + x^2/z^2)^{3/2}} =$$

$$\frac{8.52\sigma R^3}{z^2}\left[\frac{1}{(1 + x^2/z^2)^{3/2}}\right] = Kf_s(x/z),$$

where $K = \dfrac{8.52\sigma R^3}{z^2}$ is a specific coefficient for a particular calculation and gives the gravity over the center of the sphere and $f_s(x/z)$ is a normalized "form-curve" with central value of unity and is applicable to any sphere calculation. The curve for $f_s(x/z)$ is shown by the lower (solid line) curve of Figure 38.

Horizontal cylinder

The simple expression for the gravity effect of a horizontal cylinder may be used to check magnitudes for gravity anomalies which are relatively elongated. If the long dimension of a feature is more than three times greater than its short dimension a fairly close approximation to its gravity effect can be obtained from the calculations for a horizontal cylinder. The fundamental gravity effect of a horizontal cylinder with the same notation as above for the

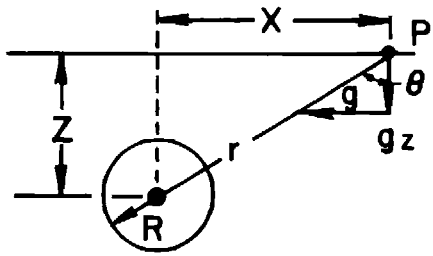


Fig. 37. Notation for gravity effect of sphere and horizontal cylinder.

sphere is given by $g_z = \dfrac{2\gamma mz}{r^2}$ where m is now the mass per unit length, which is

$$m = \pi R^2\sigma.$$

Making the same steps as above for the sphere (Nettleton, 1940, p. 106),

$$g_z = \frac{12.77\sigma R^2}{z}\cdot\frac{1}{1 + x^2/z^2} = K_c f_c(x/z),$$

where $K_c = 12.77\sigma R^2/z$ and is the gravity over the axis of the cylinder and

$f_c(x/z) = \dfrac{1}{1 + x^2/z^2)}$ is the normalized "form curve" and is given as the upper (dashed) curve of Figure 38.

Semi-infinite horizontal sheet or fault

This geometric form is very useful as an approximation to the gravity effect of a fault or step, Figure 39. At the fault a horizontal boundary between an upper layer with density d_1 and a lower layer with density d_2 is displaced vertically by the distance t. This is gravitationally equivalent to a semi-infinite layer with thickness t and density contrast $\sigma = d_2 - d_1$ terminated at the fault face.

Thin fault

For this case the material of the sheet of thickness t is considered as condensed onto a thin lamina or density sheet at its center at a depth z (Figure 39). The

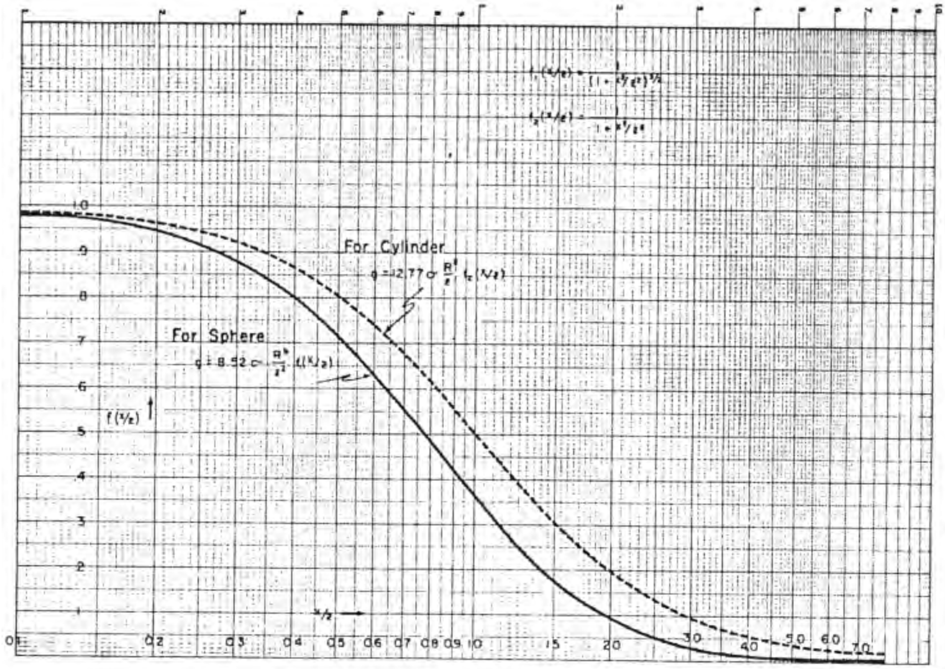


Fig. 38. Form curves for gravity from a sphere (point source) and horizontal cylinder (line source). These curves give the geometric part and are normalized to unity for maximum gravity over the center. The expressions for maximum gravity are for r and z in kilofeet. The horizontal scale is in the ratio x/z, i.e., distance to depth to center.

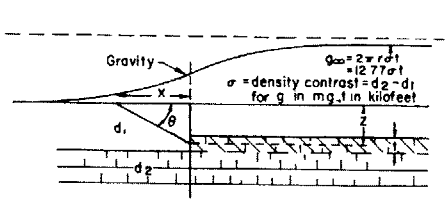


Fig. 39. Diagram and notation for gravity effect of a thin and thick fault.

gravity expression is (Nettleton, 1942)

$$g = 2\gamma\sigma t\theta = 2\gamma\sigma t\,(\tan^{-1} z/x).$$

Putting in numerical values for the gravitational constant and changing units to kilofeet,

$$g = 12.77\sigma t(1/2 + \frac{1}{\pi}\tan^{-1} x/z).$$

The approximation of condensing the thickness of the sheet onto a thin lamina is surprisingly close, even for bodies of considerable thickness. With a thickness of half the mean depth, the error at any point is less than 1 percent. For the bracket term, x is zero over the fault face, positive on the side over the up-faulted part and negative on the down-faulted side, so that the value is one-half over the trace of the fault, increases to unity at large distances to the right and decreases to zero at large distances to the left. The lower curve of Figure 40, shows the right half (values from 0.5 to unity). The left half (values 0.5 to zero) is antisymmetrical and is indicated by the scale numbers in parentheses.

Thick fault to the surface

The exact expression for a fault of finite thickness with the notation of Figure 41 (Nettleton, 1940, p. 112) is:

$$g_z = 2\gamma\sigma\left[x \ln\frac{r_2}{r_1} + \pi t - (t+d)\,\theta_2 + d\,\theta_1\right].$$

For the special case where the upper

limit of the fault block is at the surface:

$$d = 0,\ r_1 = x,\ r_2 = \sqrt{x^2 + t^2},$$

$$\theta_1 = 0, \quad t + d = t \text{ and}$$

$$\theta_2 = \tan^{-1} t/x = \pi/2 - \tan^{-1} x/t$$

$$g_z = 2\gamma\sigma\left[x \ln \frac{\sqrt{x^2 + t^2}}{x} + \pi t - t\left(\frac{\pi}{2} - \tan^{-1} x/t\right)\right] =$$

$$2\pi\gamma\sigma t\left[\frac{1}{\pi}\cdot\frac{x}{t}\ln\frac{\sqrt{1 + x^2/t^2}}{x/t} + \frac{1}{2} + \frac{1}{\pi}\tan^{-1} x/t\right]$$

$$= 12.77\sigma t\left[\frac{1}{\pi}\,\frac{x}{t}\ln\frac{\sqrt{1 + x^2/t^2}}{x/t} + \frac{1}{2} + \frac{1}{\pi}\tan^{-1} x/t\right]$$

This is of the same form as the expression for the thin lamina approximation except for the logarithm term in the bracket. This term goes to zero at $x = 0$ and at $x = \infty$.

If we take $t = 2z$ (Figure 40) we can plot the form curve (the expression in brackets above) with the same distance-to-depth ratios as for the thin lamina, as shown by the upper curve of Figure 40. We see that the maximum difference between the lamina and the thick fault to the surface is at a distance of $x/z = 0.5$ where it amounts to only about 4 percent of the total effect. Thus, in spite of the extremes of the approximation, the curves for the thin and thick faults are not greatly different; these extremes embrace the entire range of possible fault-like occurrences with vertical contact.

The total gravity effect for an infinite horizontal sheet (corresponding to the Bouguer correction) gives the order of magnitude of anomalies which is appli-

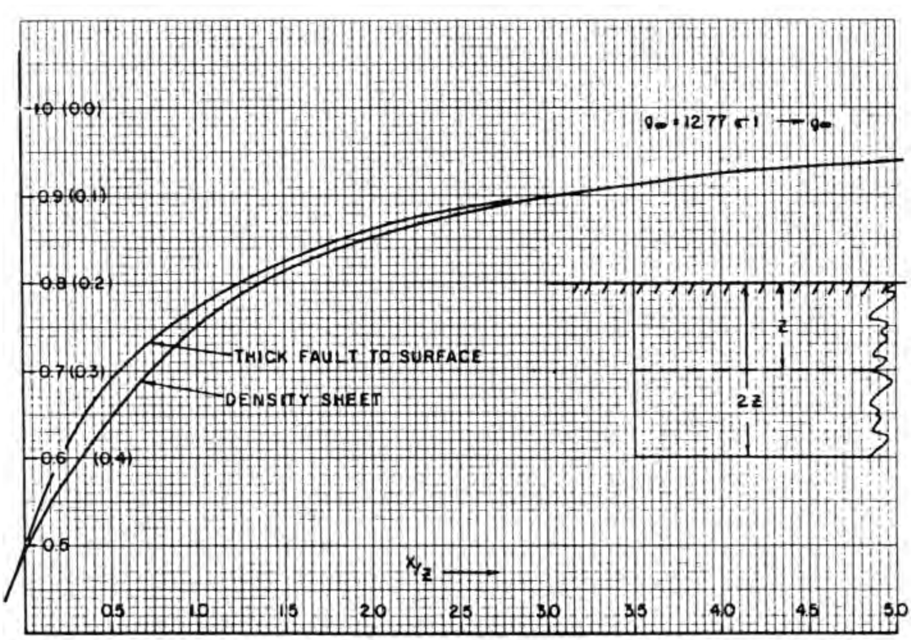


Fig. 40. Form curves for a fault or step. For the "density sheet," the mass is considered as compressed onto a thin lamina at the center, at depth z. For the "thick fault" the material is considered to extend uniformly from depth $2z$ to the surface; thus these two curves bracket the complete range which faults may have.

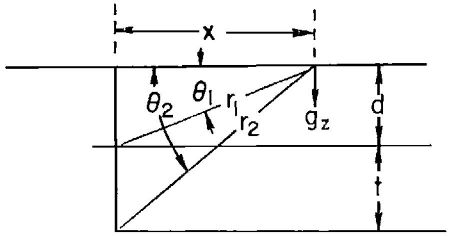


Fig. 41. Notation for fault of finite thickness.

cable to many situations. The total range of the gravity effect for $x = \infty$, i.e., the factor $g = 12.77\ \sigma t$ (for t in kilofeet) indicates the thickness necessary to account for a given gravity effect. This means that for unit density contrast it would take a thickness of 1000/12.77 = 78 ft to produce a gravity effect of one milligal. This expression applies to a sheet whose horizontal extent is infinite compared with depth or thickness. From the fault curves of Figure 40 it is evident that it takes a distance of 6 times the depth (from $-3z$ to $+3z$) to reach about 80 percent (from 0.1 to 0.9) of the theoretical total magnitude for an infinite width, and for this width it would require $\frac{78}{0.80} = 97$ ft of unit density contrast to have a gravity effect of one milligal. Thus, as a practical and easily remembered basis for estimating orders of magnitude to be expected from broad density anomalies we may use a figure of 100 ft of thickness of material with unit density contrast to have a gravity effect of one milligal (rather than the theoretical value of 78 ft for a sheet of infinite horizontal extent). Then, for a given gravity effect and density contrast, the required thickness, in feet, is $t = (100/\sigma)g$, where t is in feet, σ is the density contrast, and g the maximum gravity anomaly in milligals. As an example of application of such an estimate we may take the total gravity effect of the Los Angeles Basin from the gravity profile across the basin (Figure 33). With some allowance for a regional gravity decrease toward the right (north) end of the profile the gravity relief over the center of the basin is approximately 60 mgal. The density contrast is not known definitely, but the relatively recent Pliocene and Miocene clastic deposits would be expected to be of relatively low density with an average of perhaps 2.2 to 2.4, increasing with depth. The underlying Franciscan metamorphic basement should have a density of around 2.6 to 2.7. These values suggest that the density contrast between the sediments and the basement as a whole should be in the neighborhood of 0.3 to 0.4. If we use a figure of 0.35, we have a thickness of 100/0.35 = 300 ft per milligal so that the thickness of the sediments in the basin for the 60 mgal anomaly would be 60 x 300 = 18,000 ft. This is in quite reasonable agreement with the basement depth which might be inferred from the section (Figure 33).

APPLICATIONS OF FORMULAS FOR GRAVITATIONAL FORMS

The gravity expression for a sphere is, as mentioned above, applicable to anomalies which are roughly equi-dimensional in plan. As an example Figure 42 (Nettleton, 1962) shows a gravity map of a salt dome in the Gulf Coast (the Humble dome about 20 miles north of Houston). Figure 43 is a profile on line A-A′ of the map. A regional is removed to give the residual gravity curve shown and the analysis is carried out in this residual. By application of the simple parameters of the normalized curve for the sphere we can determine the general features of the dome. We designate the "half-amplitude" width of the residual anomaly ($x_{1/2}$) as the width of the anomaly (from its center) where the amplitude is half the maxi-

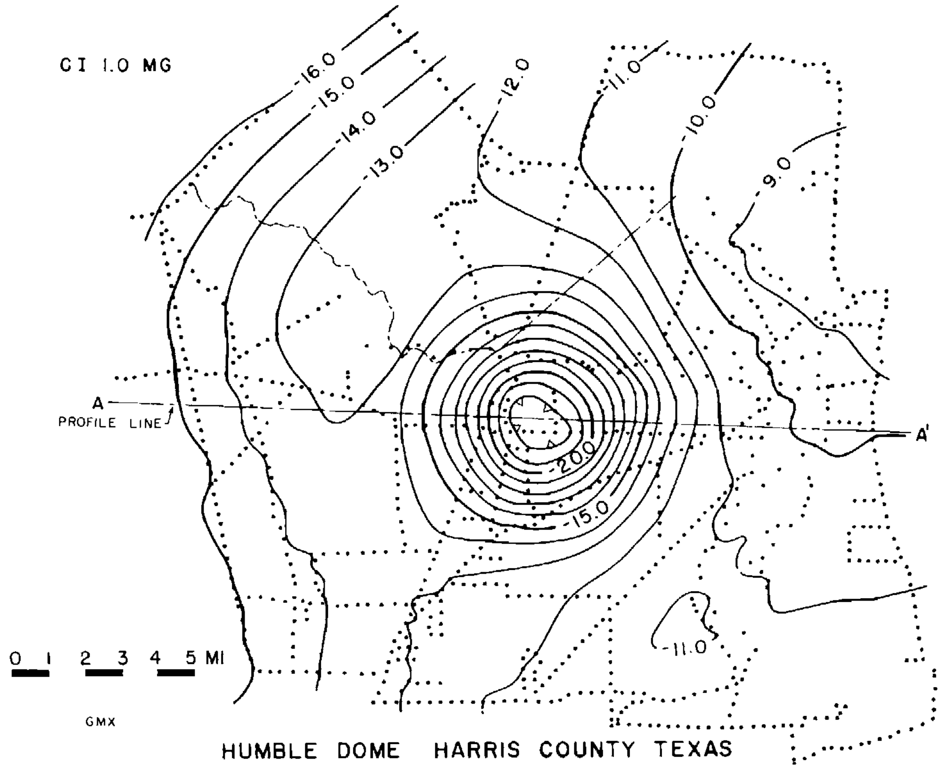


Fig. 42. Bouguer gravity map, Humble salt dome, Harris County, Texas. Contour interval 1.0 mg. Control shown by dotted lines, each dot representing a gravity station location (Nettleton, 1962, p. 1839). *Courtesy AAPG.*

mum (i.e., $g/g_0 = ½$). From the normalized curve for a sphere, this width is $0.766z$, and the depth to the center of the sphere is $x_{½}/0.766 = 1.305x_{½}$. If we measure the total width of the curve at its half-amplitude, this horizontal distance is the double half-width or $(2x_{½})$ and the depth is $1.305x_{½} = 0.652\ (2x_{½})$. The residual gravity curve of Figure 43 gives a double half-width of about 25,000 ft from which the indicated depth to the center of the equivalent sphere is 25,000 x 0.652 = 16,300 = 16.3 kft. The total amplitude of the anomaly is about 13.9 mgal from which we can estimate the radius of the equivalent sphere. From page 49, the total amplitude factor is $g_0 = 8.52\sigma R^3/z^2$ and $R^3 = g_0 z^2/8.52\sigma$. If we assume an average density contrast of 0.3, put in values of $g_0 = 13.9$ mgal and $z = 16.3$ kft, we have

$$R^3 = \frac{13.9(16.3)^2}{8.52 \times 0.3} = \frac{13.9 \times 265}{2.556} = 1445\ \text{kft}$$

and $R = \sqrt[3]{1445} = 11.3$ kft $= 11,300$ ft. The depth of the top of the dome, is $T = z - R = 16,300 - 11,300 = 5000$ ft. The gravity anomaly from the equivalent sphere can be calculated very simply as shown by the following table, with all of the numerical operations being done quickly with a slide rule.

For $z = 16.3$ kft, $g_0 = 13.9$ mgal and reading $f\ (x/z)$ from curve 1, Figure 38

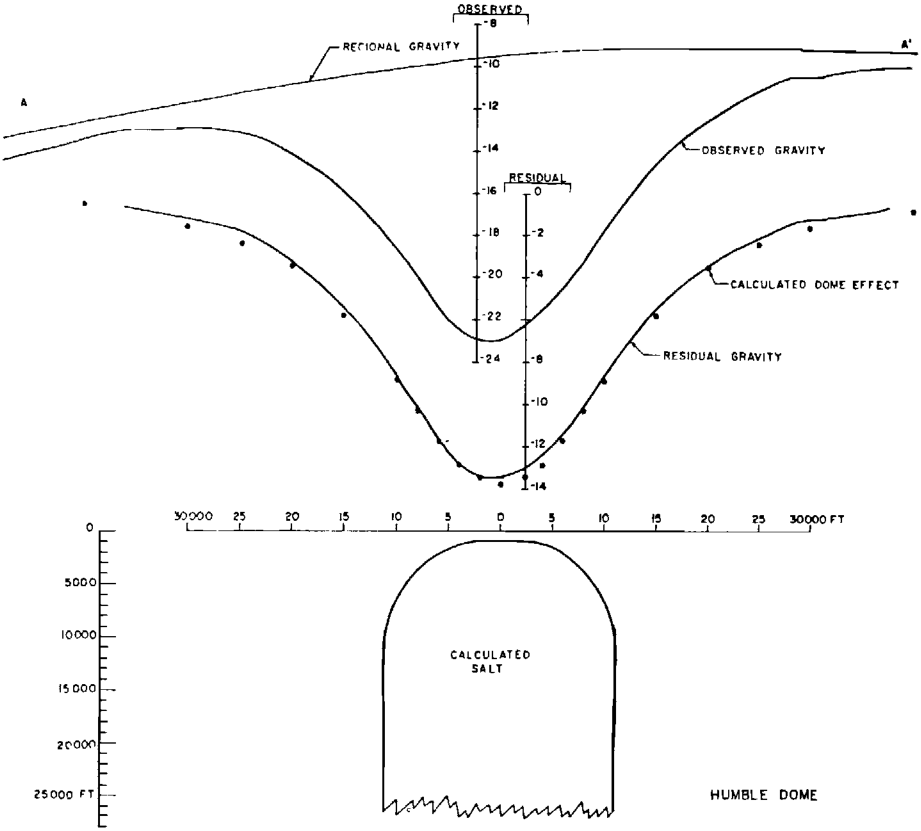


Fig. 43. Observed, regional, and residual gravity on line AA′ of Figure 40, with cross-section and calculated gravity effect shown by dots along residual gravity curve. (Nettleton, 1962, p. 1836). *Courtesy AAPG.*

we have:

x	x/z	$f(x/z)$	g
0	0	1.000	13.90
2	0.123	0.975	13.70
4	0.246	0.912	12.70
6	0.369	0.822	11.42
8	0.492	0.718	9.98
10	0.614	0.615	8.55
15	0.920	0.410	5.70
20	1.228	0.256	3.56
25	1.536	0.162	2.25
30	1.840	0.110	1.53
40	2.429	0.050	0.69

to give the calculated points shown along the residual curve.

A calculation such as that shown by the table can be carried out in five to ten minutes by anyone reasonably skilled in the use of the slide rule. The calculated values of Figure 43 are as close to the residual gravity curve as the reliability with which that curve is determined. Without more accurate information on densities and on the depth of the dome at some point, such as a drilling contact or seismograph data the simple calculation with the spherical approximation gives: (1) the position quite accurately, (2) a fair approximation of the depth, and (3) a rough measure of the diameter and therefore of the deeper flanks of the dome. For relatively shallow domes such as this, where the top of the dome is known to be about 1000 ft, the depth to the calculated top is too great because the density contrasts of the shallow sediments with respect to the salt are less than the average contrast used for the calculation (see p. 62). The point of this example is to illustrate that very simple calculations may be adequate as control on geological assumptions and that more elaborate calculations, such as may be made with dot charts or computers, are justified only when additional information is available. For more complex gravity anomalies such as those from shallow domes with cap rock, where the density and geological situation are much more complex, the gravity picture may have components which may be separated from the total anomaly and used for other interpretive calculations (see Figure 49).

A corollary of the use of relatively simple calculations is that the degree of detail of the gravity data which can be useful depends very much on the objective of the survey. The gravity contour map of Figure 42 is based on stations approximately one-half mile apart on the principal roads and trails in the area and there are some wide gaps between lines. The control is quite adequate for locating and determining the general nature of this dome. On the other hand, to determine the details of a shallow dome with an associated shallow cap rock would require much closer detail over the central part of the dome to map the relatively sharp changes there and to separate the shallow dome anomaly from the total gravity field. As a crude approximation, a gravity station spacing of less than about one-half the minimum depth to the feature or density contrast to be mapped is about as close as can contribute to determination of significant geological details. Closer spacing may be desirable when shallow inhomogeneities produce sharper effects which must be filtered out to accurately determine the anomaly to be analyzed.

METHODS FOR MORE DETAILED CALCULATIONS

The simple formula for spheres, cylinders, and faults can serve very well to give the approximate magnitudes of calculated gravity effects for comparison with the observed values and for quantitative checks of geological possibilities to account for observed gravity anomalies. There are, however, many interpretation problems in which more detailed calculations are useful. Such problems arise, for instance, when it is desired to determine the gravity effect from a geological situation which is partially defined by known stratigraphy and drilling and where the gravity expression is complex but well defined or when general effects over large areas are to be interpreted.

There are a number of ways in which more detailed calculations can be carried out. In general they may be divided into two-dimensional and three-dimensional problems. Two-dimensional calculations are applicable to situations where the contours are roughly parallel. Then a profile which can be correlated with a subsurface cross section is representative of conditions extending for relatively large distances perpendicular to the line of section. Three-dimensional methods are required when the anomalies are irregular or do not have a strong elongation in one direction. Greatly simplified methods can be used where anomalies are roughly equidimensional and can be approximated by circles. In general the three-dimensional calculations are an order of magnitude more complicated or expensive than those for two-dimensions.

Two-dimensional calculation systems

These methods can be applied to a problem where conditions perpendicular to the line or profile are essentially constant for distances on each side of the profile line of about two or three times the depth of the section calculated. While details differ, in several useful systems the mass to be calculated is considered as made up of horizontal line elements (or cylinders) of infinite length in the direction perpendicular to the section.

In one form, a chart is made up in which units of area are proportioned so that they represent elements of mass having equal effects at the origin of the chart. Thus these units increase in size with their distance from the origin. If these elements of area are made small enough each such area can be represented by a single dot at its center. Then, to determine the gravity effect it is necessary only to draw an outline of the body being considered and count the number of dots within the area. The dot count at each point is multiplied by a factor which depends on the design of the chart, the scale of the cross section and the density contrast. By repeating the operation for a number of points along the profile, the variation of gravity can be determined to whatever degree of detail is required to give the calculated gravity effect. Such a chart is illustrated by Figure 44 (Nettleton, 1940).

A number of other chart designs have been published in the geophysical literature, some of which are more convenient for certain conditions or shapes of the bodies for which gravity effects are desired. All require some sort of measurement or count within the area or around the periphery of the cross section of the body being computed and summation of effects for each point of calculation. While the operations are basically simple and can be carried out readily by untrained personnel they still become quite time consuming and tedious. The chart methods are applicable when a simple section is to be computed at a moderate number of points. These methods can be considered as intermediate between the very simple calculations made with horizontal cylinder or fault formulas discussed previously (p. 49-52) and those made with electronic computers. Where the section to be computed has several layers with different densities, it must be computed at a large number of points, and where sections must be successively modified on the basis of calculations to obtain a reasonably close fit with the anomaly being interpreted, it is almost necessary to use the computer to make the calculations feasible.

Three-dimensional calculations

When the body for which the gravity effect to be calculated is not relatively uniform in one direction and it is too complex to be approximated by simple geometric bodies, it is necessary to carry out three-dimensional calculations.

One relatively simple approach is to make "end corrections" for a two-dimensional or dot chart type calculation. This recognizes the deficiency in the effect calculated for the ideal two-dimensional condition when the length perpendicular to line of section is limited, as illustrated in Figure 45. The end correction is the ratio of the gravity effect for a two-dimensional body of a limited length, Y in Figure 45, to the effect which would be calculated for infinite length and is given by the relation

$$g/g_\infty = \left[\frac{1}{1 + (\rho/Y)^2}\right]^{\frac{1}{2}} = F(Y/\rho),$$

where Y is the length of the element perpendicular to the plane of the section and ρ is the distance from the center of the element to the point of calculation. This function is shown by the curve of Figure 45.

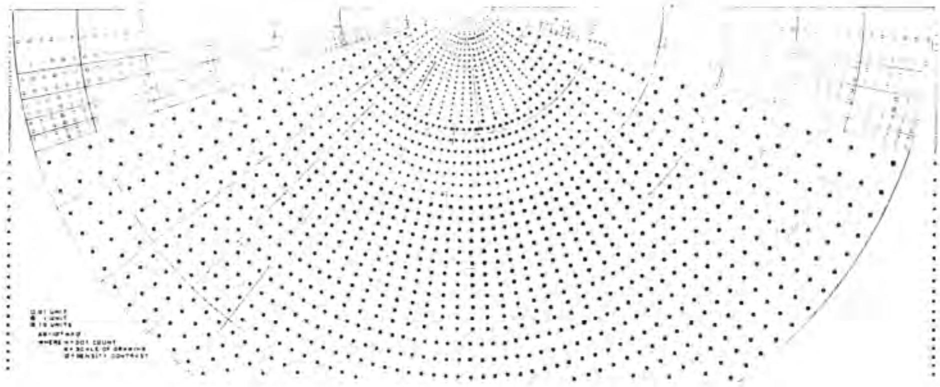


Fig. 44. Dot chart for calculation of gravity effects of two-dimensional mass forms (Nettleton, 1940, p. 116). *Courtesy McGraw Hill.*

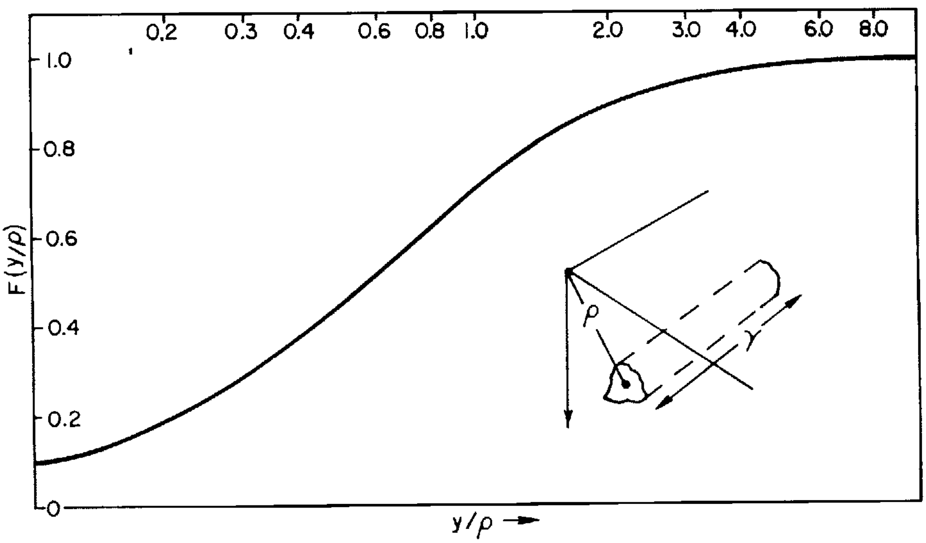


Fig. 45. End correction, $F(y/\rho)$, for reducing calculations made with two-dimensional methods to their value for finite length of body. Note that if length of body (on each side of plane of calculation) is twice the distance to the point of calculation, the effect is about 90 percent of that for infinite length.

Solid angles

It can be shown that for a horizontal lamina (Figure 46), the vertical gravity at a point O is directly proportional to the solid angle ω subtended by the lamina from that point. Thus if the mass of an irregular body of thickness t and density contrast σ is considered as condensed onto a thin sheet or lamina at its center, as indicated by the dashed line of Figure 46, its gravity effect is given by $g = \gamma \omega \sigma t$. If we put in the values for the gravitational constant and express t in units of kilofeet and g in milligals we

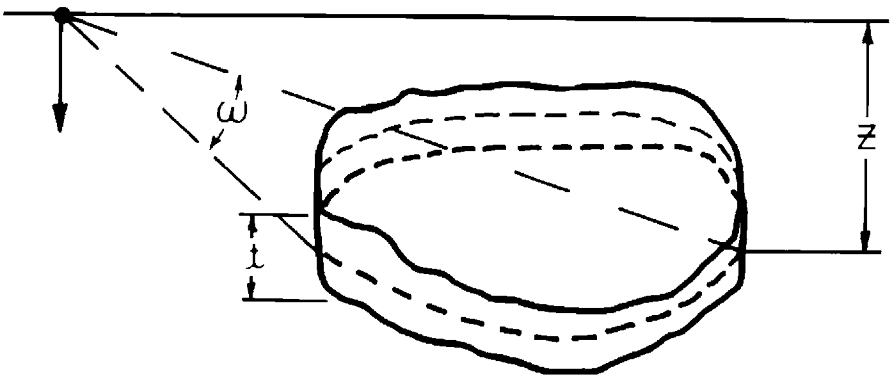


Fig. 46. Diagram and nomenclature for calculation of gravity effect of a finite horizontal tabular body by the equivalent solid angle.

have $g = 2.03\omega\sigma t$. The approximation of condensing the body of thickness t onto a density sheet is quite good. Even for a thickness equal to half the depth, the maximum error (which occurs over the center of the body) is only about 6 percent.

A simple way of using the solid angle approximations is to prepare horizontal polar dot charts (Figure 47) for which each dot represents an area with the same solid angle. Then, by drawing the outline of the body and counting the number of dots within the outline, with the center of the chart at the point of calculation, the gravity effect at that point can be determined. The chart has to be made for a certain depth (which is shown on Figure 47) and the outline of the body being calculated must be drawn at such a scale that its depth corresponds to the "unit depth" for the chart. To accommodate a considerable range of depths it is desirable to reproduce the chart photographically at a number of different scales to minimize the magnitude of the changes of scale required to match the unit depth of the chart.

It is possible to make a relatively simple chart which will give the solid angles of circular lamina (Figure 48, Nettleton, 1954). If the body has relatively great vertical dimensions and is roughly circular in section (such as a salt dome) it can be approximated by circles representing contours of various depths. The depth intervals can be relatively large (up to about half the depth to their center) and different density contrasts can be used for the different levels. The gravity effects for each such layer, with its appropriate depth, radius and density contrast, are computed for each point of a profile and added. The chart for circles can be a very effective system of calculation when the number of layers is not too large (say five or less), and the number of points of calculation is limited (say 10 or less).

APPLICATIONS OF DIGITAL COMPUTERS

The advent of high speed digital computers and their application to geophysics has made extensive and complex gravity calculations quite feasible. The equipment and the "soft ware" or programs required are considerable but are becoming available in many of the larger oil company

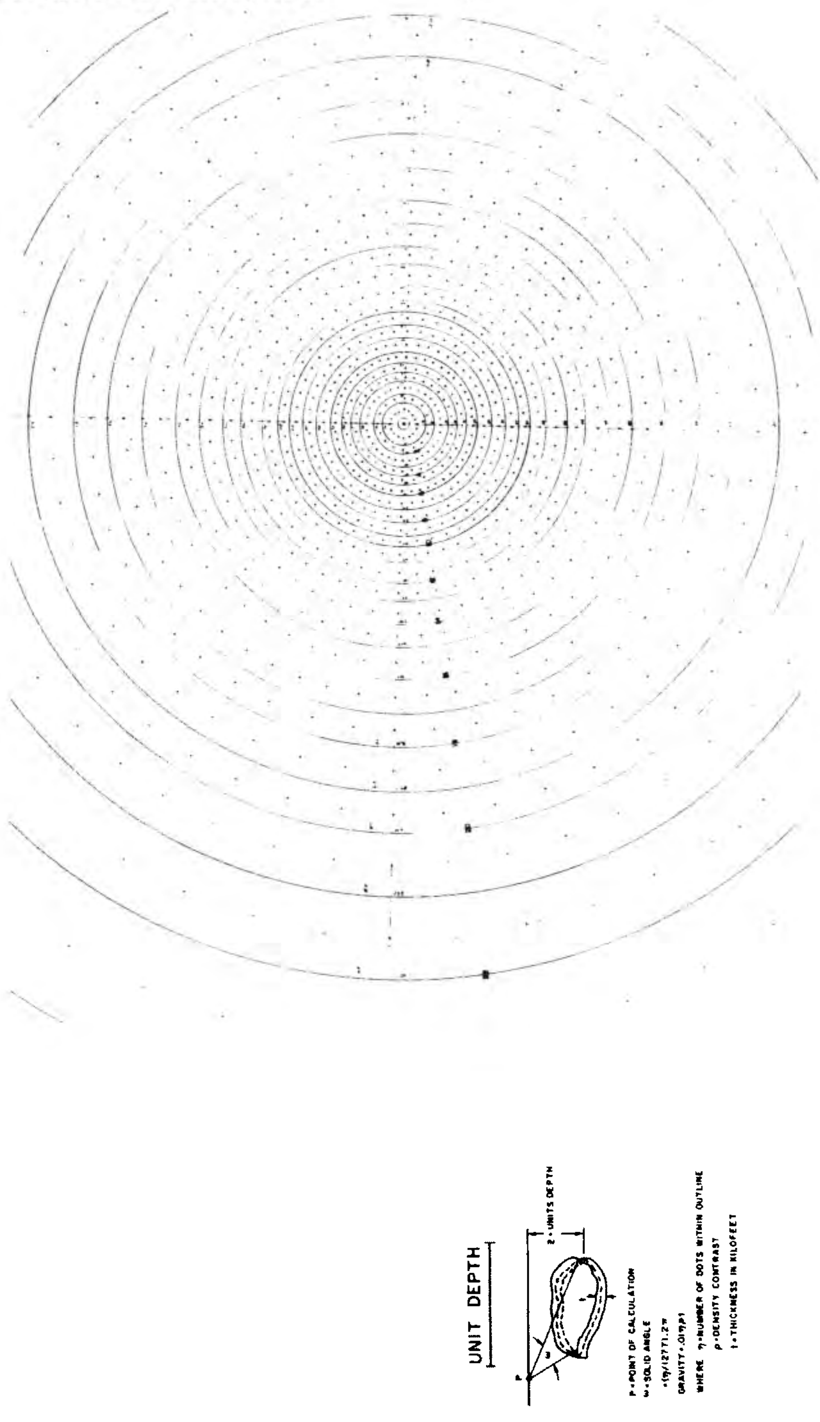


Fig. 47. Polar chart for calculating gravity effects by solid angles.

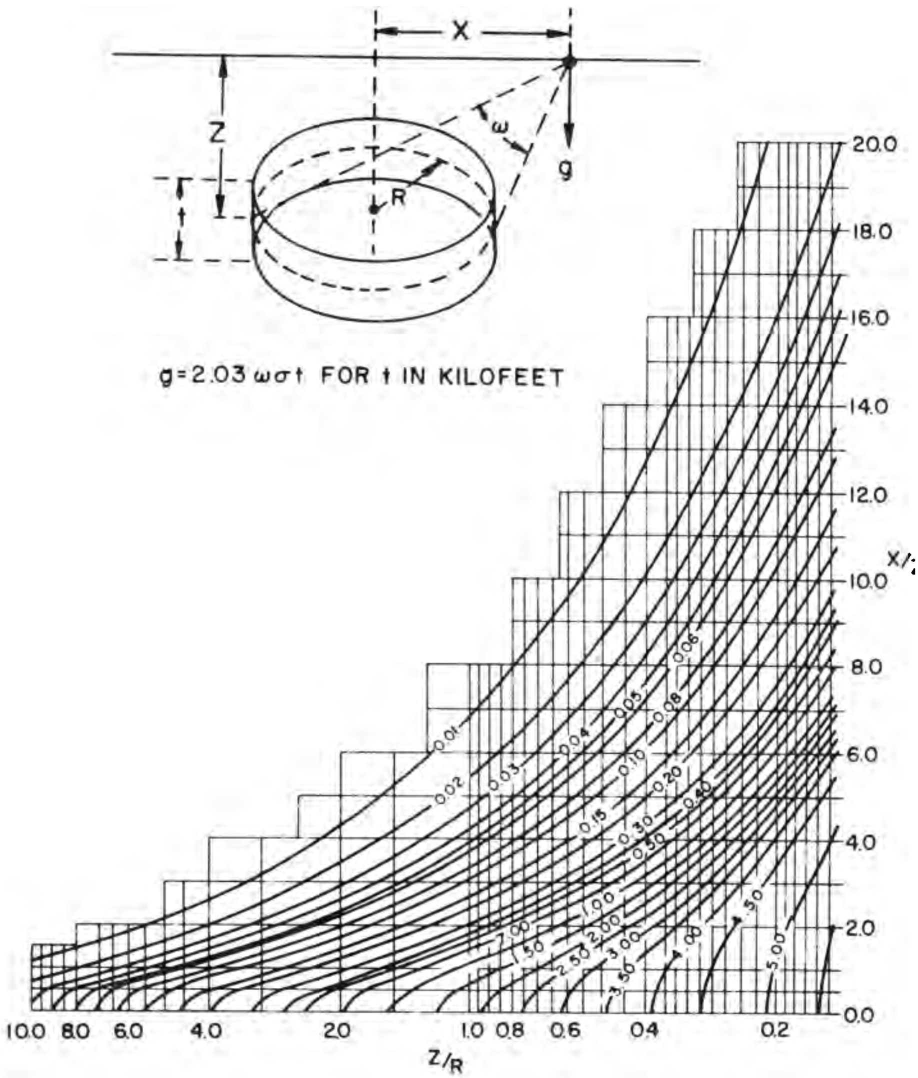


Fig. 48. Chart for calculation of solid angles of horizontal circular discs to determine their gravity effects (Nettleton, 1954). *Courtesy SEG.*

offices or research centers and universities. This means that the various approximate systems outlined above are useful primarily when relatively few calculations are to be made or where computers are not available. For any very extensive calculations for gravity (or magnetic) interpretations, the use of a computer is almost essential and permits much more detailed, complex, and voluminous calculations than are feasible without it. On the other hand the simple formulas, charts, and graphical methods can be very useful for quick approximate evaluation of interpretation questions and also may be helpful as a guide to setting the necessary boundary or density conditions which are involved in more elaborate applications or computer analysis. Furthermore, unless quite detailed data are available, together with some control on the nature, depth, and density contrasts responsible for the anomaly under consideration, the simple calculations may give all the detail which is geologically significant. There is a common tendency, because of the facility with which it can be done, to carry out computer operations to a degree of detail or closeness of fit which is not significant.

There is no geological usefulness in matching calculated to observed curves to a degree which is beyond that justified by the precision or spacing of the original data or by the certainity with which the necessary boundaries and density contrasts are known. It must be kept in mind that a perfect fit does not make a necessarily correct solution, and such a solution may be misleading if its limitations are not appreciated.

ANALYSIS OF SALT DOME GRAVITY ANOMALIES

Since its beginning in the Gulf Coast by torsion balance exploration, the gravity method has had a very intensive application to the discovery and delineation of salt domes. It is still being used very extensively for this purpose, particularly for off-shore domes.

A typical Gulf Coast salt dome consists of an intrusion of rock salt (halite) through the sediments starting at a salt layer of unknown but very large depth (probably of the order of 30,000 to 40,000 ft) and extending upward to the surface or to any intermediate depth. In plan, the salt column is commonly circular to elliptical with dimensions of the order of 1 to 5 miles. On shallow domes the salt is often covered by a cap rock, of which the lower part is anhydrite and the upper, limestone. Salt domes are economically important primarily because of the associated accumulation of oil in the deformed sediments around or over the dome or for sulphur in the cap rock or for the salt itself.

Densities and density contrasts

The density situation around salt domes is complex (Figure 49, Nettleton, 1962). The sediments increase in density with depth from values as low as 1.6 near the surface to probably 2.5 to 2.6 at great depth. The salt has a uniform density of approximately 2.20; thus the density of the salt is greater than that of the sediments at shallow depth and is less than that of the sediments below the "cross over" depth where the increasing density of the sediments is equal to that of the salt. In the Gulf Coast this "cross over" depth varies from around 2500 to as much as 5000 ft depending on the geographic location and the rate of deposition of the sediments. The cap rock materials have relatively high densities, approximately 2.9 for the anhydrite and 2.5 to 2.6 for the limestone.

Gravity anomalies

These complex density relations produce complex gravity anomalies, as illustrated in Figure 49. The large volume of salt with negative density contrast below the "cross over" depth produces a negative anomaly, and likewise the positive density contrast of any salt above the "cross over" depth will produce a positive anomaly. The strong positive density contrast of the cap rock materials also has a positive effect. Thus the total observed gravity effect is quite commonly a more or less circular minimum (Figure 42) which may have a generally circular maximum in its central part as shown by the residual gravity profile of Figure 49. The magnitude of the negative and positive anomaly components varies

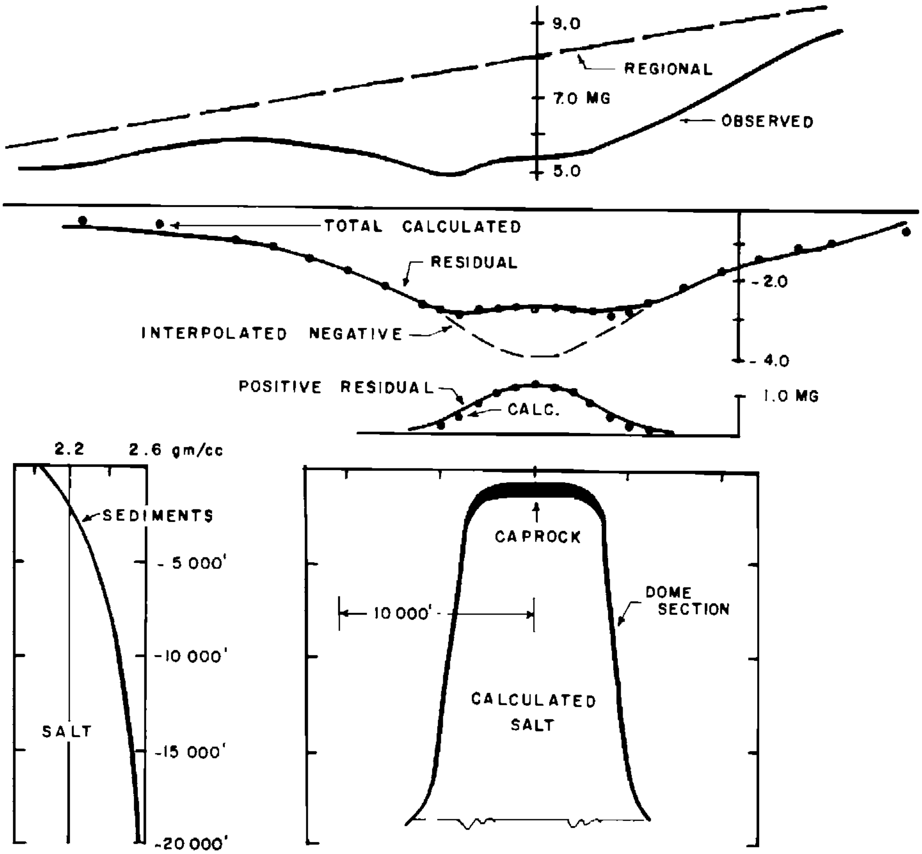


Fig. 49. Salt dome section, gravity anomaly and anomaly separation. (Nettleton, 1962). *Courtesy AAPG.*

widely. The negative anomalies may range from less than 1 mgal for a small dome with a salt column less than 1 mile in diameter to as much as 15 mgal for very large domes and even more for a few exceptional domes. The positive effect will range from none at all for domes with all their salt below the "cross over" depth and with no cap rock (as in Figure 43), to around 2 or 3 mgal for very shallow domes with no cap rock and thus with all this positive effect due to salt, to as much as about 6 mgal for domes with very thick (±1000 ft) cap rock.

Cap rock and salt effects

The gravity anomaly can be separated into positive and negative components by appropriate control calculations so that an "interpolated negative" effect, as indicated in Figure 49, can be determined; this is the gravity effect attributed to salt below the "cross over" depth. By subtraction of this interpolated negative from the total residual anomaly curve the "positive residual" can be determined as shown on the profile, in this case with a relief of a little over 1.0 mgal.

The resolution of the positive residual

into components due to salt above the "cross over" depth and cap rock cannot be made without further information or assumption. For instance, in the illustration in the figure, the 1 + mgal positive anomaly could be accounted for by salt coming to within perhaps 1000 ft of the surface. The same anomaly could be accounted for by salt coming to within 1500 ft of the surface and with about 200 ft of cap rock above the salt. Either assumption could account for the gravity expression quite completely. On the other hand if the depth to the top of the dome is determined by drilling or by reflection or refraction seismograph data, the gravity calculation can be carried out with a great deal more assurance of the separation of cap rock and salt effects and thus make the determination of the presence and probable thickness of the cap rock much more definite.

An analysis of this kind can be carried out quite readily, particularly if the gravity anomaly and dome cross section are nearly circular, with the simple calculation systems, especially the application of solid angles due to circles. If the gravity anomaly is so elliptical that the circular approximation is not valid, the calculations become much more complicated and a reasonably adequate analysis is greatly expedited by the use of a computer and appropriate programs to calculate more complex models. Also, iterative computer programs have been developed in recent years which can automatically build up a model to account for a given gravity expression (Cordell and Henderson, 1968). The model is controlled by certain parameters such as depth to the bottom of the salt, assummed or known points of control on the salt and, of course, the density-depth relations.

With the application of computerized procedures now available it is possible to carry out a systematic analysis not only on a single dome, the gravity effect of which has been isolated from a complex field, including the effects of several surrounding domes, but also to use the computer techniques to help model a field of domes with overlapping effects and thereby greatly facilitate the determination of an anomaly attributable to a single dome in a complex field.

CHAPTER VII

GRAVITY, ISOSTASY, AND THE EARTH'S CRUST

Since the very early history of gravity measurements, the relation of gravity to the shape of the earth and the nature of the earth's crust has been the subject of extensive studies. The variation of gravity with latitude was first noted when pendulum clocks were moved over great distances and their rates found to change. This, of course, results from the variation of gravity with latitude which comes about because of the combination of the effects due to the outward acceleration of the rotating earth and the consequent bulging at the equator and flattening at the poles. The shape of the earth is approximately that of a rotating liquid with a balance between its gravitational attraction on itself, which would make it a perfect sphere, and the centrifugal force due to the rotation which makes the sphere oblate. These relations have been defined by many observations of gravity over the earth and geodetic analysis to give several formulas for the variation of gravity from the equator to the pole. The 1930 "international formula," now generally accepted is

$$\gamma = 978.049(1 + 0.0052884\sin^2\varphi - 0.0000059\sin^2 2\varphi).$$

This gives γ, the gravity at sea level, as a function of the latitude φ. Detailed tables of values of gravity from the equator to the poles are available.[1]

THE DIFFERENT GRAVITY ANOMALIES

In the geodetic studies of gravity and the shape of the earth, and of the effects of topography, it is customary to compare the observed value of gravity at a given latitude and elevation with the value calculated. The "observed" gravity value is determined by relative gravity measurements, made by pendulums or gravity meters, referred to certain primary gravity base stations where absolute measurements have been made by special instruments such as the Kater reversible pendulum or, more recently, by falling body techniques. The long established datum for these measurements was the absolute gravity value at Potsdam, now in East Berlin. Later measurements at other stations which have been tied to Potsdam now indicate that the adopted value there (978.2740 gals) is about 11.8 mgal too high. Revision of this value would change the first term (978.049) of the gravity formula.

The difference between the observed

[1] Nettleton (1940), p. 139-143, gives listings to 10^{-6} gals at intervals of 10 min of latitude. Publication G53 of the U.S. Coast and Geodetic Survey, Washington, 1942 "Theoretical Gravity at Sealevel for each Minute of Latitude by the International Formula" gives values to 10^{-4} gals for each minute of latitude.

and calculated value is the "anomaly"; it is positive if the measured value is higher than that calculated and negative if lower.

The "free air" anomaly

The vertical gradient of gravity, as mentioned in the section of data reduction, p. 20, is approximately the simple effect of a station at higher elevation being farther away from the center of the earth. To a close approximation the effect is linear and independent of latitude so that the vertical gradient is simply $-Kh$, where K, the free air coefficient, is $+ 0.3086$ mgal/m or $+ 0.09406$ mgal/ft. The theoretical "free air" gravity is $g_f = \gamma_0 - Kh$. It is called "free air" because the theoretical anomaly is calculated as if the gravity measurement were made at the elevation of the station but without taking into account the attraction of material between that elevation and sea level; that is as if the gravity measuring instrument were suspended free in the air.

The "free air" anomaly is the difference between the observed value g_0 and the calculated "free air" gravity.

The Bouguer anomaly

Consider gravity observations made at points 1 through 3 of Figure 50. Station 1 is at an elevation h above sea level, station 2 is at sea level, and station 3 is on the ocean surface where the water depth is d (where gravity is measured by a pendulum in a submarine or by a shipborne gravity meter). The measurement at point 1 is subject to the attraction of the slab of earth material with a thickness h between it and sea level (which is the same as the Bouguer effect considered earlier). A station at point 3 has a deficiency in gravity because the material between the ocean bottom and the surface is sea water with density 1.03 rather than the material of the earth's crust. The Bouguer effect is a calculation to take these attractions into account.

The simple Bouguer effect is calculated as if the material under the station were horizontal (except when calculations include effects from great horizontal distances and the curvature of the earth's surface is taken into account). For a simple horizontal slab of material the attraction is: $B = 2\pi\gamma\sigma h$ where γ is the gravitational constant, σ is the density of rock and h the elevation of the station or the thickness of the slab of material between the station and sea level. When we put in the numerical values for 2π and for the gravitational constant (6.660×10^{-8}), the effect becomes $0.04185\sigma h$ (mgal/m) or $0.01276\sigma h$ (mgal/ft). The calculated theoretical gravity at the station, including free air and Bouguer effects, is $g_B = \gamma_0 - 0.09406h + 0.01276\sigma h$ (for h in ft). For station 3 the Bouguer effect is that calculated as if the water were replaced by rock of density σ and becomes (for a measurement made at the sea surface) $(g_B)_w = [0.01276(\sigma - 1.03)]\,d$. The Bouguer anomaly is the difference between the measured value at the point of observation and the theoretical value calculated for that elevation and the appropriate density of the earth's materials. An average value for the density of the crustal material within the range of the visible topography and which has been used world wide for calculation of general Bouguer effects is 2.67 gm/cm^3.

THE CONCEPT OF ISOSTASY

The Bouguer effect on gravity measurements is a very obvious one and it is necessary to make such a correction if the effects of the visible topography are to be taken into consideration. However, it was found very early in the history of gravity measurements, that Bouguer

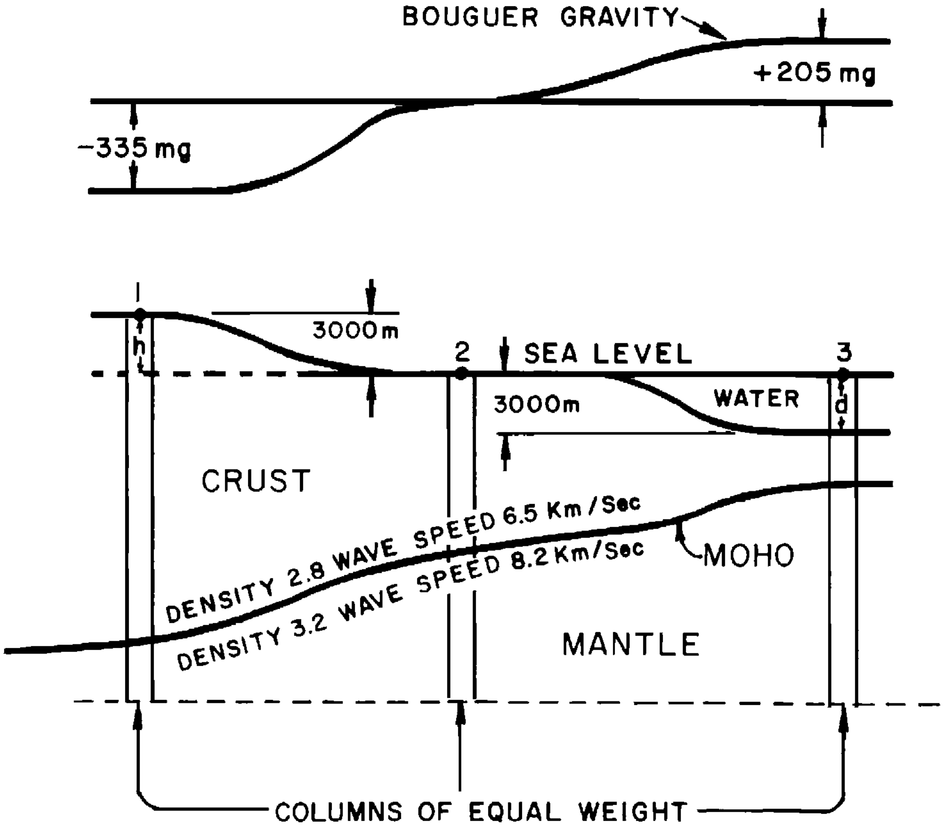


Fig. 50. Idealized diagram of crustal changes and gravity effects from deep ocean to continental areas. The magntitudes shown for gravity anomalies are for heights and depths of 3000 m, calculated for rock density 2.67, water density 1.03, and assuming ideal isostatic compensation.

anomalies have a strong correlation with topography over broad areas. In general, areas with high topography have negative Bouguer anomalies (Figure 51, Woollard, 1959); also (but not included in the figure) those with low topography (i.e., the ocean basins) usually have positive Bouguer anomalies. For broad areas of large relief these anomalies can be several hundred milligals.

The existence of such anomalies points to some deficiency in understanding the earth's crust. In the first place, it is evident from general knowledge of the earth's crust and its physical properties that the crustal material is not strong enough to support the weight of a large mountainous area. This weight corresponds to the magnitude of the Bouguer attraction averaged over the area. Therefore there must be some other property of the earth's crust which supports the topography. In the early history of such measurements two competing theories were put forth to provide a principle which would hold up the mountains:

1) The Pratt theory assumes that the density of the earth's material below sea level under areas of high topography is lower than average and

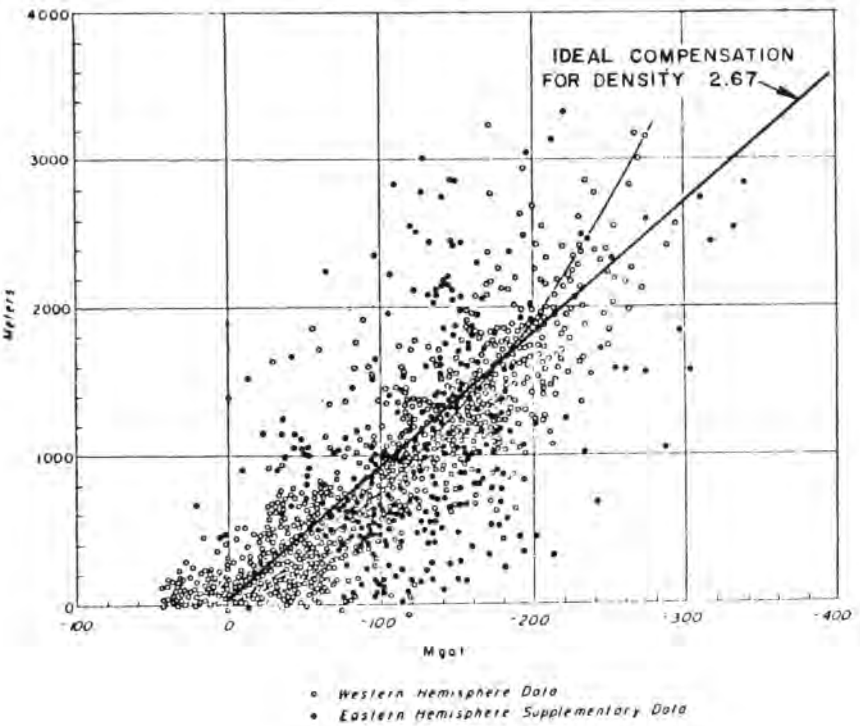


Fig. 51. Relation of Bouguer gravity anomaly to surface elevation (land stations). Straight line (added to original diagram) shows the Bouguer effect for density 2.67. If all locations were completely and ideally isostatically compensated for 2.67, they would fall on this line (modified from Woollard, 1959, p. 1523). *Courtesy AGU.*

that this low density material extends to a fixed "depth of compensation."

2) The Airy theory assumes that there is a discontinuity in density at some depth within the earth's crust and that this discontinuity is at variable depths, being deeper under high topography and shallower under low topography. Since the difference in density would be much smaller than the actual density of the topography, the vertical relief of such a surface would have to be several times greater than the relief of the topography and thus the mountains would have deep roots in a manner similar to that in which an iceberg projects down into the water to a depth several times its visible height. Thus the Airy proposal became known as the "roots of mountains" type of compensation. The system or principle by which topography is compensated by variations within the earth's crust was called "isostasy" by C. E. Dutton in 1889.

Calculations can be made on the basis of either principle to determine the magnitude of the required deficiency in density under areas of high topography and systematic "isostatic corrections" can be carried out and corresponding isostatic anomaly maps can be produced. In general it is found that these maps show very much smaller anomalies than the Bouguer maps. The foregoing statements have been made with respect to elevations of the topography and negative Bouguer anomalies but exactly the same considerations apply to depressions (i.e., ocean basins) and corresponding positive anomalies. The difference between the two is that in one case the crustal surface is bounded by air and in the other by sea water so that the effective density contrast in ocean areas is reduced by the density of sea water (usually taken as 1.03).

THE "MOHO"

The above general relations were the basis of much of the geodetic analysis of world-wide gravity surveys through approximately the first quarter of the present century. Isostatic compensation was calculated, usually on the basis of the Pratt theory because it is easier to handle analytically, for early gravity measurements over the continental United States, other continental areas and gravity measurements at sea made in submarines.

In more recent years the Mohorovičić discontinuity has become an important factor in considerations of isostasy. This discontinuity (commonly abbreviated as "Moho") was first recognized as a level at which a sharp increase in seismic velocity occurs, from an average of approximately 6.5 km/sec above to a little over 8.2 km/sec below the boundary. From general relations of velocity and density and also from considerations of the gravity requirements for the variation of density with depth within the earth's crust, there are strong indications that this surface is also one where a sharp density contrast occurs, with probable density values of roughly 2.8 above to 3.2 below the Moho (Figure 50).[2] The difference in density at the Moho serves as a natural surface to be the upper boundary of the high density material proposed by the Airy "roots of mountains" system of isostasy. The material of the upper mantle must be viscous, corresponding to Airy's asthenosphere to permit adjustment to changes in composition or topography of the crust. The basic difference between the Airy-Moho "roots" on the one hand and the constant "depth of compensation" of the Pratt theory on the other is that in the former case the contact between crust and mantle is at variable depth and in the latter it is constant.

From these general considerations we have the following expectations:

1) Under areas of high topography there should be strong negative Bouguer anomalies and the Moho should be relatively deep.
2) Under areas of low topography or coastal areas the Bouguer anomalies should average near zero and the Moho should be at an intermediate depth.
3) Over deep ocean areas the Bouguer anomalies should be strongly positive and the Moho should be relatively shallow.

This means that in all three types of areas the weight of a column of material from the surface to some uniform depth below the Moho would be the same (Figure 50). In mountain areas the greater thickness of low density material is compensated by a smaller thickness of high density material below the Moho. In ocean areas the smaller thickness of low density crustal material is compensated by a greater

[2] These values are, of course, inferred from indirect evidence. No drilling or core from the Moho is available. It was because of an intense interest in the nature and properties of the mantle (i.e., the part of the crust below the Moho) that the "Moho project" to drill a hole into the mantle was conceived and started but never completed.

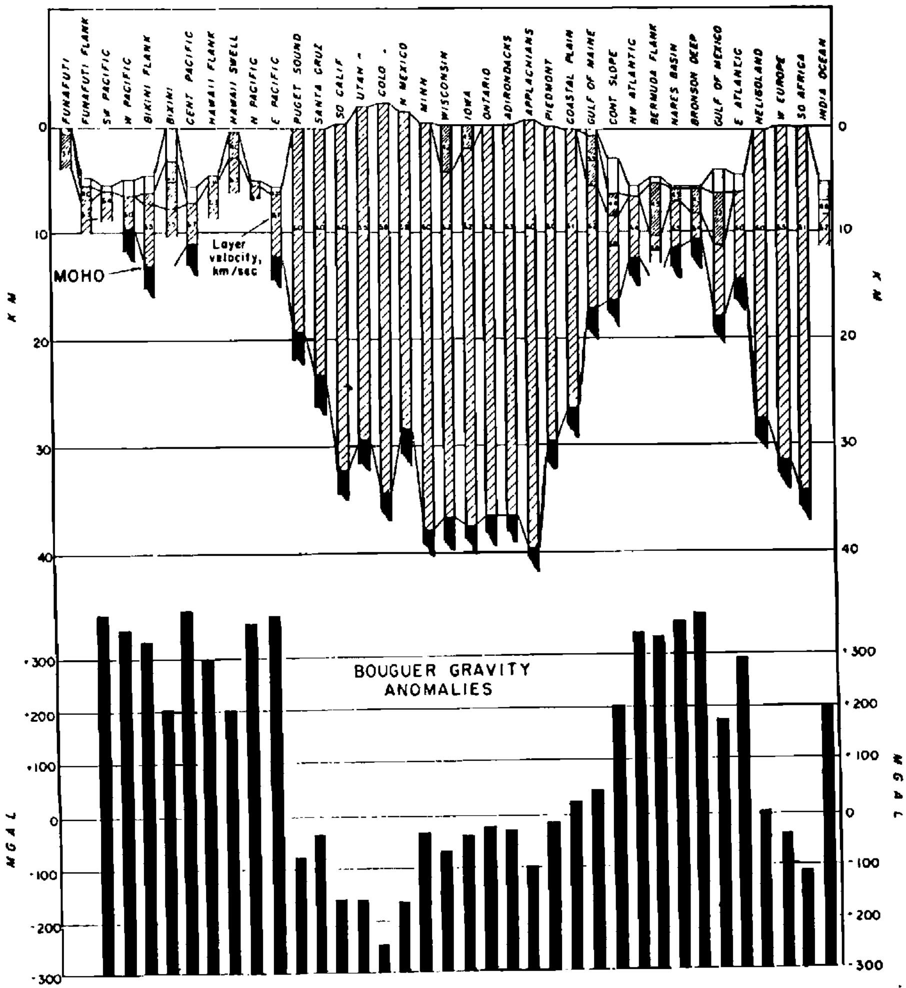


Fig. 52. Relation of crustal thickness obtained from seismic measurements to Bouguer gravity anomalies (from Woollard, 1959, p. 1526). *Courtesy AGU.*

thickness of high density material below the Moho.

Depths to the Moho have been obtained from seismograph measurements in many areas on both continents and oceans. When these depths are related to the topography and to the Bouguer anomalies (Figure 52) we find that the general relations are as indicated above (Figure 50). While the details vary considerably there is no question from these measurements that the general condition of isostasy is very real and that it is adequately explained, in broad terms, by variations in the depth of the Moho.

When Bouguer and isostatic corrections are made there remain some very substantial anomalies with magnitudes of the order of 100 mgal or more (Figure 51). These represent uncompensated masses which must be supported by the strength of the earth's crust. Among the most notable are a positive anomaly of around 100 mgal on the island of Cyprus and negative anomalies of the order of 200 mgal in the island arcs of the East Indies. It has been suggested that these are an indication of convection currents carrying low density material into the crust.

Considering isostasy as being entirely or largely accomplished by a simple change of density at the Mohorovićić discontinuity is almost certainly oversimplified. There are indications, from the nature and extent of isostatic anomalies, that at least part of the density variations holding up high topography are at depths much below the indicated depth to the Moho. On the other hand, the gross relations as suggested by Figure 50 and as indicated by the observations of Figure 52 are world wide and the simple relation between topography, Bouguer anomalies and depth to the Moho must be a major part of isostatic compensation, particularly at the continental margins.

PART II
THE MAGNETIC METHOD

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CHAPTER VIII
THE MAGNETIC FIELD

INTRODUCTION

Magnetic methods have been used since the beginning of geophysical prospecting. Previous to their application in petroleum exploration, several relatively crude types of magnetometers were used in mining prospecting, particularly in the search for iron ores. Magnetic surveys for petroleum exploration were begun in the early nineteen twenties. Contrary to the gravity and seismic methods, the magnetic method was not applicable to the search for salt domes and its early use was not in the Gulf Coast. Very extensive surveys were made in the middle and late nineteen twenties in West Texas and were applied on an intuitive basis in the selection of large lease blocks. One major oil company established very substantial reserves in the Permian Basin by simply leasing large ranches along linear features indicated by magnetic maps. Later it developed that these magnetic features were expressions of the flanks of the large Central Basin Platform which localizes accumulation in several large West Texas fields.

The magnetic method is also an application of a potential field and in many ways is similar to the gravity method. However, as will be pointed out later, the mathematics of the magnetic field are somewhat more complex than those of the gravity field, because of variations of the direction of the magnetic vector with latitude and because different instruments measure different components of the field. The geological aspects, however, are considerably simpler since, for most applications in petroleum exploration, the sources of magnetic anomalies are within the basement rocks. Therefore magnetic interpretation usually begins at the base of the sedimentary section whereas gravity effects can have their sources at any depth from the grass roots down.

FUNDAMENTALS OF THE MAGNETIC FIELD

The magnetic field is a vector in space whose defining components are indicated in Figure 53. The total horizontal intensity is the vector sum of the north and east components. It is designated as H and its angle of declination d is that between H and the astronomic north. The total intensity T is the vector sum of the three primary components and the inclination i is the angle between T and the horizontal. The magnitude and direction of the total intensity vector may be defined by various combinations of three angles and cardinal components. The positive direction of the magnetic vector is defined as that of a north-seeking magnetic pole, corresponding to the north pole of a magnetic compass. By this definition the equivalent north magnetic pole of the earth is a south pole and the south magnetic pole is a north pole.

For many years, magnetic instruments on the ground measured the vertical component V. Instruments measuring the

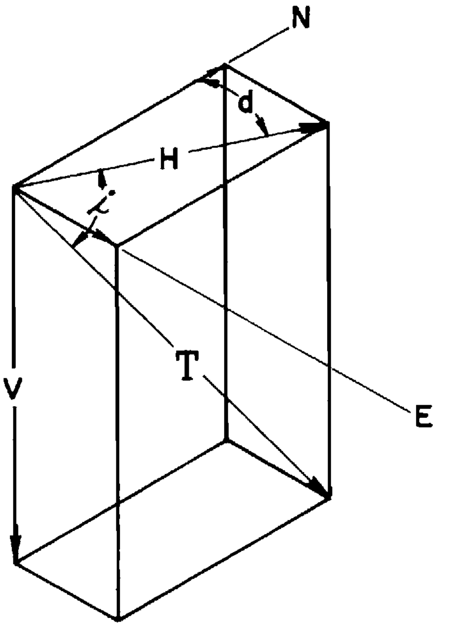

Fig. 53. Notation for components of the magnetic field vector.

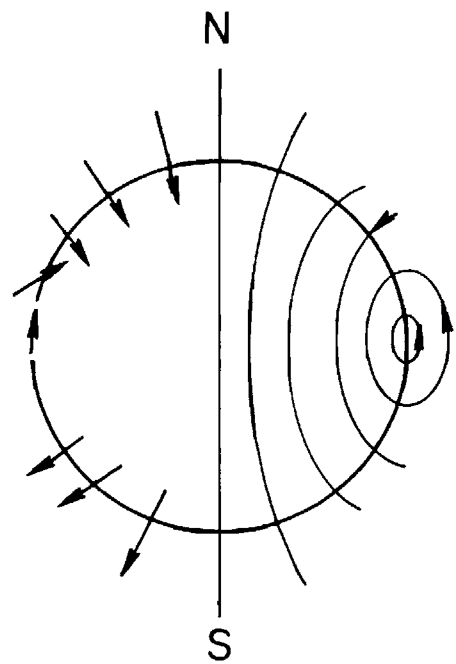

Fig. 54. The earth as a magnet; right half shows magnetic lines of force through the earth and in the external space. Left half shows direction and relative magnitude of total magnetic vector at the surface.

horizontal component *H* also were used in rather limited applications. At the present time most magnetic measurements, including all made in the air, measure the total intensity *T*. The operating principles of the various magnetic instruments will be discussed later.

MAGNETIC FIELD OF THE EARTH

To a first approximation, the magnetic field of the earth can be considered as that of a uniformly polarized sphere, Figure 54. At the north magnetic pole the field is vertical, pointing inward and at the south magnetic pole it is vertical pointing outward. The lines of force pass through the earth and close on themselves in the space outside as indicated by the right half of Figure 54. The direction of the field with respect to the surface of the earth changes from vertical, pointing inward at the north pole to horizontal pointing northward at the equator to vertical pointing outward at the south pole as indicated by the left half of Figure 54.

The variations in the direction of the field with respect to the earth's surface cause large changes in the nature of the magnetic anomaly as illustrated by Figure 55 (Nettleton, 1962). The several diagrams show the calculated variation in the total field anomaly caused by a polarized sphere with various inclinations of the direction of magnetization. The profiles are on north-south lines with north to the right. At high magnetic latitudes where the magnetic inclination is nearly vertical, the theoretical anomaly is a simple maximum with very weak symmetrical flanking minima (amplitude less than 5 percent of that of the maximum) on

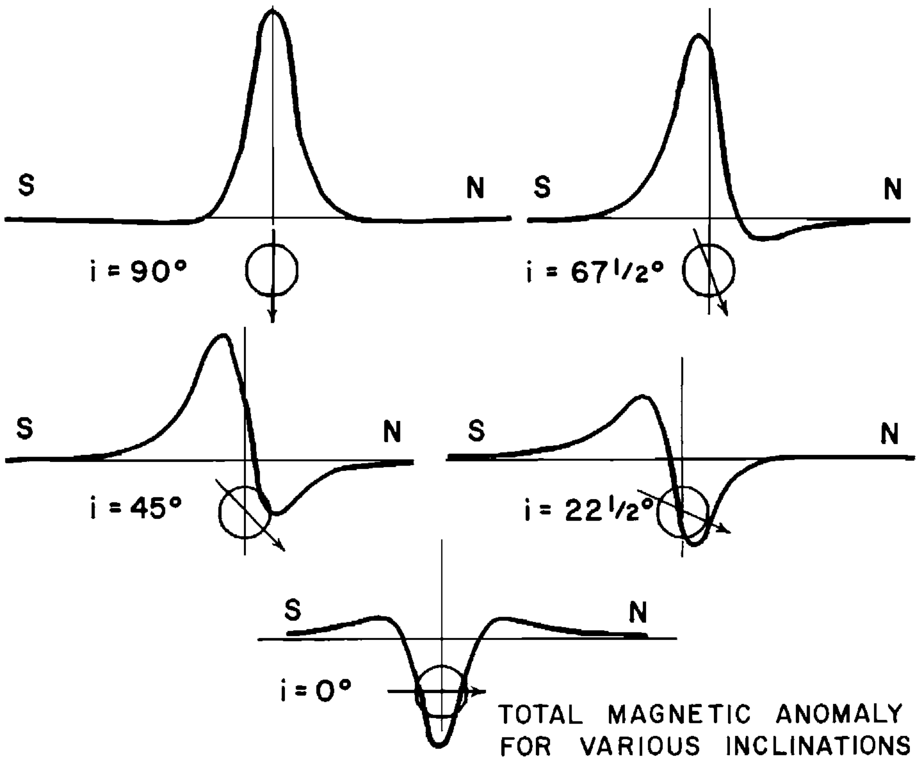

Fig. 55. Variation in form of anomaly in total magnetic intensity of a sphere with change in magnetic latitude. (Nettleton, 1962, p. 1818). *Courtesy AAPG.*

the two flanks. For the next diagram, with an inclination of 67½ degrees the figure begins to become unsymmetrical, the maximum amplitude is slightly lower, its peak is shifted slightly south of a point over the center of the sphere and a negative component begins to develop on the north side. At 45 degrees the maximum amplitude is further reduced, the peak is shifted farther to the left, and the negative anomaly is quite strong. At 22½ degrees the curve becomes nearly symmetrical with positive and negative parts of roughly equal amplitude and with the point over the center of the sphere quite close to the maximum negative amplitude. At the magnetic equator, where the direction of polarization is horizontal the anomaly has become a strong minimum with symmetrical flanking positive features. Thus the magnetic anomaly which is a simple positive anomaly at high magnetic latitudes is approximately turned inside out to become a negative anomaly at very low magnetic latitudes. The series would be repeated with left-to-right reversal if the diagram were continued to the south magnetic pole.

An example taken from an actual survey is shown in Figure 56. This is from an area where the inclination is about 20 degrees and where both a gravity and an airborne magnetic survey were carried out. The diagram on the left shows a simple, nearly circular positive gravity anomaly (the left half of the anomaly is off-shore

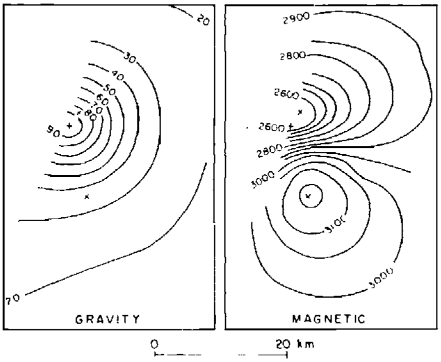

Fig. 56. Comparison of gravity and magnetic maps in same area. (Nettleton, 1962, p. 1818). *Courtesy AAPG.*

and was not observed). The magnetic map over the same area shows a strong negative anomaly to the north (up on the map) and a positive anomaly to the south. The plus sign on each of the two diagrams shows the center of the gravity anomaly and the two x's on each figure show the centers of the magnetic negative and positive anomalies. Figure 57 shows profiles of the gravity and magnetic anomalies on a north-south line together with calculated effects from a heavy sphere and a magnetized sphere. The approximate agreement in size and location of the two spheres, with reasonable assumptions for the density and magnetic contrasts, indicates strongly that these two anomalies come from a magnetized heavy mass with nearly the same boundaries for the magnetic and density contrasts.

MAGNETIZATION OF ROCKS

The degree of magnetization, i.e., the "polarization," of rocks for induced polarization is the product of their susceptibility k and the magnetizing field, i.e., $I = kH$. This is the polarization produced by magnetization in the earth's field H which has an intensity range of about 0.3 to 0.6 oersteds with a common value of about 0.5. The usual unit in magnetic prospecting is the gamma or 10^{-5} oersteds. Thus the normal earth's field is roughly 50,000 gammas. To the extent that the magnetization of the rock is caused by simple induced magnetization by the earth's field in its present direction, the anomaly forms caused by such magnetization will have the variation from magnetic pole to equator similar to those indicated by the diagrams of Figure 55. Polarization and polarization contrasts among rocks control the magnitude of magnetic anomalies just as densities and density contrasts control the magnitude of gravity anomalies except that the magnetic case is complicated by the effects of different directions and intensities of polarization in different parts of the world.

To a large extent the susceptibility of rocks is a measure of their magnetite content. There are other minerals which are ferromagnetic but none as much so as magnetite, which is by far the most common of the magnetic minerals in rocks. For low concentrations of magnetite there is an approximately linear relation between the percentage of magnetite and the magnetic susceptibility which may be

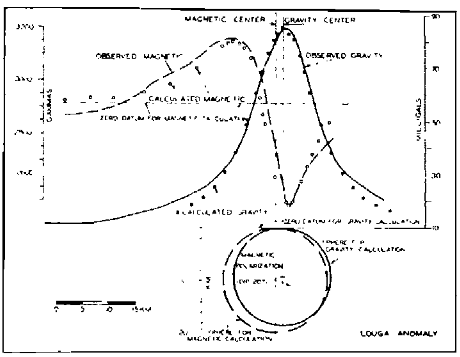

Fig. 57. North-south gravity and magnetic profiles from Figure 56 with calculated values and cross section of possible anomalous heavy, magnetized body (Nettleton, 1962, p. 1819). *Courtesy AAPG.*

expressed as $k = 0.3p$ in which p is the percentage (by volume) of disseminated magnetite and k is the susceptibility. Thus, if a rock with, say, 1 percent by volume of magnetite were magnetized in the earth's field with intensity 0.5 oersteds (i.e. 50,000 gammas) its intensity of magnetization I would be $I = 0.5 \times 0.3 \times 0.01 = 0.0015 = 1500 \times 10^{-6}$ cgs units which is only a little more than an average figure (about 0.001) commonly used for basement rocks.

The range of susceptibilities of rocks is very much greater, percentage wise, than the range of densities. Susceptibilities of common igneous rocks range from the neighborhood of 2000×10^{-6} for granites or porphyries with high quartz content to around 15,000 for diorites and diabase. On the other hand, the susceptibilities of sedimentary rocks are very low, usually in the range 0 to 50×10^{-6}. Thus, for most applications of magnetic surveying, the magnetic effect at the magnetometer may be considered as approximately the same as if the sediments were not present and the magnetic disturbances recorded have their origin at or below the base of the sediments. This is the basis for use of magnetic measurements to map the basement surface.

For the most part magnetic interpretations can be made as if the rocks were polarized by the earth's field in its present direction. It is quite common for rocks to retain a significant "remanent polarization" component remaining from a previous geologic period in a direction which may be quite different from that of the present earth's field. However, distortions of the direction of polarization by remanent magnetization are not regular or consistent enough to cause serious problems in magnetic interpretation. Calculation of anomalies from models to compare with those observed usually can be made on the assumption that polarization is in the direction of the present earth's field. This is also shown by the fact that magnetic maps, world-wide, in low magnetic latitudes, nearly always show the general pattern of negative anomalies on the north side (in the northern hemisphere) with approximately the expected ratios of amplitudes of positive and negative parts corresponding with magnetic latitude as illustrated by the examples in Figure 55.

CHAPTER IX

MAGNETIC INSTRUMENTATION

GROUND MAGNETOMETERS

The early exploration by the petroleum industry was done with a magnetic field balance. This instrument consists essentially of a horizontal magnet supported on a quartz knife edge. The torque due to the reaction of the magnetic poles with the vertical component of the magnetic field is balanced by the weight of the moving system with its center of gravity being adjusted at the correct distance from the knife edge support to achieve this balance. The position of the moving system is indicated by a scale in the eye piece which is reflected from a mirror on the moving system. As commonly adjusted this instrument can measure changes in the magnetic field with an accuracy of 5 to 10 gammas. Such instruments were used very extensively in magnetic prospecting for petroleum in the years from the beginning of such prospecting in the early nineteen twenties until they were largely superseded by airborne instruments beginning shortly after the second World War. In some later instruments, ligaments or fibers are used instead of the rather fragile quartz knife edge bearing. Also electronic instruments, using the principles from the airborne instruments described later, have been used for magnetic surveys on the ground, particularly for very detailed observations.

AIRBORNE MAGNETIC INSTRUMENTS

The fluxgate or saturable core magnetometer was the first applied to airborne magnetic measurements. This and the other airborne instruments are all entirely electronic in their operation; they do not have mechanical parts which would be affected by the motional acceleration forces of an airplane.

The fluxgate magnetometer

The fluxgate magnetometer for airborne use was first developed at the Gulf Research Laboratory shortly before the beginning of World War II. During the war the same basic system was developed, in somewhat different version, for use as a submarine detector. After the war, development of the instrument as an airborne magnetometer for petroleum exploration proceeded rapidly and extensive surveys for petroleum exploration were undertaken, beginning in about 1946.

The basic magnetically sensitive element of the fluxgate magnetometer is a small strip or bar of very highly permeable magnetic material. The magnetic material is so highly permeable that it approaches saturation in the earth's field. In one form, shown by Figure 58 (Wycoff, 1948), two parallel strips are wound in opposite directions with coils carrying alternating current at 1000 hz. Both strips

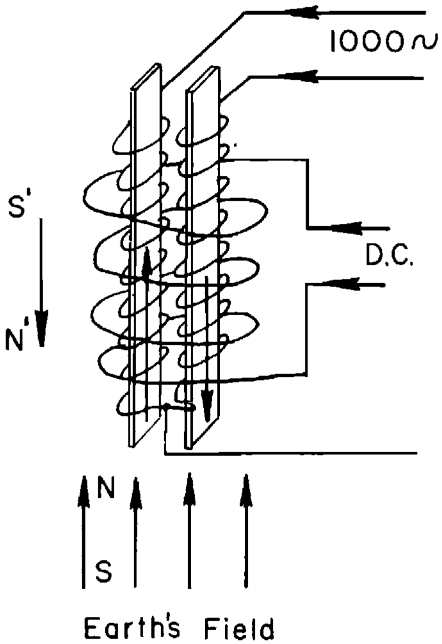

Fig. 58. Schematic diagram of operating principle of Fluxgate magnetometer (Wycoff, 1948). *Courtesy SEG.*

are within a magnetizing and measuring coil carrying direct current. If the ambient field is zero, magnetization curves of the two strips are equal and opposite in all details and there is no external field to be detected by the surrounding measuring coil. However, if the ambient field is not zero there are changes in waveform and phase relations of the induced voltages and they do not balance, there is a net output voltage and this voltage is a measure of the magnetic field. In practice the output voltage operates through a servo-system which controls the current in the magnetizing coil surrounding the two cores to give zero signal. Then variations in the current of this magnetizing coil are a measure of the changes in the magnetic field. The system is so arranged that the current which measures the major part of the field can be changed in accurately determined fixed steps so that the part recorded is the relatively small variable component. When the recording approaches the edge of the record it is automatically stepped back by one of the fixed steps to keep the pen on the paper.

The field component measured is that parallel to the axis of the measuring cores. Therefore it is necessary to orient the sensitive element so that its axis is parallel to the direction, in space, of the magnetic vector. If the core is not parallel it will measure less than the total intensity of the field.

As commonly used, the necessary orientation is achieved by using two additional similar magnetically sensitive elements arranged so that the three are mutually perpendicular. The two orienting elements are in the plane of a gimbal-mounted plate and the third, or measuring element is perpendicular to these. Servo-mechanisms controlled by each of the two orienting elements operate small motors to keep the output signal zero for each of the two orienting elements. This means that the plane of the plate that carries them is perpendicular to the earth's field and the third or measuring element therefore is parallel with the earth's field and measures the total component. In practice the gimbal orienting mechanism is very fast and will keep the measuring element in the proper orientation in spite of rapid motions of the "bird" in which the instrument is mounted. The shell or "bird", shaped like an aerial bomb, is some six or eight inches in diameter and three or four feet long. The length is desirable aerodynamically but also is required to keep the motors of the servo-mechanism at some distance from the magnetically sensitive elements; the gimbal movements are accomplished through small, highly nonmagnetic shafts, gears, strings, and pulleys.

The magnetometer element may be carried on a cable suspended several hundred feet below the airplane in order to be free from magnetic disturbances caused by the plane itself. It is also possible to use a "stinger" mounting in which the instrument is assembled in a short projection or "stinger" on the tail of the airplane. This requires very careful compensation of the magnetic parts of the airplane itself so that, at the instrument location, there are no residual magnetic effects which change as the attitude of the plane changes with respect to the magnetic field.

The proton-precession magnetometer

This magnetometer depends upon a fundamental property of the atomic nucleus. Electrons spinning about the nucleus have the property of a circular electric current and therefore each spinning electron has a magnetic field and acts as a tiny magnet. If the spin axis is deflected by a transverse magnetic field and that field is then removed, the axis precesses back toward a direction parallel with that of the ambient magnetic field. This precession frequency is $f = \gamma F/2\pi$ where F is the magnetic field and γ is the gyro-magnetic ratio. If the active element is hydrogen in water the gyro-magnetic ratio of protons is $\gamma_p = 23.4868$ gamma/hz and is a fundamental constant of the nucleus, independent of its surroundings or chemical state. Therefore if the frequency of precession f can be measured, the magnetic field is determined. This means that in the earth's field of, say, 50,000 gammas, the precession frequency is $\frac{50,000}{23.4868} \cong 2100$ hz. Thus, to measure, say, a one-gamma change in the earth's field, the precession frequency must be measured to about one part in 10^5 or around 0.02 hz.

In the airborne instrument the active element is water in a small bottle surrounded by a suitable coil. The precession frequency is dependent only on the total intensity of the magnetic field. The sensitivity is maximum if the magnetizing field is perpendicular to the magnetic vector. The electronics consist of a system which magnetizes the water bottle periodically and then measures the frequency of the resulting precession.

The general performance and quality of the recording of commercial airborne proton precession magnetometers is generally comparable to that of fluxgate instruments. With the equipment and associated electronics, well-maintained recordings can be made which are reliable to about one gamma. The record itself has small steps, because it is inherently intermittent from the necessarily separate magnetization "charges" of the water bottle and the interlude necessary to determine the precession frequency. At a charging interval of one sec and at an airplane speed of, say, 200 mph or 300 ft/sec, the space between the individual measurements is equivalent to measurements 300 ft apart on the ground.

Pumped alkali vapor magnetometer

The proton precession magnetometer depends upon the precession frequency of protons. Electrons also have a magnetic moment which may be used to determine the strength of the magnetic field by the measurement of much higher frequencies. The family of pumped alkali vapor magnetometers depends upon this phenomenon.

A suitable vapor (commonly caesium or rubidium) is used. The vapor is included in a tube which is illuminated by circularly polarized light from a lamp which contains excited vapor of the same kind. Thus, light passing through the tube has a frequency which is the same as the characteristic emission frequency of the vapor and this light raises the energy levels of certain frequency components of the spectral system and changes populations in certain subfrequencies of the spectrum. This process extracts energy from the circularly polarized beam which decreases the intensity of the light falling on the photocell at the end of the tube. The photocell current controls a servo-system in such a manner that a radio frequency applied to the coil is resonant with the precession frequency which, in turn, is dependent on the ambient magnetic field. Therefore a measurement of this resonant frequency serves as a measurement of the magnetic field. For Rb^{85} the frequency is approximately 467 Khz/oersted so that, in the earth's field of about 50,000 γ or 0.5 oersted the frequency is approximately 230 Khz.

The optimum orientation for the pumped vapor sensor is at 45 degrees with the field, and deviations affect the signal strength rather than the frequency. However, there are secondary effects of orientation which must be avoided. The measuring shell contains a constant temperature oven within which are mounted the vapor cell, pumping lamp, photocells, lenses, filters, etc. This shell is mounted on a gimbal system which permits orientation in any direction. The system is oriented by magnetic sensors to maintain the optimum orientation with respect to the magnetic field vector. The precision of orientation required is not as great as that for the fluxgate instrument.

The basic principles of electron spin have been applied to produce operating airborne magnetometers using rubidium vapor, caesium vapor, and, to some extent, helium vapor. In the laboratory, instruments using these principles can measure magnetic changes to as small as about 10^{-4} gammas. As applied in airborne operations the measurements can be one or possibly two orders of magnitude more sensitive than those with the fluxgate or proton precession instruments.

The vertical magnetic gradient

The development of the very high sensitivity optically pumped magnetometers has provided a basis for the measurement of the vertical gradient of the total magnetic field. Two separate magnetometers are flown with a vertical separation of the order of 200 ft. In some installations, the upper instrument is in a "stinger" mount at the tail of the airplane and the lower one is carried on a suspended cable. In other installations both instruments are on a cable. In either case the measurement is made by recording variations in the difference between the two instruments. This variation can be measured to a precision of around 0.02 gamma/ft. A continuous record is made of the gradient variation and, on the same recorder, a trace of the total intensity is shown also. The vertical gradient then can be mapped from these variations.

It is quite possible to calculate the vertical gradient from an accurate map of the magnetic field, and there is some disagreement as to whether or not measuring the gradient adds value corresponding to the rather considerable increase in the cost of the operation. Making the observations has one definite advantage in that the record of the vertical gradient serves to identify disturbances in the total intensity record which are caused by sudden changes in the earth's magnetic field (diurnal effects, see p. 84) so that such disturbances can be distinguished from those caused by magnetic sources at or below the earth's surface. The magnetic gradient map is much more responsive to local influences than to broad or regional effects and therefore tends to give a considerably sharper picture than does the map of the total field intensity. Thus it can serve to localize the smaller anomalies and make them more easily apparent in an area of strong background disturbances. The same general effect is obtained by a vertical derivative operation, and therefore vertical derivative maps calculated from the total intensity field tend to be very similar to maps made from field measurements of the vertical gradient. In particular, the high and low closures, nosings, or other details will generally be in nearly the same areas on the two kinds of maps.

An approximate comparison of vertical gradient and second derivative patterns for the same area is shown by Figure 59. The solid contours are from the second derivative map of the Puckett area, taken from Figure 70. The dashed lines are from a published vertical gradient map of the same field. It will be noted that the general contour pattern and the sharpening of local features is quite similar in the two sets of contours.

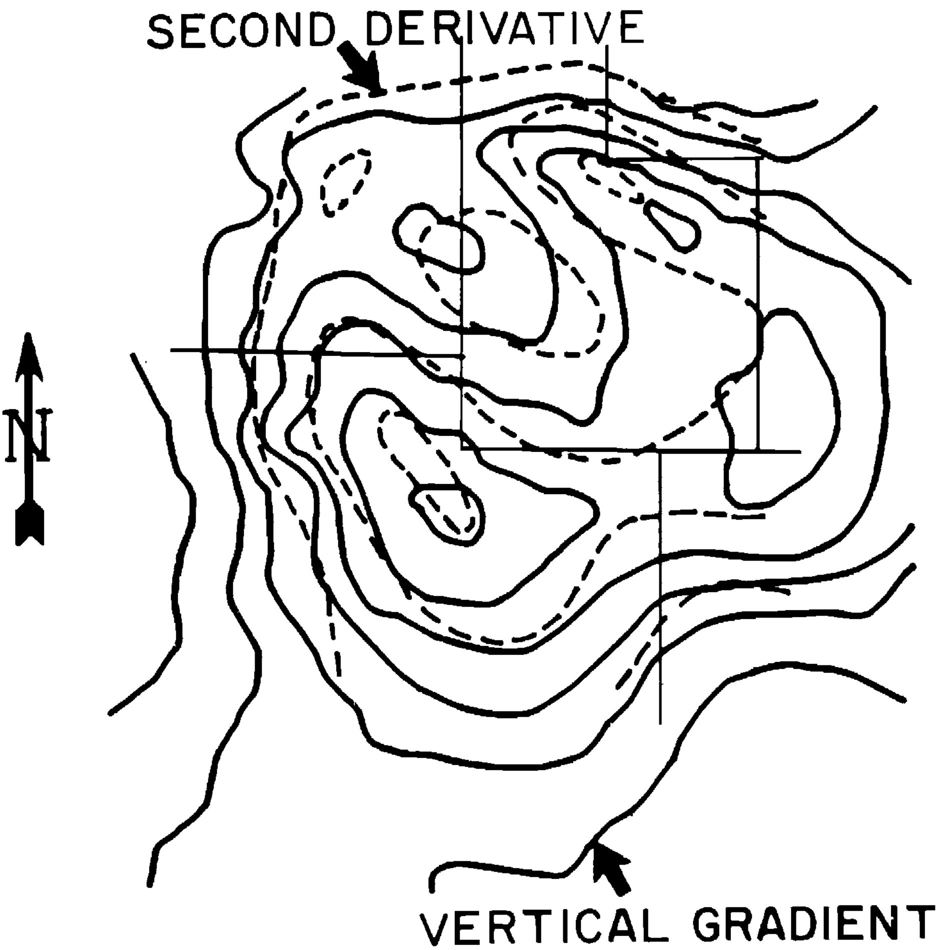

Fig. 59. Comparison of vertical gradient and second derivative maps of same area.

CHAPTER X

AIRBORNE MAGNETIC SURVEYING

Most magnetic surveying is now done from the air, measuring the total intensity of the magnetic field. Only relatively limited and special observations, such as in some mineral prospecting, are done on the ground with mechanical magnetometers measuring the vertical component or electronic magnetometers measuring the total intensity.

For any of the airborne magnetometer systems, the instrument assembly in the airplane carries the necessary electronics and a recorder on which a continuous record of the magnetic field is shown. The airplane also usually is equipped with a radar altimeter and with the necessary positioning equipment. The several items of data (magnetic field intensity, electronic or photographic positioning data, barometric elevation, and height above ground) are correlated with one another so that all can be brought together to provide the necessary information for reducing the observations to a magnetic map. In modern equipment, the data recording is on magnetic tape so that the data reduction can be carried out largely by electronic computers and machine mapping.

The volume of surveying carried out with airborne magnetometers is very large. In petroleum exploration, surveys have been made over the order of ten million miles of flight line traverse. Such surveys have covered most of the sedimentary basins of the world including very substantial offshore areas. The degree of control varies widely; some surveys have line spacing as close as one half mile, or more commonly one to two miles, but there are also broad reconnaissance surveys carried out with single lines or band flying.

Flight elevation is commonly about 1000 ft above the surface. Where there is a large range of topographic relief the survey may be carried out in two or more parts at different constant elevations, usually with some overlap along the common boundaries. A great deal of airborne magnetic work also has been done for the mining industry, usually in much closer detail than for petroleum exploration. Commonly, mineral surveys are made with a line spacing of about one-quarter mile. In many instances the work is done from helicopters to make the observations nearer the ground surface. The Canadian government has had underway for several years a systematic magnetic survey of a large part of the country including closely spaced surveys in the shield areas. These have been found useful, not only for mining prospecting and the location of individual anomalies possibly associated with mineral accumulation, but also in routine geologic mapping. It is found that, in the pre-Cambrian areas, the general trend pattern of the magnetic contours is often closely parallel with that of the outcrops, and the magnetic pattern can be used to interpolate contacts over covered areas.

DIURNAL MAGNETIC VARIATION

The magnetic field of the earth is not constant. At any fixed location there are daily changes which are roughly cyclic but with irregular variations. The source of these changes is the movement of ionized layers high in the atmosphere which act like irregular electrical currents to produce the disturbing magnetic fields. The daily cycle may have an amplitude of around 20 to 50 gammas. The short period irregularities may have amplitudes of 10 gammas or more. At times of magnetic storms, commonly during strong sun-spot activity, the magnetic changes may be so violent and the magnetometer records so disturbed that accurate removal of their effects in the data reduction procedure would not be possible, and flying operations should be discontinued. It is common practice to maintain a magnetometer on the ground near the area of flight operations to monitor the short-time changes so that the field operations can be discontinued when diurnal effects become too severe.

GENERAL PLAN OF MAGNETIC OBSERVATIONS

A common general plan of magnetic operations with a sample from an actual survey and corresponding geological interpretation is shown by Figure 60 (Nettleton, 1950). Usually the airplane is flown in parallel flight lines tied together by cross lines at an interval of several times the primary line spacing. The survey lines and cross lines form closed loops around which the magnetic changes may be adjusted for closure to eliminate effects of diurnal changes or instrument drift. The optimum line spacing depends on the depth to the basement and need not be closer than about one-half that depth. However, the depth is usually known only very generally or not at all, and it is not feasible to change the flight spacing to fit a variable basement depth. Therefore, a pattern usually is chosen empirically and an entire survey or unit of a survey is flown with the same pattern. Common spacings, for petroleum exploration, are one kilometer, one mile, two kilometers or two miles between flight lines with tie line spacing approximately six times the flight line interval. Thus, spacings of one by six miles or two by ten miles are quite common. In sectionized country where roads are one mile apart it has been common practice to plan the flight lines to be over the roads as the airplane can then be held on line quite easily by visual observation of these roads.

The orientation of flight lines can be important. If the general tectonics of an area are known, or earlier, less detailed magnetic surveys are available, the flight lines are laid out approximately perpendicular to the expected magnetic trends. Departures from the ideal direction of as much as 30 degrees are not particularly troublesome, but, when flight line traverses are less than 45 degrees from the general tectonic strike, large corrections are necessary in depth calculations and their precision deteriorates.

In exploration of very large areas where only preliminary reconnaissance control is desired, to give information as to whether or not large basin areas are present, an economical substitute for complete coverage is so-called "band flying." In this practice, bands of three (or sometimes only two) parallel lines are flown at what would be considered the normal spacing for the area. The interval between these bands may be quite

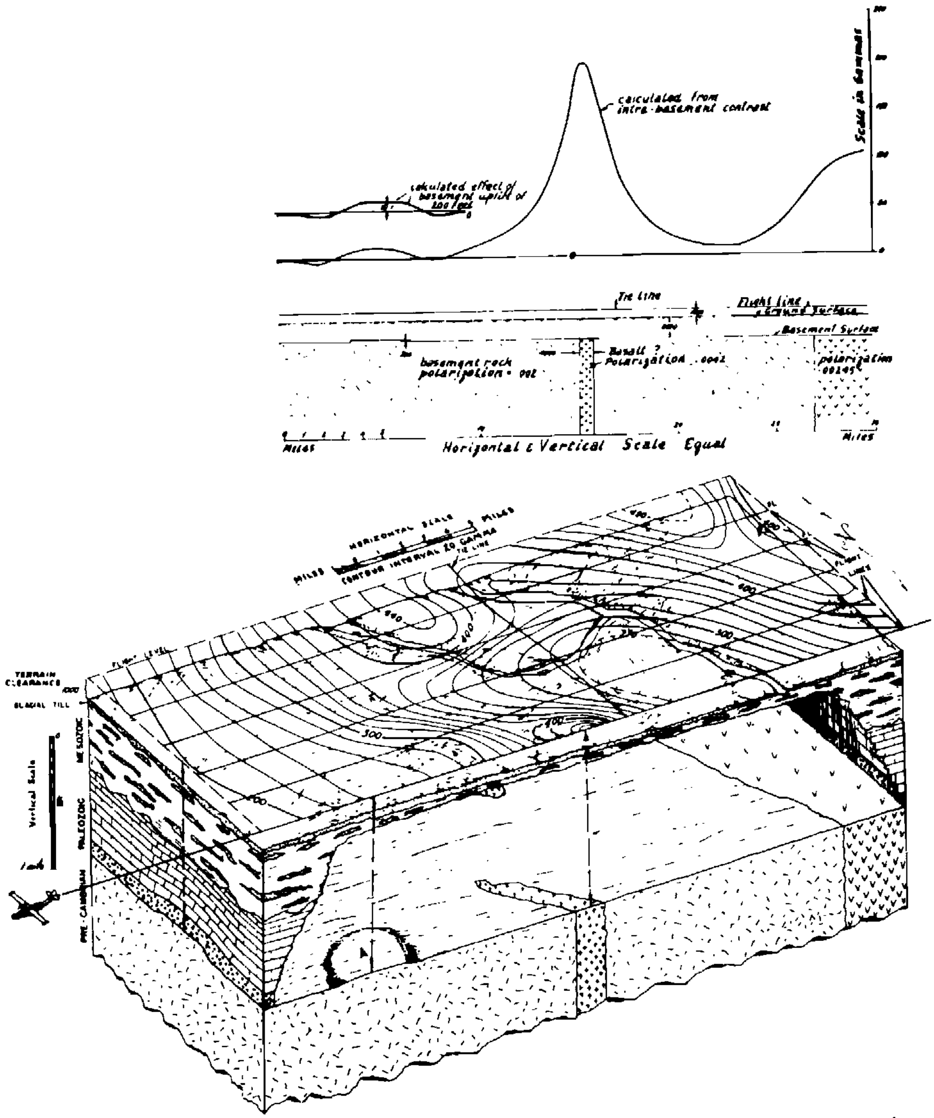

Fig. 60. Diagram of airborne magnetic survey and sources of anomalies with cross-section and corresponding magnetic profile above. The belt of steep gravity between the 300 and 400 gamma contours comes from a change in the nature of the magnetic material. The sharp closure at the near edge is from a dike-like body. Note that neither of these magnetic features represents variations in the basement surface. The hill (or uplift) of the basement surface, designated by A, causes only a low-relief broadening of the magnetic contours (Nettleton 1950). *Courtesy "Oil in Canada."*

large, often twenty-five to fifty miles. Thus the entire area is covered by the bands of two or three lines with the areas between blank except for a few tie lines. This system gives better information than would be obtained by distributing the same amount of airborne mileage uniformly over the area. This is because the bands of lines permit contouring the magnetic field within the area of each band to give the strike of the magnetic pattern so that the angle between magnetic features and the flight lines is determined and depth calculations can be appropriately corrected for this angle. Thus a pattern of uncertain depths on lines at a smaller interval is replaced by much more certain depths along the bands to give a more accurate determination of the general basin form and the thickness of sediments which it contains.

NAVIGATION SYSTEMS

It is of course necessary to map the actual positions of the airborne flight traverse with respect to the ground. The horizontal scale of the recordings must be known accurately to permit basement depth calculations and the locations must be known accurately to permit contouring of the observations to make a continuous magnetic map. Several types of location systems are used.

Photography

Much magnetic surveying has been controlled by photomosaics. The programed flight pattern is laid out on a photomosaic map of the area. This map is cut into strips of a convenient width (about one foot) and rolled up so that, as the airplane flies along, the copilot, by unrolling the strip and observing the ground, can direct the pilot to fly over or near the preassigned flight lines. The ground is photographed continuously either as 35 mm individual pictures or by a continuous strip camera, and "fiducial" or reference numbers are shown on each picture and also on the recorder chart. In the assembly map room these photographs are carefully matched with the photomosaic to determine the position of the airplane at the fiducials, which gives the actual path of the flight line. Correlation with the corresponding fiducial numbers on the magnetometer record gives the exact ground position of each detail of the magnetic field.

Electronic positioning

Electronic surveying has been used very widely for airborne magnetics. Shoran, Raydist, Lorac, Toran, and some other electronic positioning systems all have been used for positioning magnetic surveys both over land and over water. These systems all require shore based antenna towers and accompanying electronic installations and power supplies and therefore are relatively expensive to establish.

Doppler navigation

In recent years, considerable application has been made of Doppler navigation for airborne magnetics. The Doppler equipment measures two components of speed of the airplane, one parallel and one perpendicular to the plane axis. Two pairs of beams of relatively high frequency radiation are directed to the ground and the phase differences between the reflected signals from the two beams of each pair are measured. Thus the phase difference between signals from one beam pointed forward and one backward gives the component of speed along the axis of the airplane while those between one beam to the right and one to the left give the lateral speed component. By reference to a gyrocompass, the north-south and east-west components of speed are determined. The speed components are integrated to give distances and thus the equipment can plot the track of the airplane or register its position on magnetic tape. The Doppler system has the great advantage of being entirely self-contained within the airplane and not depending on communication with fixed ground stations as do all of the various other electronic position systems. The precision is of the order of one percent of the distance traveled. Errors are generally linear and can be adjusted by reference to rather widely spaced control locations. The system is not so satisfactory over water because reflections from the surface are not reliable when the water is smooth, and because of errors introduced by currents in the water.

CHAPTER XI

SOURCES OF MAGNETIC ANOMALIES AND INTERPRETATION

As mentioned earlier, the source of magnetization of rocks is primarily from magnetic induction by the earth's field of particles of magnetite within the rocks. Usually, the lighter colored rocks such as granite and porphyries have much less magnetite than the dark basic rocks such as gabbro and diabase. If we consider the magnetizable material in the rock as disseminated sources, the nature of the magnetic anomaly will depend on the way the sources are distributed and the distance from the rock surface at which measurements are made. This is illustrated by Figure 61 which shows magnetic curves calculated from point sources in a north-south vertical plane and with vertical magnetization.

The upper curve with high relief and much detail is calculated for an elevation of 0.1 unit corresponding to the depth to the first row of dots. The lower curve, which is much smoother and of lower relief, is calculated for an elevation of 0.5 unit. At very low flight levels, the influence of individual sources is distinguishable as small variations or waves in the total effect. At greater distances above the surface these effects are merged together and the magnetic field is practically the same as it would be if the sources of the magnetic disturbances were simple blocks (or sheets) of magnetized material with boundaries corresponding to the divisions between areas with different degrees of concentration of sources. In a way this is analogous to the manner in which a picture made up of small dots (as in a coarse-grained newsprint half tone) changes when viewed from different distances. Very close up the individual particles or dots are separately distinguishable. At greater distances certain of these merge into the smaller details of the picture. At still greater distances these, in turn, merge into larger components of the picture. In a similar way as the magnetic field is observed from successively higher flight levels (or conversely at successively greater basement depths) the sharp details which are apparent at low levels merge into broader and broader features at higher levels (or greater depths). Thus it is usually possible by simple inspection of a magnetic map over a broad area covering a wide range of depths to separate the map into areas of shallow, intermediate, and large basement depths by noting the areas of sharp, intermediate, and broad magnetic anomalies. It is the refinement and quantitative evaluation of these characteristics of the magnetic field which permit calculation of depths to the sources of the measured magnetic effects. The resulting mapping of the basement surface gives a quantitative picture of the thickness

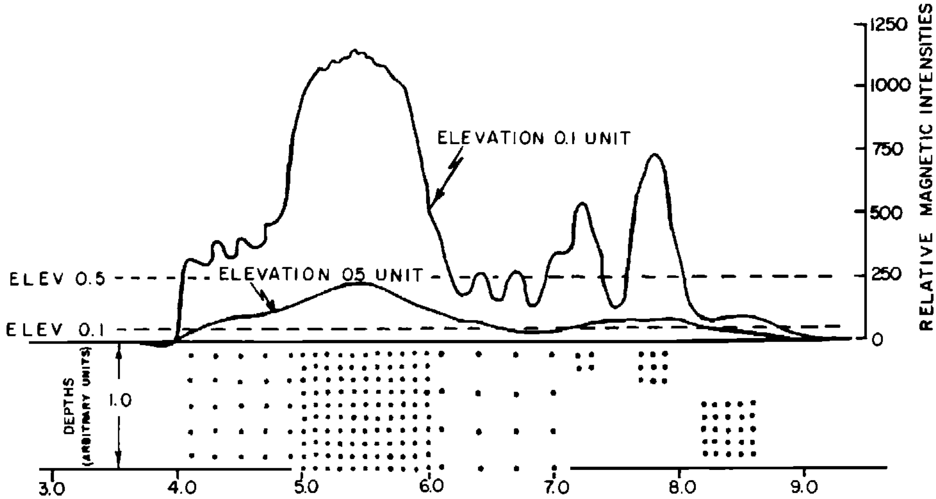

Fig. 61. Calculated magnetic effects of point sources. Each dot represents a unit dipole magnetized vertically ($I = 90°$). The two curves are for simulated north-south "flight lines" at elevation of 0.1 and 0.5 distance units. Note that the various groups of poles act as blocks of magnetized material. At the very low flight level the effects of individual sources are detectable; at the higher level their effects are much smaller and merge into a very smooth curve.

of the sedimentary section. It may be reiterated here that this characteristic of magnetic rocks depends on the geological "fact of life" that sedimentary rocks generally are nonmagnetic, and the principal features of the magnetic field are the same as they would be if the sediments were not present at all.

THE MAPPING OF BASEMENT DEPTHS

Several somewhat different approaches have been made to the interpretation of magnetic maps and mapping of the basement surface. In general these may be divided into curve-fitting and spot calculation systems.

In the curve-fitting approach, a form for the magnetizing body is assumed, its effect calculated and compared with that observed and the process repeated until an acceptable fit is obtained. Alternatively the observed field or profile may be compared with fields calculated from a series of models. This process is based on the magnetic map over the entire area of a selected anomaly and therefore is dependent on the precision of the mapping, including adjustment of inconsistencies between flight lines. The analysis gives relatively few carefully determined depth values as each anomaly used gives only one depth and only relatively few anomalies can be clearly isolated from their environment and have forms for which it is feasible to make such calculations simply.

DEPTH DETERMINATION FROM MAGNETIC PROFILES

The spot calculation approach uses the characteristics of the magnetic profiles, usually the flight record as made on the airplane. Certain parameters of these records are related to the depth to the top of the source and therefore to the depth to the basement surface. The individual depth values from this approach may or may not be as accurate as those obtained by curve-fitting, but the method has the great advantage that it gives a much larger number of depth values. The restrictions on a suitable anomaly are less severe and calculations can be made on each flight line crossing an anomalous feature rather than deriving a single depth from the feature as a whole. In a commonly used system the "slope" and "half slope" parameters are used as depth criteria.

The measurements are made independently on the individual flight line records and therefore do not depend on the precision with which these records are assembled to make a magnetic contour map. These depth parameters are determined as indicated in Figure 62 which is taken from an actual airborne magnetic record. The record shown is machine plotted from a digitally recorded rubidium vapor magnetometer.

In Figure 62 the distance d is the interval over which the curve is substantially a straight line and the ends of the interval are the points at which an appreciable departure from this straight line begins. This is not a parameter that can be defined mathematically in terms of the calculated effects from magnetized bodies, but it turns out in practice that this is a very useful empirical measurement. The straight line distance is linearly related to the depth to the surface of the contact which produces the change in the magnetic intensity.

The "half slope" parameter is indicated on the figure by the distance p. This is the distance between the points at which a straight line with half the maximum slope is tangent to the curve be-

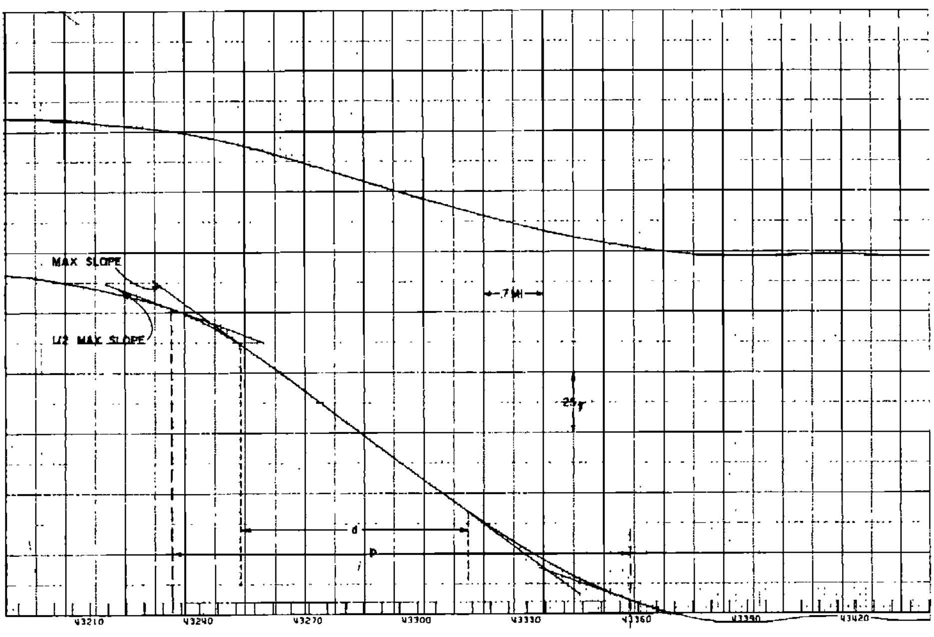

Fig. 62. Example of airborne record showing measurements of the slope parameter d and the half-slope tangent parameter p. For ideal anomalies "p" is approximately twice "d." These parameters are useful for basement depth determination.

low and above the straight line segment. The distance p also is empirically related linearly to the depth by a factor which is approximately half as great as the parameter for the distance d. (For a complete discussion of this method and the magnitudes of the depth parameters see Vacquier et al, 1951.)

CORRECTIONS

After a depth figure has been determined by establishing the distances d and p on the magnetic record, certain corrections have to be made.

Correction for scale

The scale of the record, usually in terms of inches on the record per mile on the ground, has to be determined. This is done by simply measuring distances on the record corresponding to known distances between certain points so that the distance measured on the record can be converted to feet or miles.

Correction for azimuth

In general the flight line is not perpendicular to the contours and the distances determined from the record are greater than they would be if it were. This means that distances measured on the record must be shortened by multiplying by the cosine of the angle between the direction of the contours and the flight line. The contour direction of the particular feature being used for the depth calculation is usually better determined from a second derivative map than from the total intensity map.

Correction for flight elevation

The basement maps are usually made with depths in feet below sea level, while the parameters measured on the record apply to the flight elevation. Therefore, it is necessary, after determining the depth of a given point from the record, to subtract the flight line elevation to give the depth on a sea level datum.

The general mapping procedure is illustrated by Figure 63. The background contours indicate a large positive magnetic anomaly. The slopes on the flanks of this anomaly correspond to the straight line portion of a record such as is illustrated in Figure 62. Depths (below sea level) are determined for each flight line and on each side of the anomaly. The underlinings of one, two, or three lines are a quality indication corresponding approximately to *poor* for one, *fair* for two, and *good* for three lines. This evaluation is based, in part, on the consistency of the depth indication between the slope and half-slope measurements and, in part, in the quality of the record and its freedom from local disturbances. After these depth numbers are placed on the map the basement is contoured as indicated by the depth contours shown. Note that in this example the very large magnetic anomaly does not conform to any local relief on the basement surface as the contours pass regularly across it. Thus this is an "intrabasement" feature having its origin entirely in magnetic contrasts below the basement surface.

INTRABASEMENT AND SUPRABASEMENT ANOMALIES

In general, there are two types of anomalies having their origin in magnetic contrasts at or below the basement surface. The *intrabasement* anomalies are those for which the causative body has a thickness which is of the order of or greater than the depth to its surface. The *suprabasement* sources are those which are thin compared to their depth. The differences are illustrated by the cross-section and magnetic record of Figure 60. The magnetic curve corresponds quantitatively with magnitudes calculated from the section and polarization contrasts shown. In the section, the strong

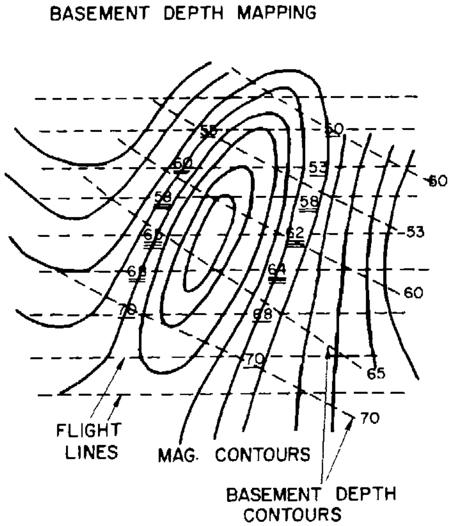

Fig. 63. Schematic diagram of basement depth mapping showing east-west flight lines, total magnetic intensity contours, depths calculated from flight line records and contours on the basement surface.

rise of magnetic intensity of about 100 gammas at the right-hand end is caused by a thick body with a polarization of 0.00245 contrasted with a general polarization of basement rocks of 0.002. The sharp peak in the center is caused by a basalt dike with polarization of 0.0042 contrasted with the general basement polarization of 0.002 and this produces an anomaly of some 200 gammas. On the contour map the intrabasement contact near the right side of the figure produces a broad band of magnetic increase from the 300 to about the 440 gammas contour. The dike-like body, which is shown tapering out to the north, produces a sharp anomaly at the south edge of the diagram. These are "intrabasement" features. On the contrary the small low relief anomaly at the left side of the profile with a relief of only 10 gammas is produced by the topographic feature on the basement surface designated by the letter *A*. This is of such low magnetic relief that it produces only a gentle broadening of the contour map between the 240 and the 260 gamma contours. This is a "suprabasement" feature as it does not involve a contrast of magnetic material within the basement but rather an irregularity at the basement surface.

As an approximate rule of thumb, the magnetic relief produced by a suprabasement anomaly, with polarization of 0.001 (which is a fair average value) is about 20 gammas when the vertical relief is 10 percent of the depth. For smaller or larger relief (less than 30 percent) the anomaly relief is proportionally smaller or larger. By careful inspection and analysis of the magnetic records it is possible to recognize the suprabasement anomalies and use them also for depth calculations. They are used in much the same way as the larger anomalies, i.e., by slope or half-slope measurements. Recognizing and isolating these suprabasement features from the very much stronger effects due to the intrabasement sources can be difficult and uncertain. Although the quality of depth figures obtained from them is generally inferior, they are often very useful because they usually are much more numerous and may give points of depth control in large areas where no suitable intrabasement anomalies are available.

The suprabasement anomalies may be important as indications of basement surface deformation (such as the feature *A* of Figure 60) which may have affected the overlying sediments. As mentioned above, the intrabasement contacts quite often are simply the peneplained surface of basement areas with rocks of different composition and different magnetic polarization on each side of the contact. They may produce very strong magnetic anomalies, but usually they are not indicative of corresponding deformation of the basement surface. On the other hand the suprabasement anomalies may be caused by local relief of that surface and therefore may indicate the type of feature for which deformation of the basement surface occurred after deposition of overlying sediments or by compaction over previously existing topography. For this reason a low relief suprabasement anomaly may be much more important, from the standpoint of petroleum exploration, than the much stronger and much more readily apparent intrabasement features.

ISOLATION OF ANOMALIES FOR MAGNETIC INTERPRETATION

It has been mentioned in an earlier section, in discussions of gravity analysis that certain operations such as second derivatives and filtering procedures are useful in separating anomalies from broader background disturbances. The second derivative operations in particular are quite useful in magnetic interpretation.

Depth calculations depend on slope and half-slope measurements which are multiplied by certain empirical factors to give depths to the basement surface. The magnitudes of these factors are dependent, to a small extent, on the form and particularly the length-to-width ratios of the basement body causing the anomaly (Vacquier et al, 1951). Also the azimuth correction for the angle between the flight line and the magnetic contrast boundary can be determined better if the local anomaly is more sharply defined; a second derivative map is very useful for this purpose. Furthermore the suprabasement anomalies with very weak magnetic expression are usually much more readily apparent on a second derivative map and may be important in indicating local relief on the basement surface which can cause local structure in the overlying sediments.

The second derivative calculations themselves are carried out in the same way as for gravity maps. Usually somewhat sharper coefficients can be used effectively because the magnetic measurements, being made at some distance above the nearest magnetic sources, are much smoother than most gravity maps and therefore are not as likely to have errors enhanced by the sharpening effects of the derivative calculation. For all of these reasons second derivative maps are helpful in determining magnetic basement depths, not so much for direct quantitative use, but rather as an aid in sharpening the anomalies and establishing parameters to be used for depth calculations.

AUTOMATIC INTERPRETATION

Some very sophisticated systems of magnetic interpretation are now coming into use. They depend on elaborate computer programs which process digitized airborne magnetic recording. One system uses the total intensity record and also a measured vertical gradient record and a horizontal gradient record which is the horizontal derivative of the total intensity.

It can be shown analytically that certain parameters of a causative magnetic body affect these records in ways that depend on the depth. Inversely, by analysis of the records, the magnetic body parameters can be determined. The analysis requires solution of several simultaneous equations. The parameters for the solution are determined by values on the magnetic curves at a series of points. By repeating the analysis with various intervals between the points. solutions favoring different depths are obtained. These depths are plotted automatically in the process. The final interpretation depends on selecting the most probable of the depths and assembling them as a map of the basement surface. Also, in many cases, other sources either in the sediments or below the basement surface may be indicated and included in the mapping.

CHAPTER XII
EXAMPLES OF BASEMENT MAPPING

There is a small volume of published examples of magnetic surveys and basement depth maps. Some are in areas where the interpretations have been followed by drilling, which permits a reasonably objective basis for evaluating the precision and usefulness of the results. The following examples include instances of broad-scale mapping of the basement surface to determine general basin features and also those where surveys and interpretations have been made over local basement features closely related to oil field structures.

SENEGAL

An interesting example of the application of the basement depth mapping procedures outlined above is available from a magnetic survey made for the French government in Senegal. Figure 64 (Nettleton, 1962) is a generalized geologic map of the area. From this map alone we can determine very general features only. It can be inferred that much of the map is over a basin area, because there are outcrops of older rocks to the east and there are some outcrops of igneous rocks (not indicated on the figure) in the vicinity of Dakar. This surface geologic information alone would give very little basis for evaluation of the area for its petroleum possibilities.

A rather loose reconnaissance magnetic survey was carried out over the area. This was not sufficiently closely controlled to give a detailed map of the basement surface but approximate basement depths could be determined throughout the area. Figure 65 shows generalized contours, at an interval of 1000 m, on the basement surface. While this reconnaissance has provided little detail there is now a much more specific and quantitative picture of the gross features of the basin from which to begin an evaluation of its petroleum prospects. If one were an exploration executive having a choice of certain options for selecting limited portions of the area for further exploration, one would have a much more definite basis for choice than from the surface geologic map alone.

The general conditions here may be compared with those of the Permian Basin of West Texas. Note that the area covered by the geologic map is approximately 500 km or nearly 300 miles square, which is generally comparable with the broad extent of the Permian Basin from the Central Mineral Region of Texas to eastern New Mexico. The magnetic survey covers roughly the western half of this area and is of dimensions comparable with the large western part of the Permian Basin. From the geologic map alone (Figure 64), just as in the case of the surface geologic map of the Permian Basin, there is no suggestion of the very large structure corresponding to the Central

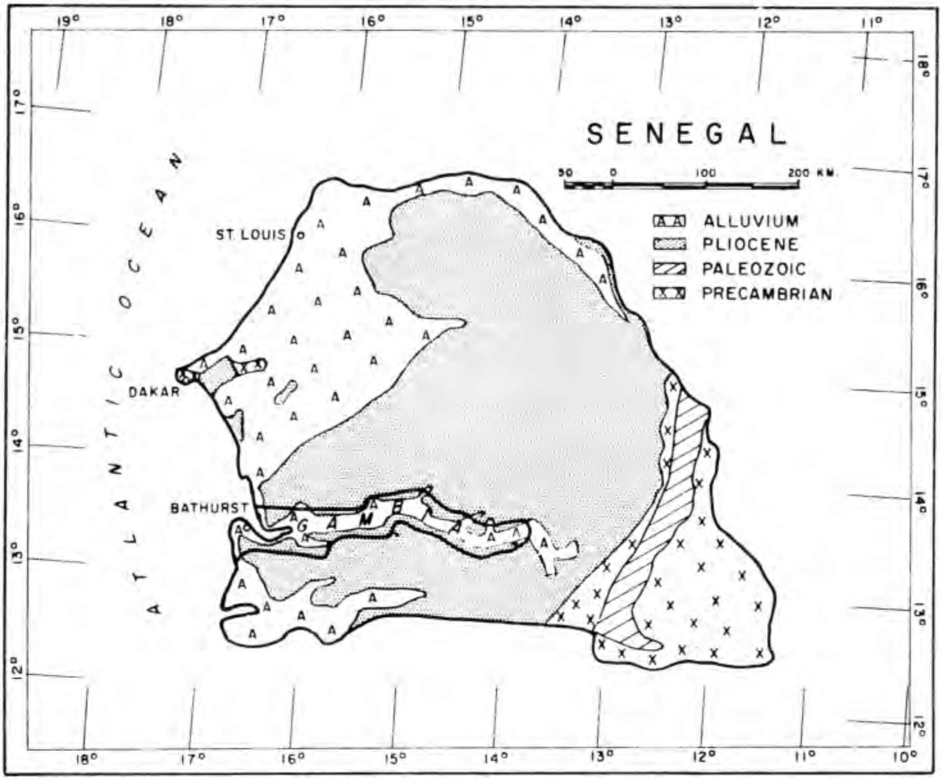

Fig. 64. Geologic map of Senegal (Nettleton, 1962, p. 1821). *Courtesy AAPG.*

Basin Platform of West Texas. In a further extension of the parallel between the two areas the magnetic basement map of Senegal also shows a sort of central basin platform, as outlined by the dashed line on Figure 65, which is about 120 km wide and 250 km long or roughly 75 by 150 miles. These dimensions are comparable with those of the Central Basin Platform. The basin at the west with maximum depth of 7000 m, with its axis running northeasterly from north of Dakar is rather comparable in horizontal dimensions and depth with the Delaware Basin. Another shallower basin on the southeasterly side of the platform is crudely comparable with the Midland Basin; and the large basin at the southwest part of the map, again with a depth of over 7000 m (well over 20,000 ft) is rather comparable in size and depth, but not in location, with the Valverde Basin. Thus we see that even a very general basement map derived from a relatively crude "once over" magnetic survey can lend a great deal of general information at small cost to an otherwise almost completely unknown geological province covered by alluvial and recent deposits so that surface geologic studies are almost useless.

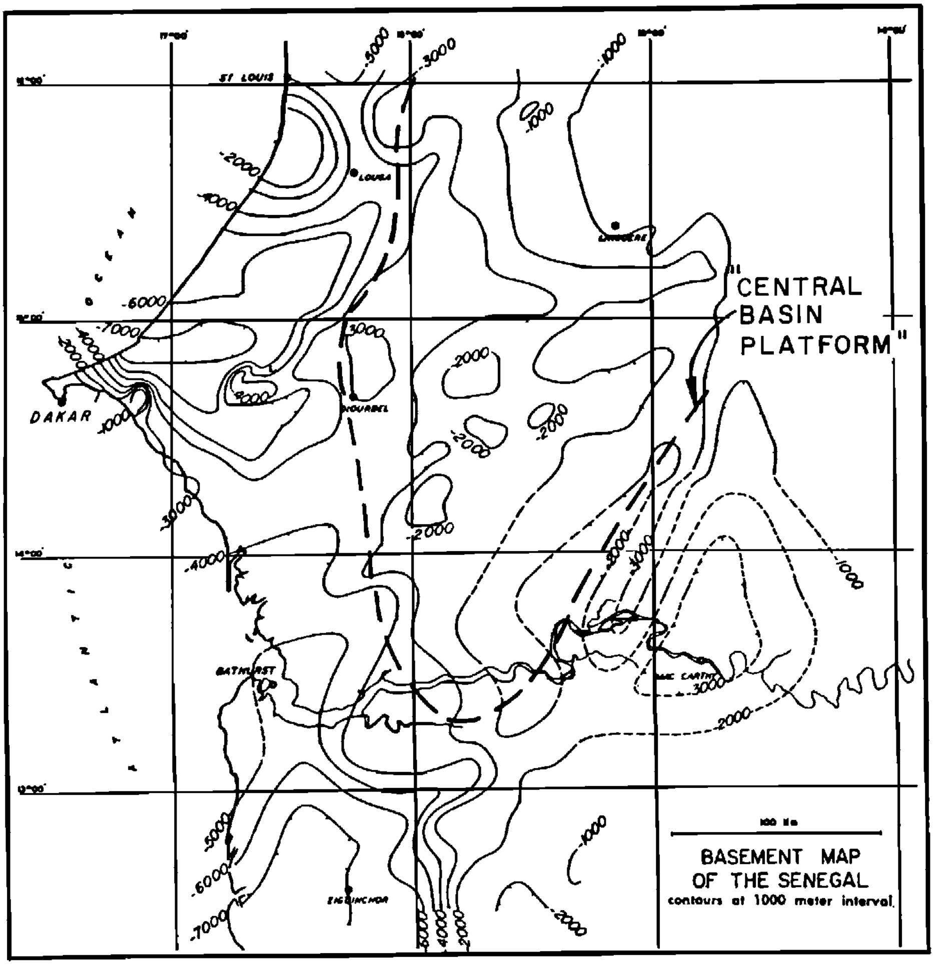

Fig. 65. Basement depth map, Senegal, calculated from reconnaissance aeromagnetic survey; contour interval, 1000 m (Nettleton, 1962, p. 1822). *Courtesy AAPG.*

PEACE RIVER: STATISTICS OF RELIABILITY

Since the procedures by which basement depth determinations are made are largely rather empirical and, to a considerable extent, are based on experience and judgment of the interpreter, it is highly desirable to have an example by which an interpretation can be objectively compared with factual information on the depth to the basement. An example for such a comparison is provided by a survey in the Peace River area of northwestern Alberta made in 1950. An area of some 27,000 square miles was covered by a fairly detailed and adequate magnetic survey and a basement map constructed. At the time of the survey, a few drilling contacts on the basement were available for very general control and checking of the interpretation. After the survey and interpretation were carried out, the area became an important oil and gas province and a great many wells have been drilled, either actually into the basement or so far into the sedimentary section that a reasonably reliable extrapolation of the depth to the basement could be made. There were 169 new wells, and each such well provided a comparison between the actual depth to the basement and that predicted from the magnetic analysis. Figure 66 (Steenland, 1963) is a histogram from this comparison and shows the distribution of differences between the basement depth from magnetic analysis and the actual depth found by drilling. The differences are expressed as errors in percent of the total basement depth from the magnetometer flight level.

The histogram is unsymmetrical with more values on the right of center (magnetic depths too shallow) than would be expected from a normal error distribution curve. Also there is some tendency for a peak a little left of the center. This analysis was made early in magnetic interpretation experience and the cause of these errors is now known to be certain systematic errors in depth parameters. These sources are now much better understood, and with modern interpretation techniques a considerably better set of statistics would result. In general it may be said that with modern methods magnetic basement maps can be made for which 50 percent of the depths are within 5 percent or less of the actual depth below flight level.

ORINOCO BASIN, VENEZUELA

An evaluation of an extensive aeromagnetic survey and interpretation from the Orinoco Basin of eastern Venezuela is available from published material (Jacobsen, 1961). In this case the survey was made, in part, to evaluate the method. The results of a basement depth interpretation are shown by the map of Figure 67. As control on the interpretation, basement depths were provided at

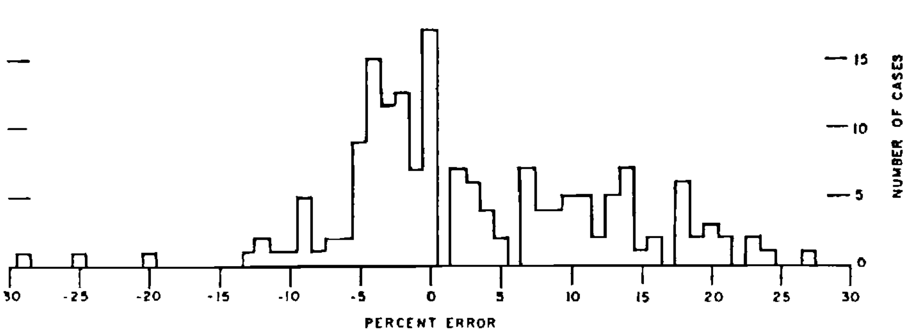

Fig. 66. Histogram of errors in basement depth estimates, Peace River, Alberta at 169 wells drilled after magnetic survey and interpretation (Nettleton, 1962, p. 1823). *Courtesy AAPG.*

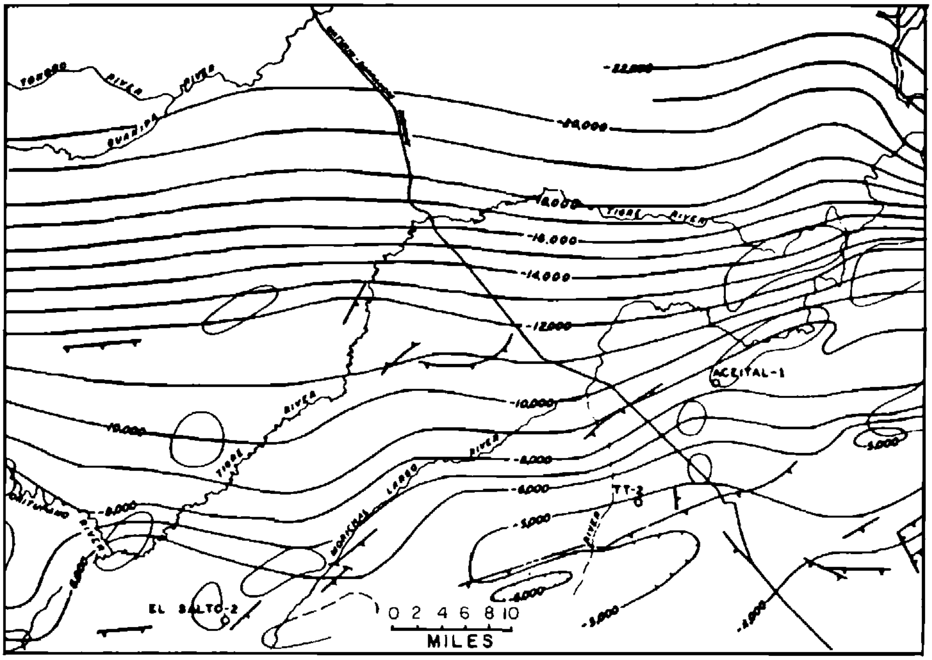

Fig. 67. Basement map, eastern Venezuela Basin, from magnetic interpretation (Nettleton, 1962, p. 1824). *Courtesy AAPG.*

the three drilling locations shown. The survey itself was not well planned as the primary control was on east-west lines with north-south tie lines in an area where the tectonic strike and magnetic trends are nearly east-west. As a result, the basement depth determinations depended, to a considerable extent, on the tie lines, which were much more nearly perpendicular to the tectonic trends.

The principle features shown by the basement map are: (1) a northerly dip of the basement surface from depths of around 4000 ft in the southeast part of the map to 22,000 ft at its northern edge; (2) a fault with about 1000 ft displacement in the southeastern part of the map and with its downthrown side to the south, i.e., it is an antithetic fault downthrown in the direction of the regionally rising basement surface; (3) a number of smaller fault indications shown by lines with hachures on their downthrown sides; and (4) a number of subcircular or elongated outlines indicating areas of possible uplift on the basement surface.

The actual basement surface is shown by Figure 68. These contours were derived from extensive drilling in the southern half of the basin and from refraction and reflection seismograph work which extended the basement surface northward beyond the area where it was defined by drilling. This information was not available to the interpreters making the map of Figure 67. This map, of course, is not identical with the one from the magnetic interpretation, but, when depths are compared, the differences are rather small. Some specific features which may be compared are:

1. Where the 17,000 ft contour from the basement interpretation crosses the highway, the actual basement contour is at about 16,000 ft.

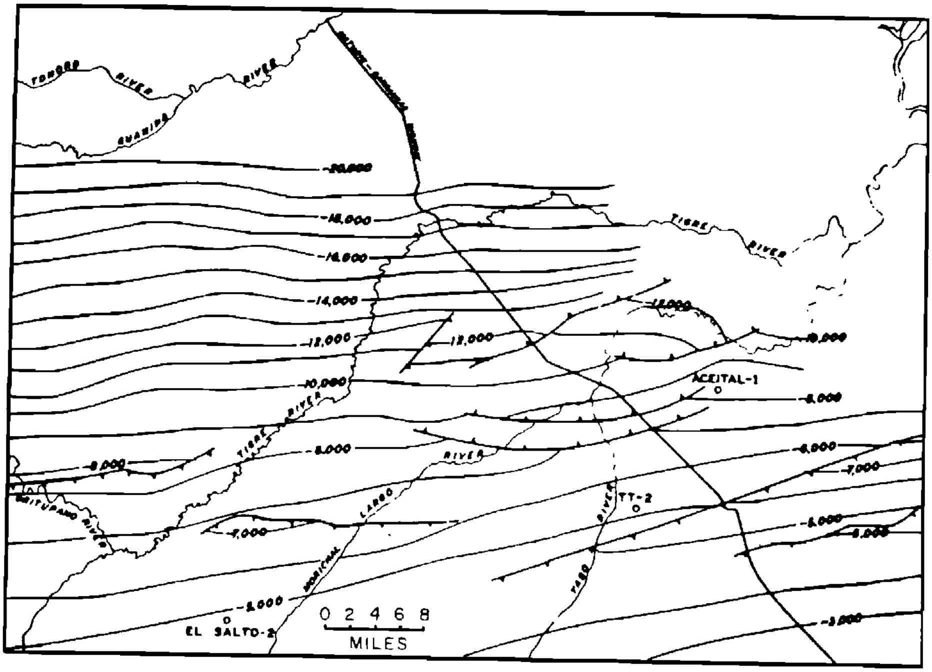

Fig. 68. Basement map, eastern Venezuela Basin, from subsurface and seismograph control (Nettleton, 1962, p.1825). *Courtesy AAPG.*

2. The fault just south of the TT-2 well has almost the identical position where it crosses the highway, nearly the same strike, and approximately the same 1000 ft of displacement to agree quite closely, except in minor detail, with the fault interpreted from the magnetic survey.

3. The more local features (faults and local uplift) generally are not confirmed definitely by the basement map although about half of the faults are near, parallel with, and have displacements in the same direction as faults shown by the map of the actual basement surface.

Since this map is generalized it may be that details corresponding with some of the magnetic features have been omitted.

MAPPING LOCAL OIL FIELD STRUCTURES

In favorable circumstances it is quite possible to map details of basement relief which have rather simple and direct control on overlying sedimentary structure. The next examples are from areas where such mapping has been proven by drilling and the development of producing structures. Illustrations are all from Steenland (1965).

The Puckett field, Pecos County, Texas

This area has very large magnetic disturbances caused by relief of the basement surface superimposed on still larger disturbances having their sources in intrabasement magnetic contrasts.

Figure 69 is the airborne magnetic map of the area from a flight line pattern of approximately one mile spacing with northeast to southwest lines. This shows a very extensive and steep gradient on the westerly side and an irregular high nose in the central part of the map.

The second derivative map (Figure 70) develops the local anomalies and shows a large closed maximum in the southwesterly part of the map and a smaller closed positive anomaly to the northeast.

The basement contour map developed from the analysis of this survey is shown by Figure 71. The very large change in depth on the western side of the map is from the differences between the depth figures of around 12,000 ft over

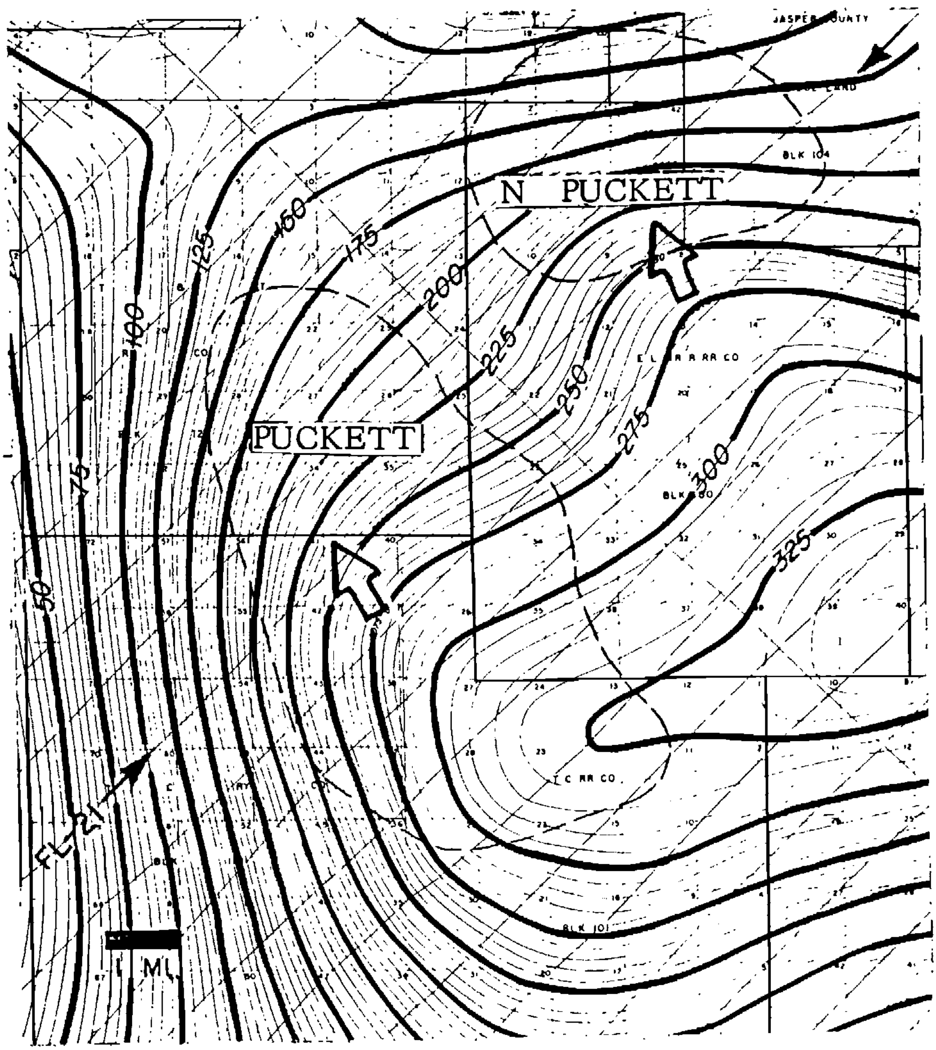

Fig. 69. Puckett oil field observed aeromagnetic map (Steenland, 1965, p. 733). *Courtesy SEG.*

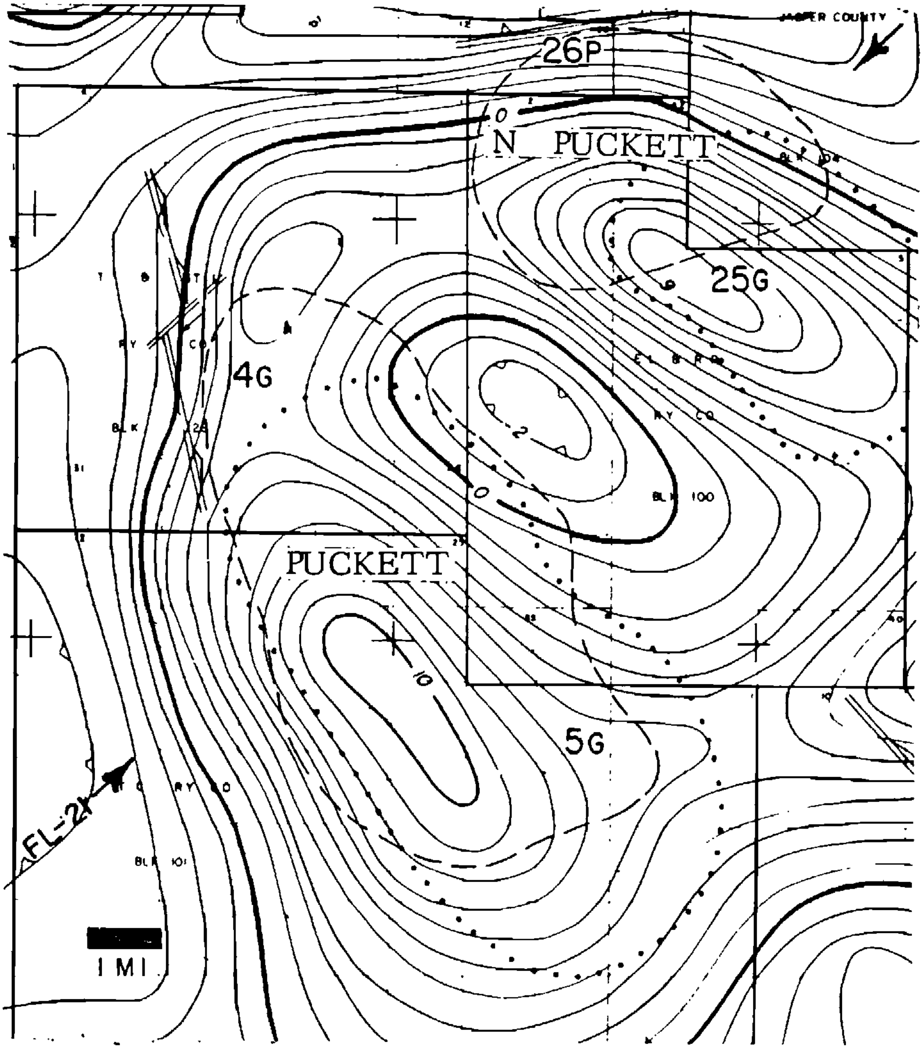

Fig. 70. Puckett oil field, second vertical derivative map, contour interval 1 x 10^{-15} cgs (Steenland, 1965, p. 734). *Courtesy SEG.*

the central structure and depths of over 20,000 ft just west of the area shown. This magnitude of structural relief on the basement surface is calculated from the local, suprabasement anomaly over the structure as is illustrated by the sample record shown by Figure 72. The resolution of the magnetic profile into the components due to intrabasement contrasts and those due to the structure itself (suprabasement contributions) is indicated on this profile.

Finally Figure 73 shows the actual structure of the area as developed by the drilling with contours on the Ordovician-Cambrian Ellenburger Limestone which is deep in the sedimentary section and probably reflects fairly well the

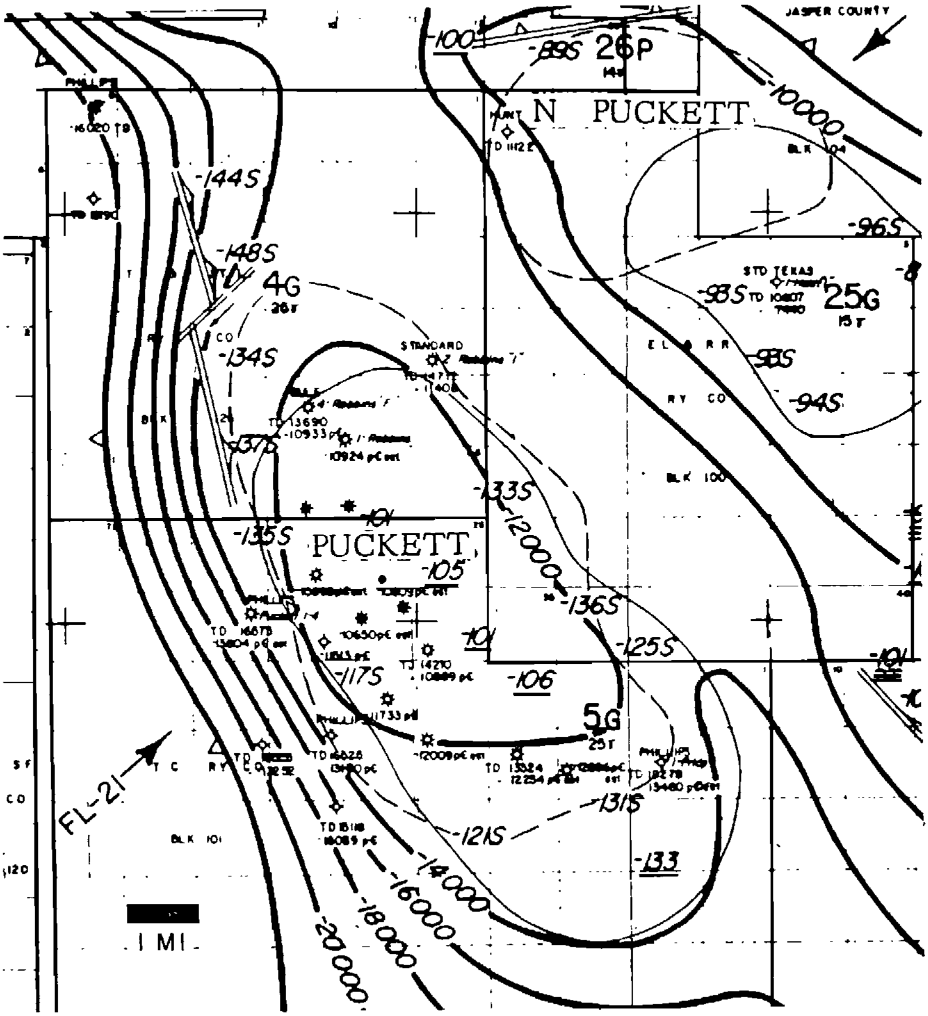

Fig. 71. Puckett field, magnetic basement, contour interval 2000 ft on the Precambrian, October, 1963 (Steenland, 1965, p. 735). *Courtesy SEG.*

actual deformation of the basement surface. The location of the Puckett structure itself and the large fault, with displacement of some 10,000 ft, on the west side are quite close to the predictions from the magnetic analysis. The North Puckett structure is also shown generally although the agreement in location is not as good as that of the main Puckett field. The general configuration with the terracing between the two producing areas and the fault on the west side of the North Puckett field are rather similar to the indications from the magnetic basement contour map.

This is a good illustration of the difference in magnitude of contributions from intrabasement and suprabasement magnetic anomalies. From the magnetic profile, Figure 72, we see a general rise

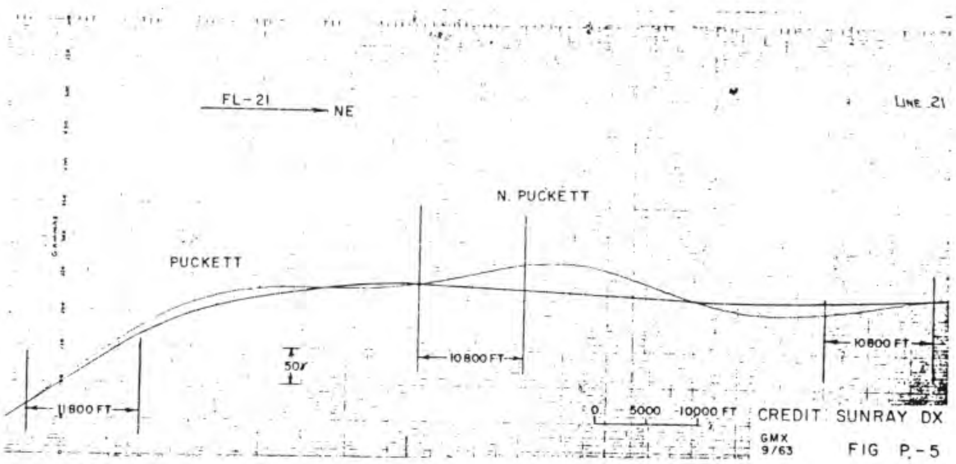

Fig. 72. Total magnetic intensity record, flight line 21 of Figure 71, with resolution into intrabasement and suprabasement components (Steenland, 1965, p. 737). Courtesy SEG.

of some 200 gammas or more due to an intrabasement magnetic contrast. Superimposed on that is a rise of only some 30 gammas as the expression of the suprabasement effect from the great Puckett structure with the relief of the order of 10,000 ft. This is a further example of the differences in magnitude of intrabasement and suprabasement contrasts illustrated by the schematic section and magnetic profile of Figure 60.

Swanson River field, Alaska

This example illustrates a case where the suprabasement effect is so strong that the magnetic map without much processing or analysis gives a fairly close picture of the local deformation of the basement surface and of the overlying structure and oil field. Figure 74 is the observed magnetic map and shows a local anomaly of approximately 100 gammas relief. This anomaly location and shape are further refined and more definitely outlined by the second derivative map Figure 75. The basement depth map derived from a combination of depth estimates from both intrabasement and suprabasement components is shown by Figure 76. Finally, Figure 77 shows the actual structure on the Hemlock zone, approximately 1000 ft above the basement, which is generally quite conformable with the basement surface derived from the magnetics but has somewhat less relief.

Tioga field, North Dakota

This example is somewhat the inverse of that at Swanson River where the simple observed magnetic anomaly has a form and magnitude consistent with the local basement structure. At Tioga the axis and general location of a very large magnetic anomaly correspond rather closely with the oil field structure but the magnitude and extent do not. At Tioga a very large magnetic anomaly, only partially shown by Figure 78, includes the area of the producing structure and extends considerably farther both north and south along strike. The second derivative map, Figure 79, develops some sharper details on the magnetic anomaly but nothing that corresponds very clearly with the oil field structure. The basement depth map, Figure 80, shows a broad uplift on the basement surface with the oil field struc-

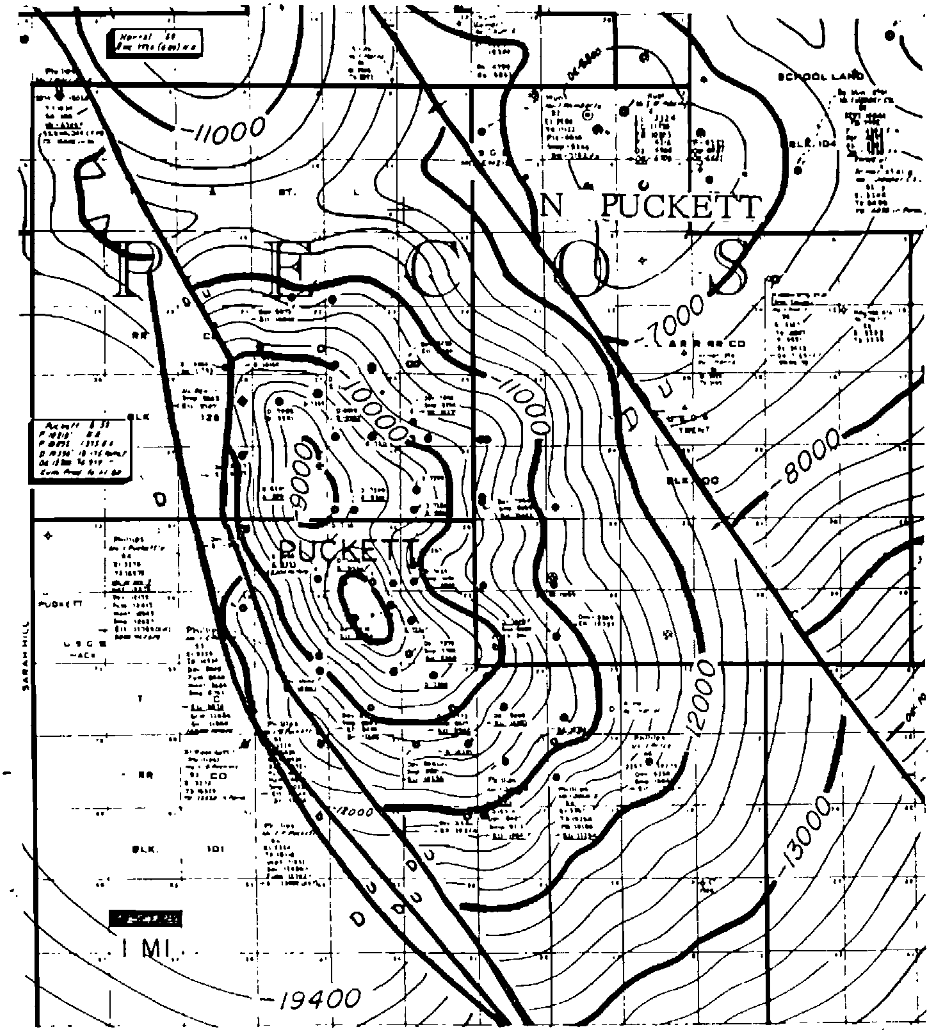

Fig. 73. Puckett field, subsurface contours on the Ellenburger, showing structure from drilling (Steenland, 1965, p. 736). *Courtesy SEG.*

ture on its easterly flank. The producing structure, Figure 81, has much less relief than the broad basement structure. Apparently it was generally influenced but not closely controlled by movement of the basement.

The very large magnetic anomaly with the relief of some 300 gammas clearly has its origin in magnetic contrasts deep within the basement rocks. Minor readjustments or movements within this block of basement material must have occurred after the deposition of the overlying Paleozoic sediments including the producing horizon. In this case the correspondence of the structure which made the oil field with the very large magnetic anomaly is very largely fortui-

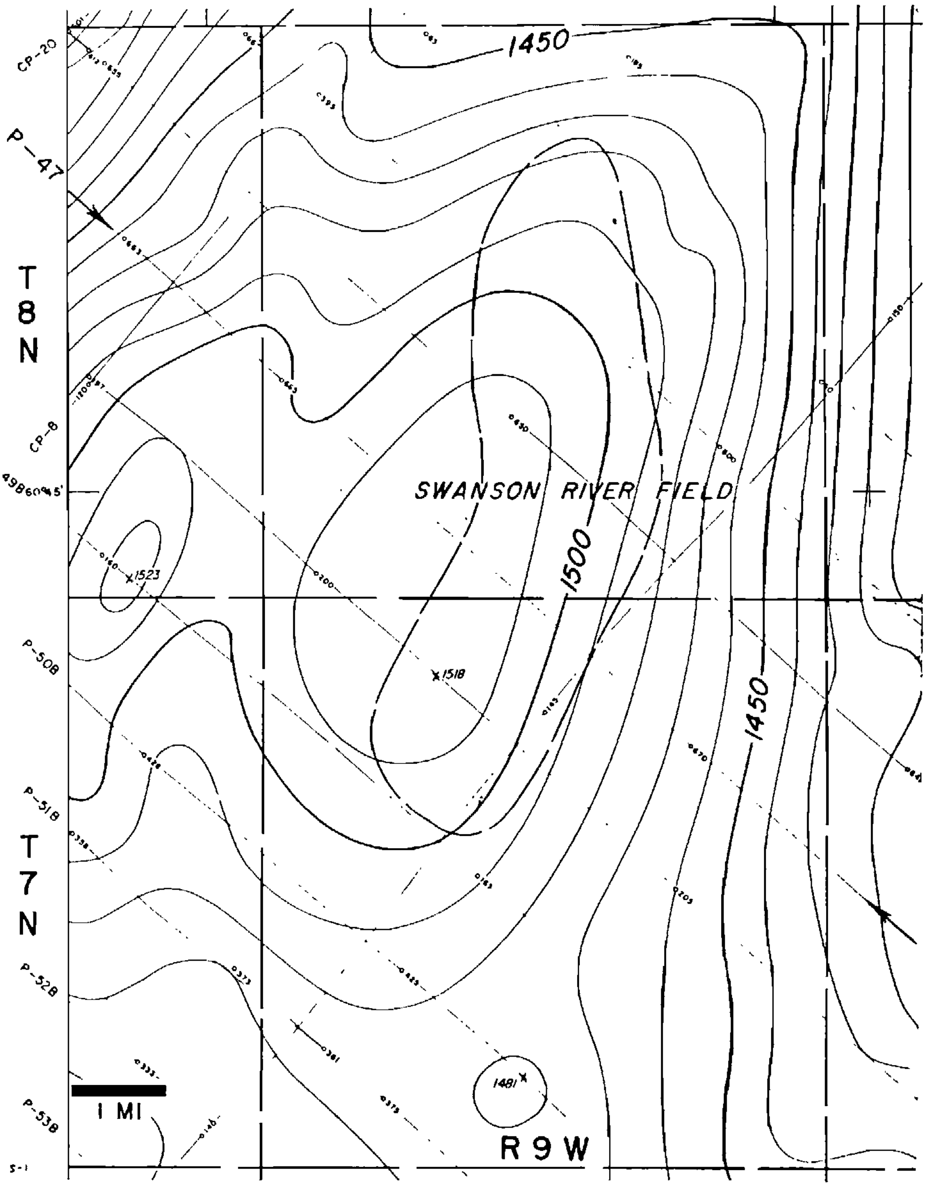

Fig. 74. Swanson River field, Alaska, observed aeromagnetic map, contour interval 10 gammas (Steenland, 1965, p. 713). *Courtesy SEG.*

tous and exploration based on such large magnetic (or gravity) features as indications of local sedimentary structure would be very unproductive.

There are other examples of situations of this kind which are the exceptions that prove the rule. The Hobbs field in southeastern New Mexico, discovered in

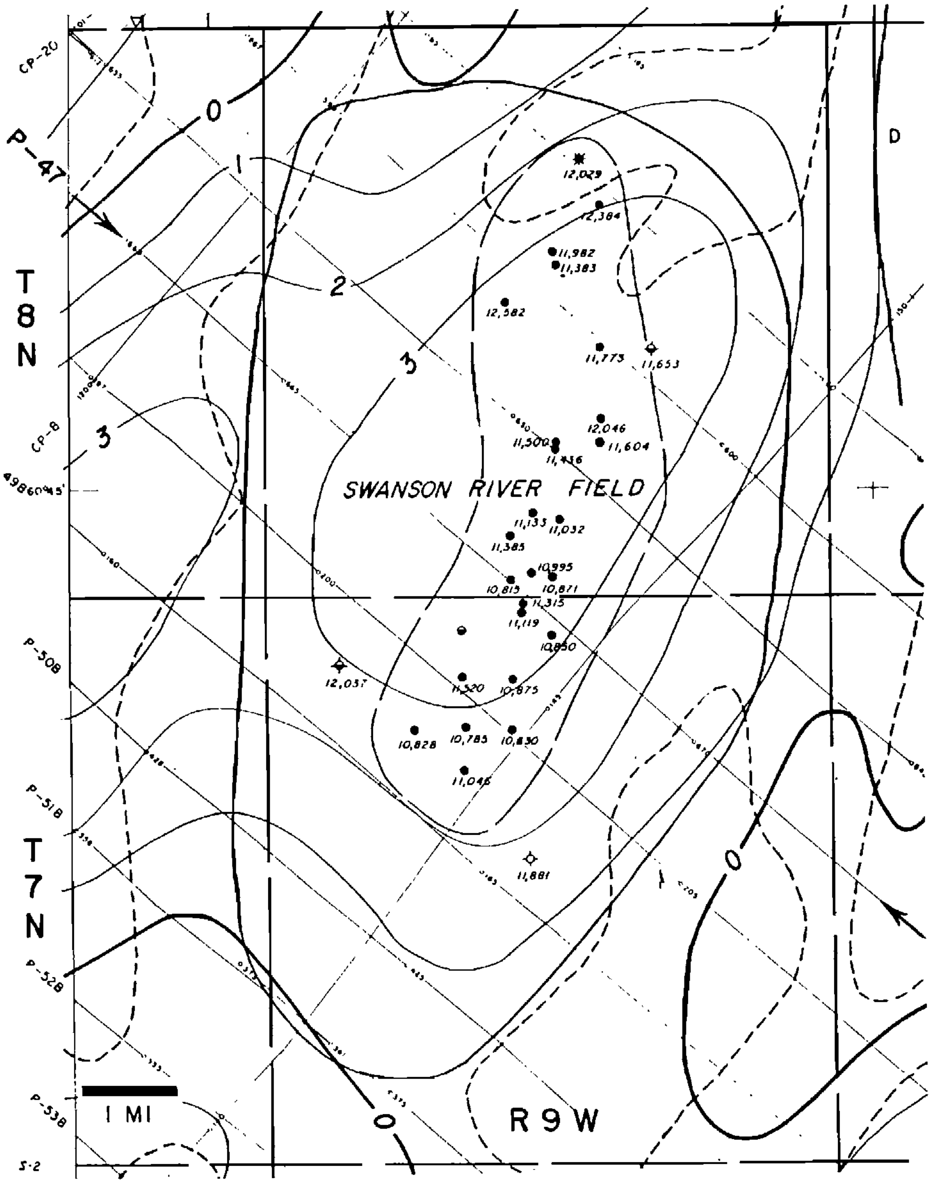

Fig. 75. Swanson River, Alaska, second vertical derivative map, contour interval 1×10^{-15} cgs (Steenland, 1965). *Courtesy SEG.*

1928, has been attributed to drilling on a magnetic feature and, indeed, there is a very large magnetic anomaly, the crest of which is within the Hobbs producing area. Following this discovery extensive magnetic surveys were made in West Texas and tests were drilled on many large magnetic anomalies (some of

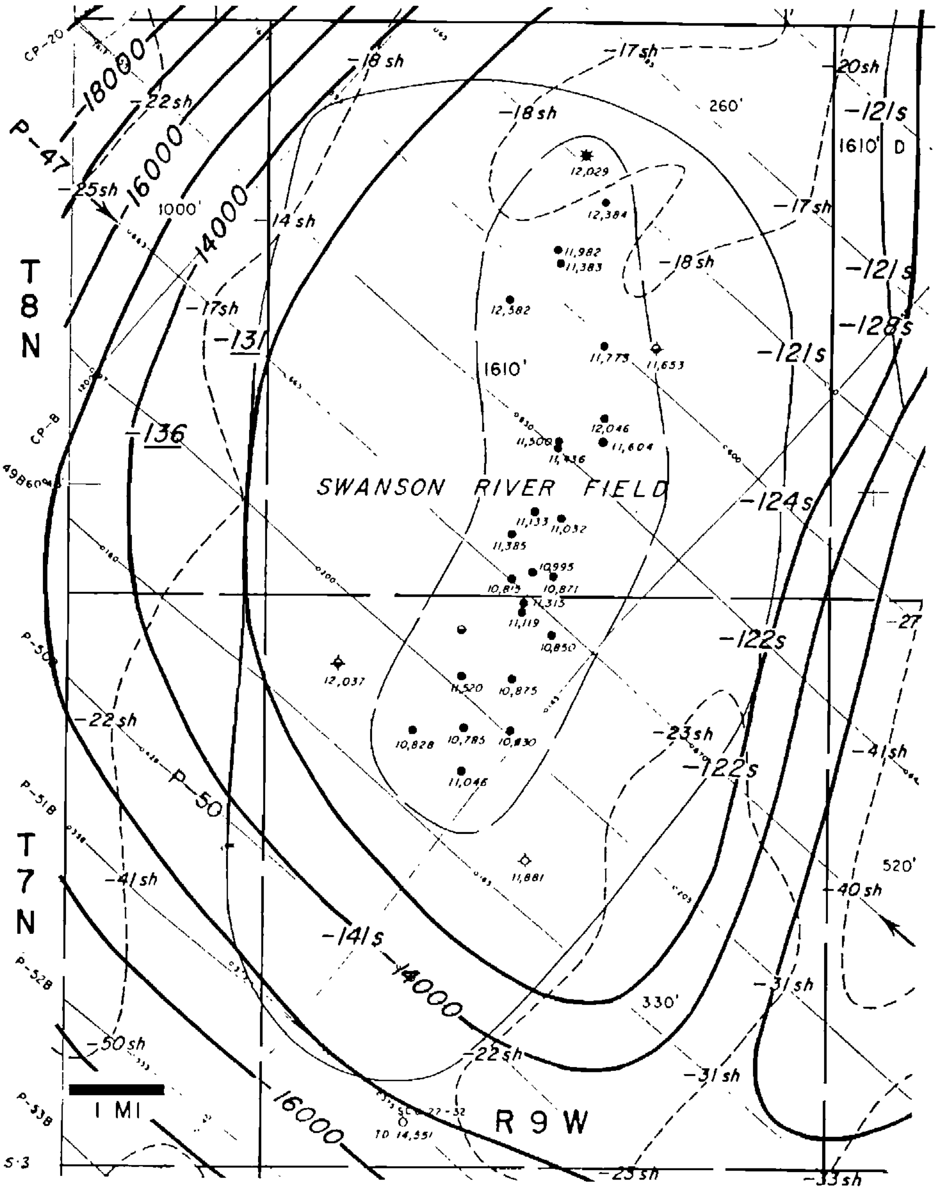

Fig. 76. Swanson River field, Alaska, contours on magnetic basement, contour interval 1000 ft (Steenland, 1965, p. 715). *Courtesy SEG.*

which are also strong gravity anomalies) but none of these found production. Later development of the Hobbs field shows it to extend considerably beyond the magnetic feature and also shows that it does not conform in outline or pattern with the magnetic anomaly. Quite clearly this and the other large

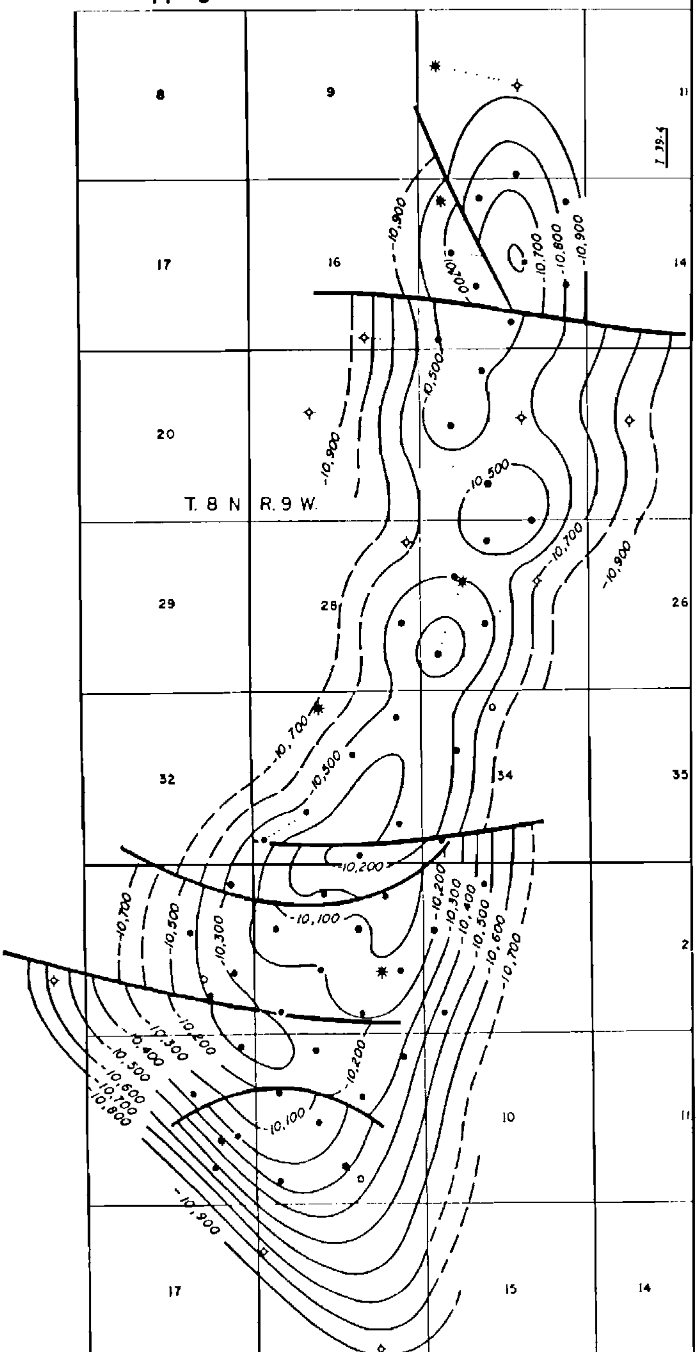

Fig. 77. Swanson River, Alaska, subsurface contours on Hemlock Zone (Steenland, 1965, p. 716). *Courtesy SEG.*

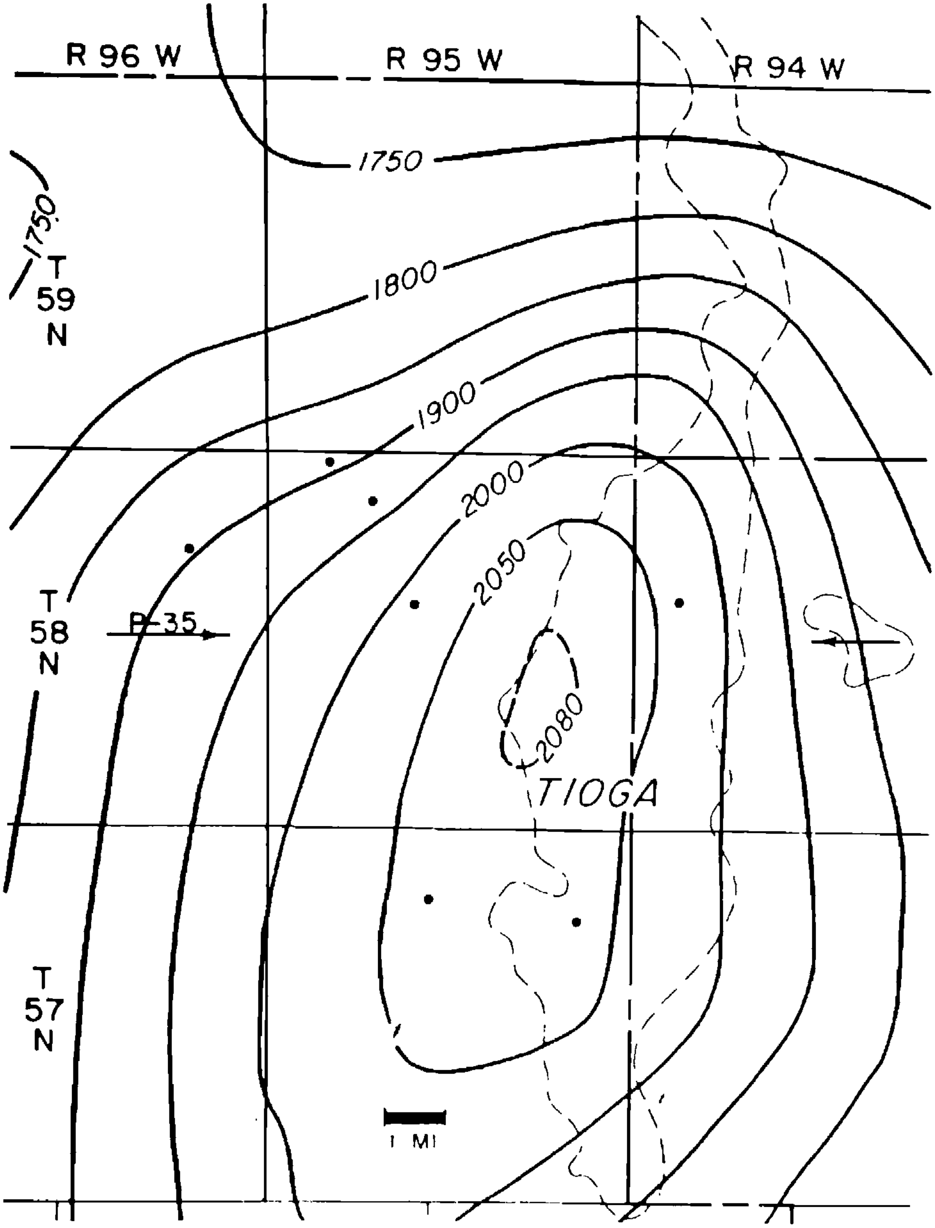

Fig. 78. Tioga field, North Dakota, observed aeromagnetic map, contour interval 50 gammas (Steenland, 1965, p. 724). *Courtesy SEG.*

magnetic anomalies of southeast New Mexico and West Texas have their origin in magnetization contrasts deep within the basement rocks which generally, have no relation to local irregularities of the basement surface. A particularly striking

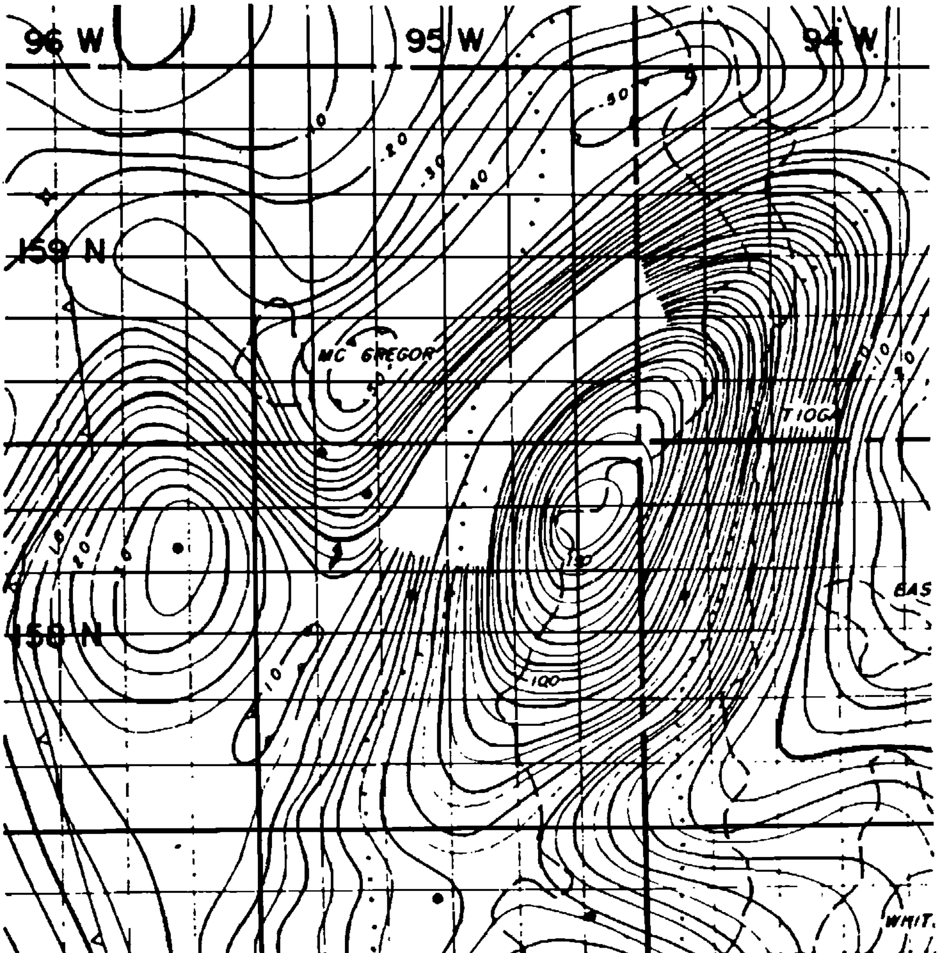

Fig. 79. Tioga field, North Dakota, second vertical derivative map, contour interval 0.5 x 10 $^{-15}$ cgs (Steenland, 1965, p.725). *Courtesy* SEG.

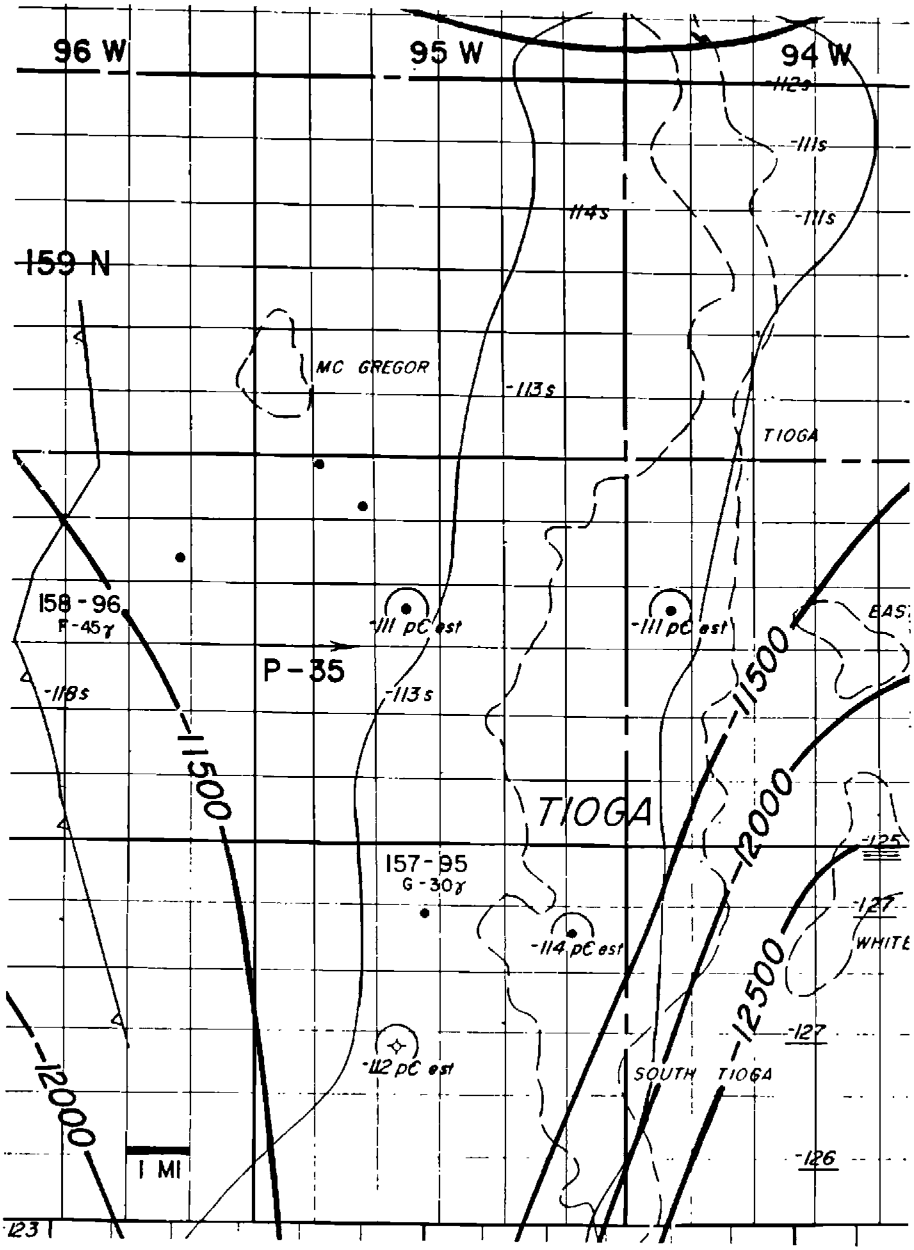

Fig. 80. Tioga field, North Dakota, magnetic basement map, contour interval 500 ft (Steenland, 1965, p. 726). *Courtesy SEG.*

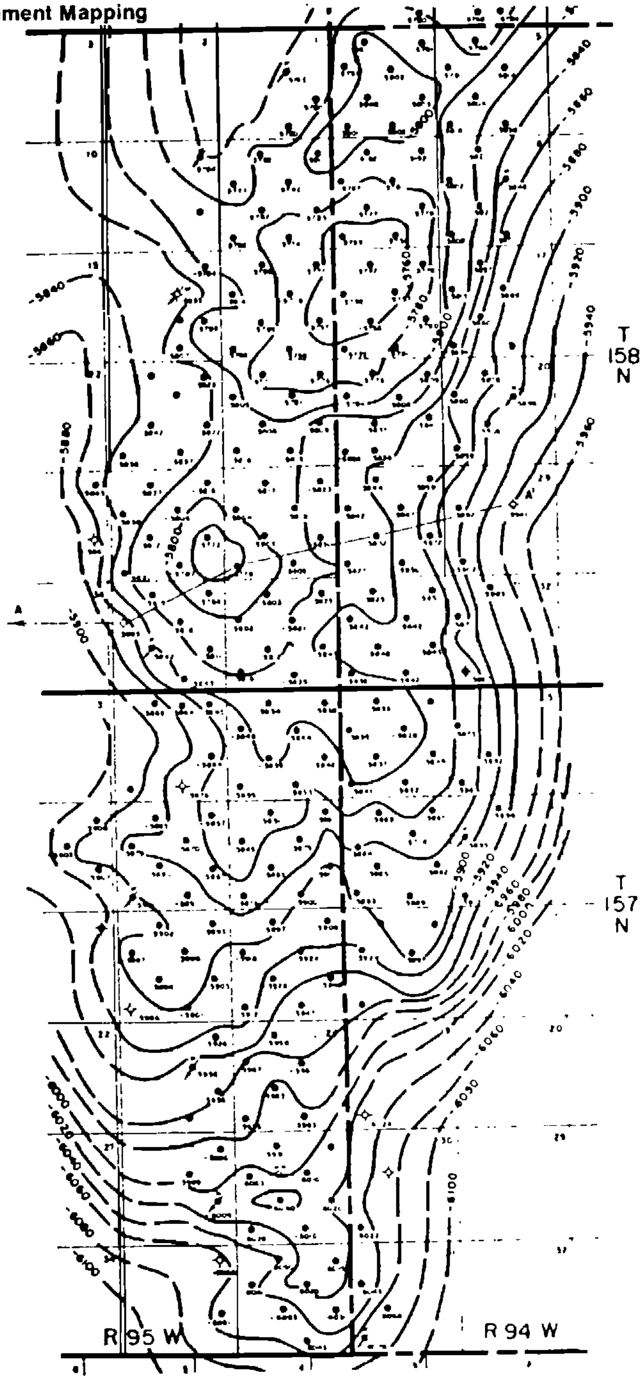

Fig. 81. Tioga field, North Dakota, subsurface contours on top of Nesson sand (Steenland, 1965, p. 727). *Courtesy SEG.*

example is the very large gravity and magnetic anomaly centering in southeastern Crosby County Texas. This is a nearly circular feature with very strong and rather closely corresponding gravity and magnetic anomalies with local relief of the order of 45 mgals and 1500 gammas respectively. A number of wells have been drilled to the basement in the area and the basement surface is quite monotonous without any evidence of local uplift or disturbance of the basement surface. Apparently these very large intrabasement density and magnetic contrasts were formed some time in the pre-Cambrian and the surface was rather completely peneplained before deposition of the sediments and no later movement or adjustment of this mass has occurred to produce structure in these sediments.

CHAPTER XIII

REVIEW AND CONCLUSIONS

The foregoing text and illustrations have attempted to point out the possibilities and limitations of gravity and magnetic methods in petroleum exploration. Some readers may be more impressed by the limitations, particularly the ambiguity and lack of precise results in terms of form, depth, and geological definition which result from applications of these methods. On the other hand it is hoped that the possibilities of the methods may be better appreciated and that a greater participation of "the other five percent" in the overall petroleum exploration picture may be encouraged.

Since petroleum exploration is now very much dominated by the reflection seismic method it may be well to point out more specifically than is generally appreciated the relations between the potential and the seismic methods and their complementary contributions to overall understanding of the subsurface geology.

The ideal order of application of the three geophysical methods would be first, reconnaissance with airborne magnetics, followed by gravity surveys in a degree of detail depending on the geological objectives and the facility of getting over the surface, and finally by reconnaissance and detailed reflection-seismograph surveys with the detailing being directed to those areas where gravity or magnetics or both point to a feature of particular interest. This idealized overall exploration approach is rarely used, partly because of land or concession or competitive considerations, and partly because of a lack of appreciation of the complementary nature of the different geophysical methods. In many instances the argument is used that an area probably will be covered eventually by the reflection seismograph method and that having such a survey the potential methods would not contribute enough to justify their cost.

There are many geological situations where the potential methods can contribute independent geological information which will add critical or decisive information to the geologic knowledge. As a very simple example there have been seismic surveys in areas where domal uplifts are found, but where their geologic nature is not known, and where they might be caused by a sedimentary structural uplift, a salt dome, or an igneous intrusion. Gravity and magnetic surveys should answer these questions unequivocably. A negative gravity anomaly with no magnetic anomaly would indicate a salt dome. A low relief positive gravity anomaly with no magnetic anomaly or with a very low relief magnetic anomaly would indicate a sedimentary uplift with or without (depending on the magnetic expression) an underlying basement uplift. A strong positive gravity anomaly with a strong positive magnetic anomaly would indicate an igneous intrusive.

Seismic interpretations in faulted areas often show a fault by a zone of no reflections and, especially in rather poor record areas, there may be no definite basis for determining the direction of displacement of the fault. Gravity information can very often find corresponding fault indications and will, of course, indicate the direction of displacement.

Magnetic surveys have their most obvious and economically important application in the exploration of new areas, particularly offshore areas. Rapid and relatively inexpensive airborne magnetics can determine the depth to the basement, usually very reliably and to a precision of the order of 5 percent or less of the depth below the flight level. Also, as has been demonstrated in some of the illustrated examples, it is often possible to determine local relief of the basement which has corresponding oil field structure in the overlying sediments. In a new area where the thickness of the sediments is unknown, a magnetic basement map can determine those basin areas where the sediments are thick enough to make reasonable petroleum prospects and to guide later exploration there.

A question may be raised as to whether the magnetic basement and the effective geologic basement are the same. This is usually true, but there are exceptions as where large thicknesses of nonmagnetic metamorphosed rocks are present and where the "geologic basement" is the top of the metamorphics and the magnetic basement is at their base. An example is a great thickness of nonmagnetic quartzite which is found in the vicinity of the Uinta Mountains in Utah. However, from worldwide experience in the application of the magnetic method to basement mapping, the instances where the basement depth can be grossly wrong are rare, and usually careful consideration of all the facts will reveal clues which can suggest that a gross discrepancy is present. Therefore, on the whole, magnetic basement mapping has proven to be a very reliable and useful part of the overall exploration picture.

In addition to the usefulness in locating local structures, gravity surveys are often helpful in evaluating larger geologic units. An example has been given (page 52) of the simple relations of the basin depth to the gravity anomaly in the Los Angeles basin. Similar relations generally apply to the smaller basins with horizontal dimensions of a few tens of miles, such as the inter-montane basins of the basin and range area of Utah, Nevada, and eastern California. The nature of larger basins often can be inferred. For instance in the great valley of California (as in other basins in the Rocky Mountains and in other parts of the world) an asymetrical basin with a fore-deep on one side is distinctly indicated by a regional gravity minimum over the area of thickest sediments. For large basins, however, information on the location and form of gross features within the basin, and quantitative information on the thickness of the sediments can be much more reliably determined from magnetic than from gravity data. This is illustrated by the example from Senegal (page 95). All such definite information of a regional nature, obtainable from relatively low-cost magnetic and gravity surveys, contributes to the regional geologic picture which needs to be taken into account in broad-scale exploration planning.

Perhaps the simplest approach that can be made to the "first look" at a new and unknown area is aerial photography for photogeologic mapping and airborne magnetics for mapping the basement. Then the entire sedimentary section is "bracketed" between some knowledge of its surface and a definite knowledge of its total thickness. The surface part, of course, is not available in the broad water covered areas which are becoming such an important part of exploration planning, but the airborne magnetics and basement depth determination are still possible and can be very helpful.

The overall exploration problem may be compared with a detective story. In a well-constructed story the "whodunit" has many possibilities at the beginning. As more and more clues are developed the number of possibilities becomes more and more limited until finally only one remains. In the exploration analog, the "criminal" to be found and identified is the true geological situation. With a single exploration method (surface geology or one of the geophysical methods) the number of geological possibilities is relatively large. As more "clues" are brought to bear by application of more geological and geophysical techniques the number of possibilities is reduced until finally, hopefully, the actual geological situation is revealed by finding the set of geological circumstances which is consistent with all the information available.

BIBLIOGRAPHY

AMBAC Industries, 1967, The D 4c temperature controlled accelerometer system: Rev. B., August 28.

Coons, R. L., Woollard, G. P., and Hershey, G., 1967, Structural significance of the mid-continent gravity high: Bull. AAPG, v. 51, no. 12, p. 2381-2399.

Cordell, L., and Henderson, R. G., 1968, Iterative three-dimensional solution of gravity anomaly data using a digital computer: Geophysics, v. 33, p. 596-601.

Fuller, Brent D., 1967, Two-dimensional frequency analysis and design of grid operators: Mining Geophysics, v. II, Theory, The Society of Exploration Geophysicists, p. 658-708.

Graf, A., 1958, Das seagravimeter: Zeit. f. instrumentenkunde, v. 60, no. 8, p. 151-161.

Grant, F. S., and West, G. F., 1965, Interpretation theory in applied geophysics: New York, McGraw-Hill Book Co. Inc.

Howell, L. G., Heintz, K. O., and Barrs, A., 1966, The development and use of a high precision downhole gravity meter: Geophysics, v. 31, p. 764-772.

Jacobsen, P., Jr., 1961, An evaluation of basement depth determinations from airborne magnetometer data: Geophysics, v. 26, p. 390-399.

LaCoste, L. J. B., 1967, Measurement of gravity at sea and in the air: Reviews of Geophysics, v. 5, p. 477-526.

LaCoste, L. J. B., Clarkson, N., and Hamilton, G., 1967, LaCoste & Romberg stabilized platform shipboard gravity meter: Geophysics, v. 32, p. 99-109.

Nettleton, L. L., 1940, Geophysical prospecting for oil: New York, McGraw-Hill Book Co. Inc., p. 442.

——— 1942, Gravity and magnetic calculation: Geophysics, v. 7, p. 293-310.

——— 1950, Interpretation of aeromagnetic surveys in Western Canada: Oil in Canada, July 31.

——— 1954, Regionals, residuals, and structures: Geophysics, v. 19, p. 1-22.

——— 1962, Gravity and magnetics for geologists and seismologists: Bull. AAPG, v. 46, no. 10, p. 1815-1838.

Slack, H. A., Lynch, V. M., and Langan, L., 1967, The geomagnetic gradiometer: Geophysics, v. 32, p. 877-892.

Steenland, N. C., 1963, An evaluation of the Peace River aeromagnetic survey: Geophysics, v. 28, p. 745-755.

——— 1965, Oilfields and aeromagnetic anomalies: Geophysics, v. 30, p. 706-739.

Vacquier, V., Steenland, N. C., Henderson, R., and Zietz, I., 1951, Interpretation of aeromagnetic maps: Geol. Soc. of America, Memoir 47, p. 151.

Wing, C. G., 1967, An experimental deep-sea-bottom gravimeter: J. of Geophys. Res., v. 72, no. 4, p. 1249-1257.

Woollard, G. P., 1959, Crustal structure from gravity and seismic measurements: J. of Geophys. Res., v. 64, no. 10, p. 1521-1544.

Wycoff, R. W., 1948, The Gulf airborne magnetometer: Geophysics, v. 13, p. 182-208.

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